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深度学习驱动的埃塞俄比亚阿拉比卡咖啡疾病分类和阶段预测

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A Dissertation Submitted in Partial Fulfillment of the Requirements for the Master Degree in Engineering

Deep Learning-Driven Classification and Stage Prediction of Diseases in Ethiopian Arabica Coffee

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摘 要

咖啡叶部病害（如 Phoma 叶斑病和咖啡潜叶蛾）导致埃塞俄比亚阿拉比卡咖啡产业减产 20-30%，威胁超过 1500 万小农户的生计。传统人工诊断方法存在 65% 准确率上限、流程耗时且无法评估病害严重程度，难以为农村地区提供及时干预。本研究开发了一种适用于埃塞俄比亚农村的轻量化深度学习系统，可分类病害类型（健康、Phoma、潜叶蛾）及严重程度（无症状、轻度、重度）。

研究采用卷积神经网络（CNN），结合卷积块注意力模块（CBAM）与多任务学习（MTL），基于从 Jimma、Sidama 和 Yirgacheffe 当地农场采集的 890 张图像数据集进行训练。方法包括数据增强、100 轮次训练、CBAM 移除与 MTL 单任务替换的消融实验，以及 VGG16、ResNet50 和 MobileNetV2 的迁移学习。MobileNetV2 因优异的严重程度分类性能和轻量化架构被选为 CBAM 集成对象。

提出的 CNN+CBAM+MTL 模型实现病害分类 97.75% 和严重程度 93.26% 的准确率，MobileNetV2+CBAM+MTL 则分别达到 97.75% 与 94.38%，精确率、召回率和 F1 分数验证了其有效性。消融实验证实 CBAM 将严重程度预测性能提升 12.36%。

部署采用 2.5 MB 的 TensorFlow Lite 模型支持离线移动端与网页应用，单图推理时间 67.73 毫秒。该方案优于传统方法，可实现早期病害检测以减少减产和农药滥用，支撑占埃塞俄比亚 GDP 5-10% 的咖啡经济。CBAM 的创新集成将诊断精度提升至超越既往缺乏严重度分期的农业 AI 模型。未来工作将扩展数据集并探索无人机成像技术。

关键词：深度学习；注意力机制；多任务学习；咖啡叶病害；埃塞俄比亚农业

Abstract

Coffee leaf diseases, such as Phoma leaf spot and Coffee Leaf Miner, cause 20-30% yield losses in Ethiopia's Arabica coffee sector, threatening the livelihoods of over 15 million smallholder farmers. Manual diagnostics, limited by 65% accuracy, labor-intensive processes, and inability to assess disease severity, are impractical for rural farmers, delaying interventions. A lightweight deep learning system designed for Ethiopia's rural regions is developed to classify disease types (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe).

Convolutional Neural Networks (CNNs), enhanced by Convolutional Block Attention Modules (CBAMs) and Multi-Task Learning (MTL), were utilized with a dataset of 890 images collected from Jimma, Sidama, and Yirgacheffe local farm regions. Techniques included data augmentation, 100-epoch training, ablation studies on CBAM removal and single-task substitution for MTL, and transfer learning with Visual Geometry Group 16 (VGG16), Residual Network 50 (ResNet50), and Mobile Network Version 2 (MobileNetV2). MobileNetV2, selected for its superior severity classification performance and lightweight architecture, was chosen for CBAM integration.

The proposed CNN+CBAM+MTL model achieves 97.75% disease and 93.26% severity accuracy, while MobileNetV2+CBAM+MTL attains 97.75% and 94.38%, respectively, validated by precision, recall, and F1-scores. Ablation studies confirm CBAM improves severity prediction by 12.36% over baselines.

Deployment employs a 2.5 MB TensorFlow Lite model for offline mobile and web applications, with 67.73 ms inference time per image. This solution outperforms traditional methods, enabling early detection to reduce yield losses and pesticide overuse, supporting Ethiopia's coffee economy, which contributes 5-10% to gross domestic product. The novel integration of CBAMs enhances diagnostic precision beyond prior agricultural AI models lacking severity staging. Future work involves expanding the dataset and exploring drone-based imaging.

Key words: Deep Learning; Attention Mechanisms; Multi-Task Learning; Coffee Leaf Disease; Ethiopian Agriculture

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List of Abbreviations

Acc - Accuracy
AI - Artificial Intelligence
API - Application Programming Interface
CBAM - Convolutional Block Attention Module
CLD - Coffee Leaf Disease
CNN - Convolutional Neural Network
DL - Deep Learning
EA - Ethiopian Agriculture
F1 - F1-Score
GDP - Gross Domestic Product
GPU - Graphics Processing Unit
HRNet - High-Resolution Network
IoT - Internet of Things
IPM - Integrated Pest Management
LCBAM - Lightweight Convolutional Block Attention Module
MD - Mobile Deployment
MLP - Multilayer Perceptron
MobileNetV2 - Mobile Network Version 2
MTL - Multi-Task Learning
P - Precision
PIL - Python Imaging Library
R - Recall
ResNet50 - Residual Network
SLIC - Simple Linear Iterative Clustering
TF - TensorFlow
TFLite - TensorFlow Lite
VGG16 - Visual Geometry Group 16 Neural Network
YOLO - You Only Look Once

1 Introduction

This chapter lays the foundation for the study by presenting the background and significance of coffee leaf disease management in Ethiopia's Arabica coffee sector. An exploration of the historical, economic, and cultural context of coffee production is provided, followed by an identification of the challenges posed by diseases such as Phoma leaf spot and Coffee Leaf Miner. The problem statement is articulated to highlight the limitations of current practices, leading to the formulation of research objectives and questions that guide the investigation. The chapter concludes with an outline of the study's scope, its potential contributions, and the organization of the thesis, setting the stage for subsequent chapters

1.1 Background and Context

Ethiopia is widely acknowledged as the birthplace of *Coffee Arabica*, a species integral to the nation's historical narrative and its influence on global coffee culture[1]. The origins of coffee are traced back to the 9th century, tied to the legend of Kaldi, a goatherd from the Kaffa region who is said to have discovered the stimulating effects of coffee cherries.



Figure 1.1 Image showing Ethiopian Arabica Coffee

This legacy has established Ethiopia as Africa's leading coffee producer and the world's fifth-largest exporter. Coffee cultivation is concentrated in key regions such as Sidama, Yirgacheffe, and Oromia, where favorable altitudes, fertile soils, and climatic conditions create an ideal environment for growing high-quality coffee[2–4]. These regions not only define Ethiopia's coffee production geographically but also anchor its historical significance in the global market.

The economic and social importance of coffee in Ethiopia cannot be overstated. Coffee contributes approximately 5-10% to the country's gross domestic product (GDP) and accounts for around 30% of its foreign exchange earnings, solidifying its status as a top export commodity ①. More than 95% of this production is driven by smallholder farmers, who cultivate coffee on small, fragmented plots averaging less than two hectares.

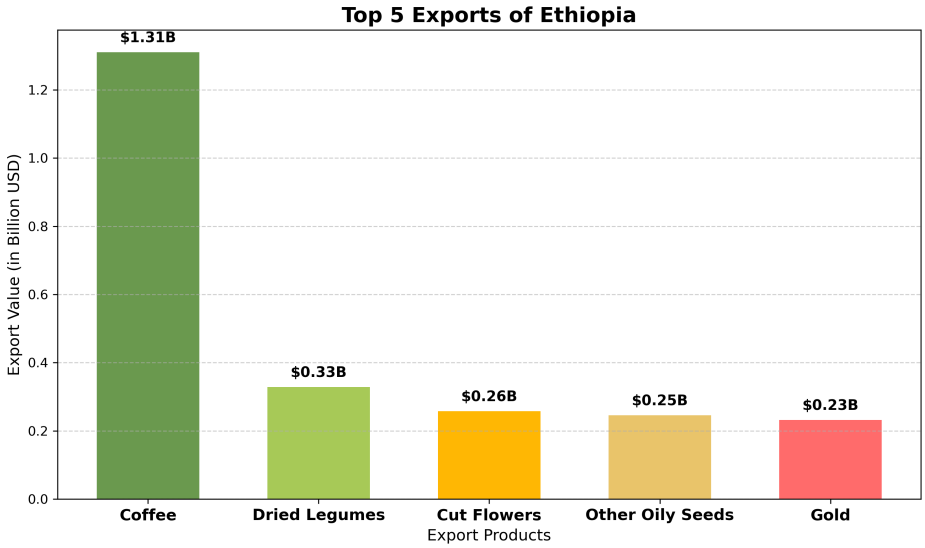


Figure 1.2 The Top Five Export Products of Ethiopia By Value In Billions of USD. Data sourced from The Observatory of Economic Complexity (OEC) Available at: https://oec.world/en/profile/country/eth[5]

This sector sustains the livelihoods of over 15 million individuals, providing a primary source of income for rural households and supporting food security[6]. Beyond its economic role, coffee holds profound cultural significance, exemplified by traditional coffee ceremonies that serve as a cornerstone of Ethiopian hospitality and community bonding,

① The Observatory of Economic Complexity (OEC), Global Trade Data, Available at: https://oec.world/en/profile/country/eth

reinforcing its deep-rooted presence in society.

Despite its importance, Ethiopia's coffee industry faces significant agricultural challenges, particularly from coffee leaf diseases such as Phoma leaf spot and Coffee Leaf Miner infestations. These diseases are estimated to reduce yields by 20-30% in severely affected areas, compromising both the quantity and quality of coffee output[7]. Phoma leaf spot, caused by fungal pathogens, and the Coffee Leaf Miner, a pest that damages leaf tissue, thrive under conditions worsened by climate change, including erratic rainfall and rising temperatures[8]. These challenges are compounded by limited access to resources such as pesticides and disease-resistant coffee varieties, as well as inadequate extension services, which leave farmers ill-equipped to manage outbreaks effectively.

Current disease management practices in Ethiopia rely predominantly on traditional methods, such as manual visual inspection by farmers or occasional support from extension agents, supplemented by limited chemical treatments[9]. These approaches, however, are hindered by several limitations. Visual inspections often suffer from inaccuracy and inconsistency due to varying levels of farmer expertise and fatigue, while the low agent-to-farmer ratio estimated at 1:1,000 in some regions restricts timely intervention[10]. Chemical treatments, when available, are applied sporadically due to cost and accessibility constraints, rendering these methods insufficient for large-scale or proactive disease control across Ethiopia's dispersed coffee-growing areas.



Figure 1.3 Image showing Current disease management practices in Ethiopia

In response to such challenges, the global agricultural sector has increasingly turned to technological innovations, particularly artificial intelligence (AI) and deep learning[11–13]. Techniques like Convolutional Neural Networks and Attention Mechanisms have proven effective in automating disease detection for crops such as wheat, rice, and citrus, often achieving diagnostic accuracies above 95%[14–17]. These advancements offer a transformative potential for Ethiopia, where traditional methods fall short. However, their application must account for local constraints, including limited internet connectivity, unreliable electricity, and the need for affordable, offline solutions that can be deployed in resource-scarce rural settings.

This study is positioned at the intersection of Ethiopia's rich coffee heritage, its economic reliance on the crop, and the pressing need for innovative disease management solutions. It leverages AI, specifically deep learning techniques, to develop an automated classification system for Phoma and Miner diseases in coffee leaves, tailored to the realities of Ethiopian smallholder farmers. By addressing the diagnostic shortcomings of current practices, this Dissertation seeks to enhance disease management, reduce yield losses, and bolster the resilience of Ethiopia's coffee sector, a vital component of its economy and cultural identity.

1.2 Problem Statement

The Ethiopian Arabica coffee sector is recognized as a vital economic and cultural pillar, yet it is confronted with significant challenges due to the widespread prevalence of coffee leaf diseases, such as Phoma leaf spot and Coffee Leaf Miner. These diseases are exacerbated by fluctuating climatic conditions and restricted access to advanced agricultural tools, leading to substantial crop losses estimated at 20-30% annually in heavily affected regions like Sidama and Yirgacheffe. Such declines are experienced directly by over 15 million smallholder farmers, whose livelihoods and food security are heavily dependent on coffee production[18].

Current disease detection practices are predominantly based on manual observation by farmers or scarce agricultural extension agents, a method characterized by subjectivity, inconsistency, and delays[9]. With an extension agent-to-farmer ratio as low as 1:1,000 in

certain areas, timely diagnosis and intervention are rendered impractical, particularly across Ethiopia's fragmented smallholder farming landscape. These limitations underscore a critical need for an automated, accurate, and scalable system to classify coffee leaf diseases and predict their progression stages, thereby ensuring effective disease management and preserving the sustainability of Ethiopia's coffee industry.

This pressing issue is addressed through the development of a deep learning-driven approach designed to automate the classification and stage prediction of diseases in Ethiopian Arabica coffee. A dataset of 890 coffee leaf images, collected from local farms and preprocessed to optimize model performance, is utilized to train a neural network model capable of identifying key diseases, including Phoma and Miner, and predicting their severity stages from early onset to advanced deterioration. This solution is crafted to leverage advanced deep learning techniques, despite delivering high diagnostic accuracy and real-time applicability tailored to the resource-constrained settings of rural Ethiopia, solving pesticide overuse. Through this technological integration, support is provided to empower farmers with an efficient tool for disease detection and management aimed at reducing yield losses and enhancing the long-term resilience of the coffee sector. Further elaboration on the methodology, model architecture, and results is reserved for subsequent chapters.

1.3 Research Objectives

The primary objective of this dissertation is

- To develop an automated deep learning-driven system for the classification of coffee leaf diseases and the prediction of their severity stages, aimed at improving the management of the disease for Ethiopian smallholder farmers cultivating Arabica coffee.

The specific objectives of this study are described below.

- To collect and pre-process a dataset of coffee leaf images that accurately represents diseases such as Phoma leaf spot and Coffee Leaf Miner, reflecting the conditions encountered by Ethiopian farmers.
- To Design and implement a CNN-based model incorporating the Convolutional Block Attention Module (CBAM) within a multi-task learning framework to classify coffee

leaf diseases and assess severity levels.

- To assess the impact of integrating attention mechanisms (CBAM) and multi-task learning on the model's ability to distinguish subtle variations in disease stages, particularly under the resource constraints faced by smallholder farmers in Ethiopia.
- To evaluate the model's performance against traditional manual detection methods and other deep learning approaches, ensuring its effectiveness and reliability in real-world scenarios.
- To deploy the model in user-friendly formats, such as mobile and web applications, ensuring accessibility for smallholder farmers and extension agents in Ethiopia's resource-constrained rural settings.

Justification for these objectives is provided to underscore their significance: collecting a representative dataset is essential because it ensures the model is trained on images mirroring real-world conditions[19], improving its accuracy and practical utility; designing and training an effective model is critical to enable precise and timely disease identification, addressing the subjectivity of manual methods; assessing CBAM and multi-task learning is vital to enhance the model's sensitivity to subtle disease stage differences, optimizing performance in limited-resource settings[20]; evaluating the model's performance is necessary to validate its superiority over existing practices, supporting its adoption; and deploying the model accessibly is crucial to bridge the technological gap, empowering farmers to manage diseases efficiently.

The scope of these objectives is defined such that the model is trained to classify Phoma and Miner diseases and their severity stages, while other coffee plant diseases or environmental factors affecting coffee production are excluded from consideration. These objectives are formulated to align with subsequent research questions, addressing the model's accuracy, its comparative performance, and its practical deployment within the Ethiopian context.

1.4 Research Questions

This study is guided by the following research questions, which are intended to explore the technical performance of the deep learning model and its practical implications

for Ethiopian smallholder farmers cultivating Arabica coffee. These questions are formulated to align directly with the research objectives, ensuring a focused investigation into the development, evaluation, and deployment of the proposed solution.

1) What levels of accuracy, precision, and recall are achieved by the deep learning model in classifying Phoma leaf spot and Coffee Leaf Miner diseases and predicting their severity stages for Ethiopian Arabica coffee?
2) How are the model's development and deployment affected by contextual challenges, such as limited dataset availability, environmental variability, and technological constraints in Ethiopia's rural coffee-growing regions?
3) What is the impact of integrating attention mechanisms (CBAM) and multi-task learning on the model's ability to distinguish subtle variations in disease stages, especially under the resource constraints faced by smallholder farmers in Ethiopia?
4) How can the model's classifications and stage predictions be applied to improve disease management practices among Ethiopian smallholder farmers, given their resource-limited conditions?
5) What broader contributions to deep learning applications in agricultural disease detection are offered by this study, particularly for other crops or regions beyond Ethiopian coffee production?

1.5 Significance of the Study

The significance of this study is underscored by its potential to transform disease management for Ethiopian smallholder farmers, who cultivate over 95% of the nation's Arabica coffee, a crop vital to both economic stability and cultural identity. An automated deep learning-driven system is provided to classify coffee leaf conditions as Healthy, Phoma, or Miner and predict their severity stages Asymptomatic, Mild, and Severe offering a precise, accessible tool that mitigates yield losses estimated at 20-30% annually while addressing the pervasive overuse of pesticides[21].

This reduction in chemical reliance is anticipated to enhance soil health, farmer well-being, and coffee quality, directly supporting the livelihoods of over 15 million individuals

dependent on this sector. Economically, the study's impact is extended to Ethiopia's coffee industry, which accounts for approximately 30% of foreign exchange earnings, by ensuring sustainable production and maintaining the competitiveness of premium export grades such as Yirgacheffe [②].

Beyond practical benefits, the research is positioned to advance academic knowledge in deep learning applications for agriculture. A novel contribution is made through the integration of attention mechanisms (e.g., CBAM) and multi-task learning, enhancing the model's ability to distinguish subtle disease stage variations, a capability particularly significant in the resource-constrained Ethiopian context.

The prediction of Arabica coffee severity stages within this specific setting further distinguishes this study, filling a gap in context-specific AI solutions for developing countries. These advancements are expected to inform broader agricultural technology practices, potentially scalable to other crops or regions, and align with global sustainability goals, such as reducing hunger and promoting economic growth. By achieving its objectives, this study delivers both practical tools for farmers and scientific insights, reinforcing Ethiopia's coffee sector resilience and contributing to the evolution of precision agriculture.

1.6 Scope and Limitations

The scope of this study is confined to Ethiopian smallholder farmers cultivating Arabica coffee in regions such as Sidama, Yirgacheffe, and Oromia, with a focus on classifying coffee leaf conditions as Healthy, Phoma, or Miner and predicting their severity stages as Asymptomatic, Mild, or Severe. A deep learning model is developed and intended for deployment via mobile and web applications, tailored to resource-constrained rural settings. Limitations are acknowledged in the dataset, which comprises 890 images collected from local farms and may not fully capture disease variability across seasons or regions. Technological constraints, such as limited internet access and smartphone availability in rural Ethiopia, are noted as potential barriers to deployment, while the exclusion of other coffee diseases beyond Phoma and Miner is recognized as a boundary of this research.

② MarkWide Research, Ethiopian Coffe Market Analysis, online: https://markwideresearch.com/ethiopia-coffee-market/, Published: January, 2025.

1.7 Thesis Organization

This section outlines the structure of the thesis, providing a roadmap for the research process. Chapter 2 reviews existing literature on coffee leaf diseases, deep learning in agriculture, and Ethiopia's coffee sector, identifying gaps addressed by this study. Chapter 3 details the methodology, including data collection from Ethiopian farms, model design with deep learning techniques, and evaluation methods. Chapter 4 presents the results of classifying Phoma, Miner, and Healthy conditions and predicting Asymptomatic, Mild, and Severe stages, discussing their implications. Chapter 5 concludes the study, summarizing findings, contributions, and recommendations for future work.

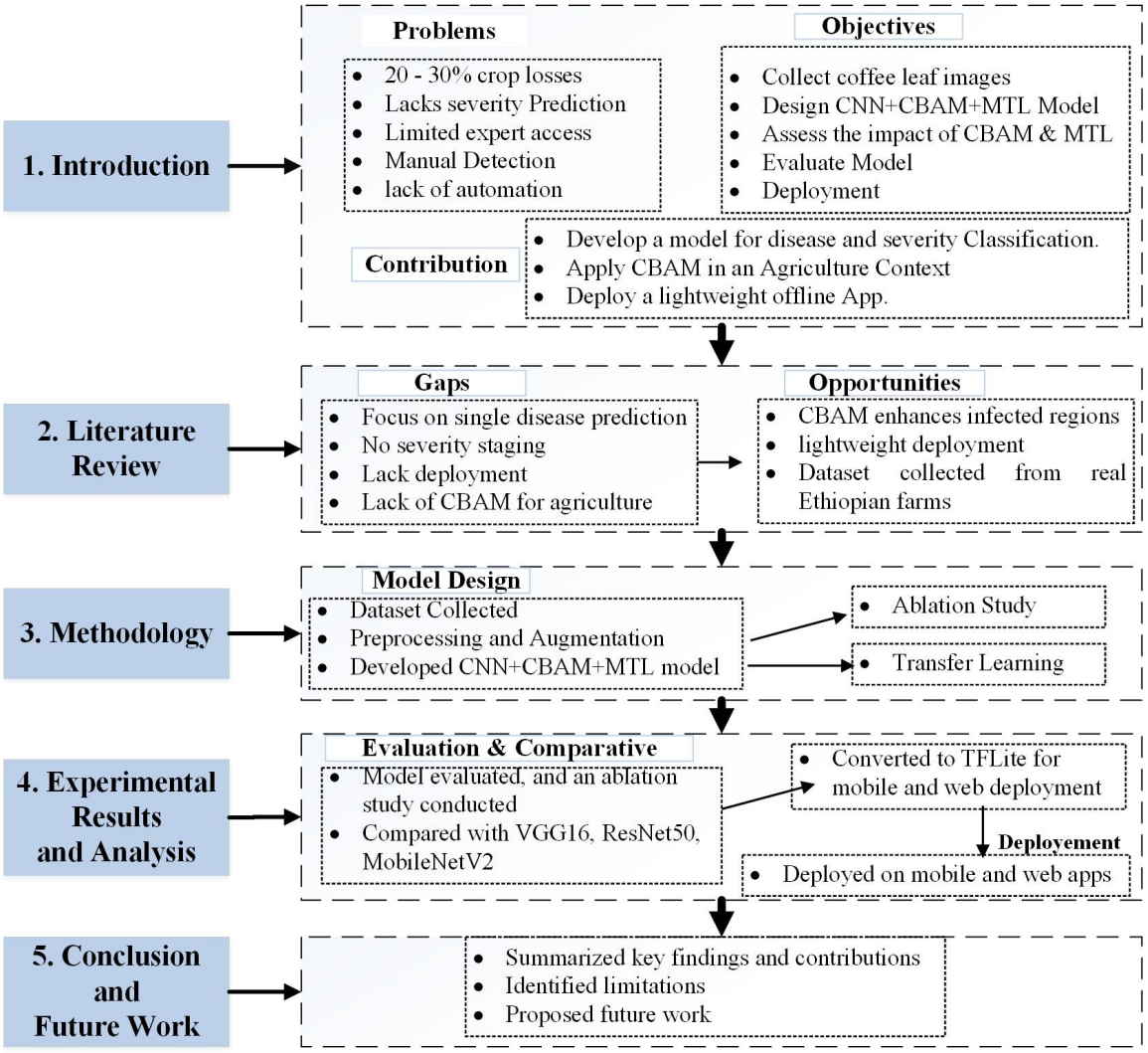


Figure 1.4 Thesis Organization

2 Literature Review

This chapter reviews the existing literature on coffee leaf diseases, deep learning applications in agriculture, and challenges within Ethiopia's coffee sector to identify the gaps addressed by this study. The scope of the review is aligned with the research objectives outlined in Section 1.3, encompassing the collection of relevant data; the development and evaluation of a deep learning model for classifying Phoma, Miner, and Healthy coffee leaf conditions and predicting their Asymptomatic, Mild, and Severe stages; and the deployment of this solution for Ethiopian smallholder farmers. Through this exploration, a foundation is established to justify the integration of attention mechanisms and multi-task learning in a resource-constrained context, setting the stage for the methodological advancements proposed in subsequent chapters.

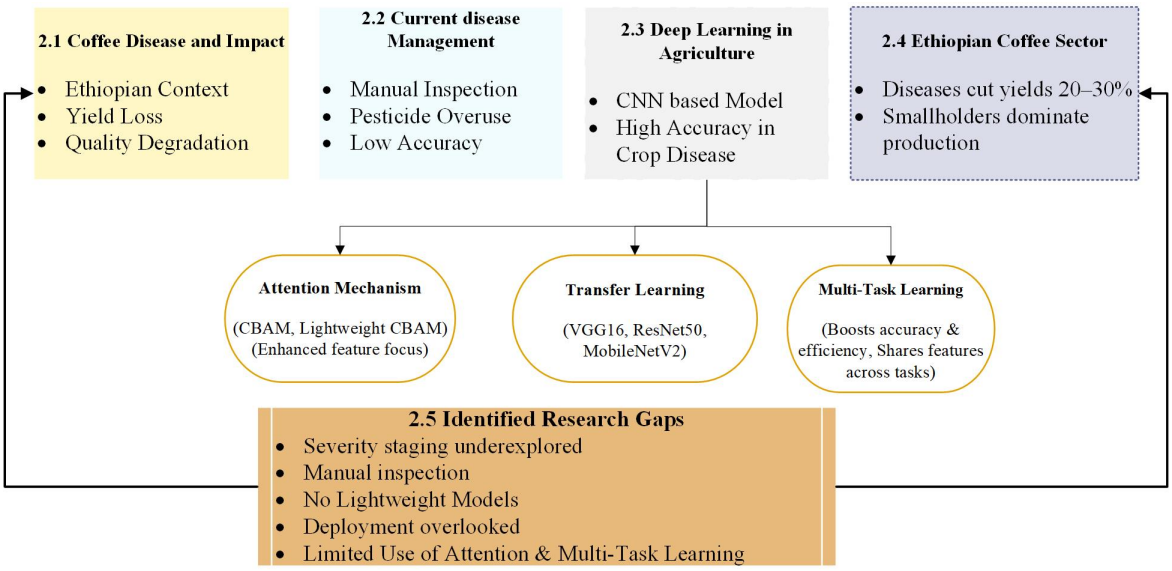


Figure 2.1 Literature Review Workflow

2.1 Coffee Leaf Diseases and Their Impact

Coffee leaf diseases are widely acknowledged as a persistent global threat to coffee production. However, their impact is particularly pronounced in Ethiopia, the origin of Arabica coffee. Phoma leaf spot, caused by the fungal pathogen *Phoma spp.*, and Coffee Leaf

Miner, a pest identified as *Leucoptera coffeella*, are consistently highlighted as prevalent in Ethiopian regions such as Sidama, Yirgacheffe, and Oromia.

Consequently, these diseases have been documented to reduce yields by 20-30% in severely affected areas, compromising both the quantity and quality of Arabica coffee, which is vital for its economic value and flavour profile[22,23]. Moreover, quality degradation, such as smaller beans and altered taste, is frequently reported, threatening Ethiopia's export competitiveness[24]. Severity stages, ranging from asymptomatic, where symptoms are absent, to mild, with initial leaf damage, and severe, marked by extensive necrosis, are recognised as pivotal for effective management. However, their systematic staging remains underexplored[25].

For instance, recent studies have emphasised that Coffee Leaf Miner infestations exacerbate defoliation, reducing photosynthetic capacity by up to 25% in severe cases[26]. Similarly, the spread of Phoma leaf spot under humid conditions accelerates yield losses, particularly in unshaded plantations[27]. This pervasive impact underscores the urgent need for precise detection and severity assessment within Ethiopian smallholder coffee systems.

2.2 Current Disease Management Practices

Current practices for managing coffee leaf diseases in Ethiopia are predominantly centred on traditional methods employed by smallholder farmers, who account for over 95% of Arabica coffee production. Manual visual inspection by farmers or extension agents is widely utilised to identify Phoma leaf spot and Coffee Leaf Miner infestations and is often supplemented by pesticide applications when resources permit[28].

However, these methods are constrained by significant limitations, including inaccuracy and inconsistency owing to subjective assessments and limited farmer training, with detection accuracies as low as 65% in some regions[29]. The overuse of pesticides, driven by a lack of precise diagnostics, exacerbates soil degradation and pest resistance, posing environmental and health risks[30]. Basic technological interventions, such as handheld imaging devices, have been explored but are hindered by high costs and limited scalability in Ethiopia's rural, resource-scarce settings[31]. These shortcomings underscore the need for automated and accurate solutions to enhance disease management in smallholder coffee farms.

2.3 Deep Learning in Agriculture

Deep learning techniques are increasingly used in agriculture to revolutionise crop disease detection, with Convolutional Neural Networks established as a foundational approach owing to their proficiency in extracting spatial features from complex leaf imagery[32–34].

Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) have become a cornerstone in the application of deep learning to plant disease detection, offering unparalleled accuracy in identifying and classifying various plant ailments. By leveraging their hierarchical architecture, CNNs can automatically learn and extract intricate features from leaf images, such as colour variations, texture patterns, and lesion shapes, which are indicative of specific diseases.

This capability enables the development of automated systems that assist farmers in early disease detection, facilitating timely intervention and potentially reducing crop losses[35]. Studies have demonstrated that CNN-based models can achieve high precision in distinguishing healthy and diseased leaves across multiple plant species, thereby enhancing the efficiency of agricultural diagnostics[36].

The implementation of CNNs in plant pathology has been bolstered by the availability of extensive image datasets and advances in computational power. Researchers have developed sophisticated CNN architectures that not only classify diseases but also provide insights into the decision-making process of the model, increasing transparency and trust in automated systems. For Instance, Semantic dictionaries generated by CNNs can mimic the diagnostic methodology of human experts by emphasising the characteristics that are most important in identifying particular plant diseases[36]. CNN integration in agriculture shows the potential for more reliable, scalable, and easily available plant disease management systems as deep learning techniques advance, ultimately leading to increased agricultural yields and food security.

Research demonstrates that CNN-based models consistently achieve accuracies exceeding 95% in diagnosing diseases across diverse crops, such as wheat, rice, and citrus, often relying on extensive labeled datasets and robust computational infrastructure[37–40].

Transfer Learning

Transfer learning has become an essential technique in plant disease detection, addressing challenges associated with limited labelled datasets and the need for efficient model training. By leveraging models pre-trained on large-scale datasets, such as ImageNet, researchers can adapt these models to specific agricultural tasks through fine-tuning, significantly enhancing classification accuracy even with constrained data. For example, a study used transfer learning with ResNet50 to automate the identification of plant diseases, refining the model for the effective classification of healthy and diseased plant leaves[41].

This approach not only accelerates the development process but also ensures that high-performance models can be deployed in resource-limited environments, making advanced agricultural diagnostics more accessible. For instance, a study on rice blast detection reported an accuracy of 98% using ResNet architecture, highlighting the potential of deep learning in precision agriculture[42].

Moreover, deep learning architectures such as VGG16 and MobileNetV2 are extensively employed in agricultural disease detection, providing robust benchmarks for this study's model comparisons. For instance, VGG16, a deep convolutional neural network, was utilised to identify tomato leaf diseases through transfer learning, leveraging pre-trained weights to analyse environmental and climatic factors and achieving high precision and accuracy in classifying diseased plants[43,44]. These studies demonstrate VGG16's effectiveness, though its computational complexity limits its real-time deployment in resource-scarce settings.

In contrast, MobileNetV2, optimized for efficiency, was applied to potato leaf disease identification with notable success. A study improved a pre-trained MobileNetV2 model with channel devolution and attention mechanisms, yielding 99.67% accuracy in the PyTorch framework after data preprocessing[45]. Similarly, another approach froze the base layers of MobileNetV2, fine-tuning the top 23 layers to classify potato lesions and achieving 99.83% accuracy and suitability for portable devices[46]. These findings highlight Transfer Learning models, offering a critical comparison for this study's proposed CBAM-integrated model tailored to Ethiopia's constraints.

Attention Mechanisms

Attention mechanisms have emerged as critical enhancements in deep learning models for plant disease detection, enabling networks to focus selectively on relevant regions

within leaf images, thereby improving classification accuracy. By dynamically highlighting important features while suppressing less informative features, attention modules facilitate the extraction of discriminative patterns associated with various plant diseases.

A study showed that the integration of the Convolutional Block Attention Module (CBAM) into CNN architectures has demonstrated notable improvements in identifying leaf diseases in pear trees, achieving higher classification accuracy than models without attention mechanisms[47]. Similarly, the incorporation of attention modules into lightweight networks, such as MobileNetV2, has enhanced the model's ability to focus on disease-specific regions, resulting in more precise apple leaf disease identification. These advancements underscore the effectiveness of attention mechanisms in refining deep-learning models for more accurate and efficient plant disease diagnosis.

Subsequently, attention mechanisms have been explored to refine deep learning models in various domains, with emerging applications in agriculture highlighting their potential for feature enhancement. For instance, a graph neural network enhanced with an attention mechanism was proposed to address few-shot learning challenges in image classification, leveraging pre-trained residual networks for feature extraction, and iteratively updating node and edge features to achieve robust performance with limited data, although its agricultural application remains unspecified[48].

In another study, a Lightweight Convolutional Block Attention Module (LCBAM) was integrated into the HRNet model for human posture estimation, adaptively capturing global features to improve keypoint detection accuracy (for example 5.2% enhancement for the left wrist) while maintaining a lightweight design, offering insights for crop disease contexts[49].

Similarly, the CBAM attention module is incorporated into the YOLOv5 backbone for object detection in road scenes, boosting significant feature extraction and improving the mean average precision (mAP) by 1.1% on the KITTI dataset, demonstrating its efficacy in complex environments[50]. These studies underscore the versatility of attention mechanisms, providing a foundation for this research's CBAM-enhanced approach to classification and severity prediction of coffee disease.

Multi-task Learning (MTL)

Multi-task Learning (MTL) has emerged as a powerful approach for plant disease de-

tection, enabling models to perform multiple related tasks, such as identifying infection sites, classifying disease stages, and estimating severity levels. By sharing representations across these tasks, MTL leverages commonalities and differences, leading to improved generalisation and efficiency.

Additionally, MTL reduces the need for separate task-specific models, saving computational resources and training time. The shared feature learning across tasks also enhances robustness, as knowledge from one task can benefit others. This makes MTL particularly effective in scenarios with limited labeled data for individual tasks. For instance, research on corn leaf disease detection introduced an MTL system that achieves higher accuracy and faster convergence by jointly learning disease classification and severity estimation tasks[51]. These examples underscore MTL's potential to provide comprehensive and accurate plant disease diagnostics by integrating multiple predictive tasks within a unified model.

Furthermore, multi-task learning approaches are investigated to optimise multiple related tasks within a single model, offering insights applicable to agricultural contexts. For example, a study on biometric recognition employed convolutional neural networks to simultaneously identify traits such as face, palmprint, and palm vein, designing a multi-task model that leverages shared features across modalities to enhance recognition efficiency and achieve notable performance in intelligent authentication systems[52].

In a distinct approach, the relationship between feature sets and multiple outputs is explored, where a multi-task model is trained using feature intersections for correlated outputs (e.g., improving performance over single-task models) or unions for uncorrelated outputs (e.g., yielding suboptimal results compared to single-task baselines), providing guidelines for feature selection based on output correlation[53].

These investigations highlight the potential of multitask learning to streamline complex predictions; however, its application to agricultural disease severity staging, such as in coffee, remains limited, informing this study's novel integration of multitask learning with CBAM for Ethiopian coffee disease analysis. This gap in coffee-specific, low-resource deep-learning applications underscores the need for tailored innovations, such as those proposed in this study.

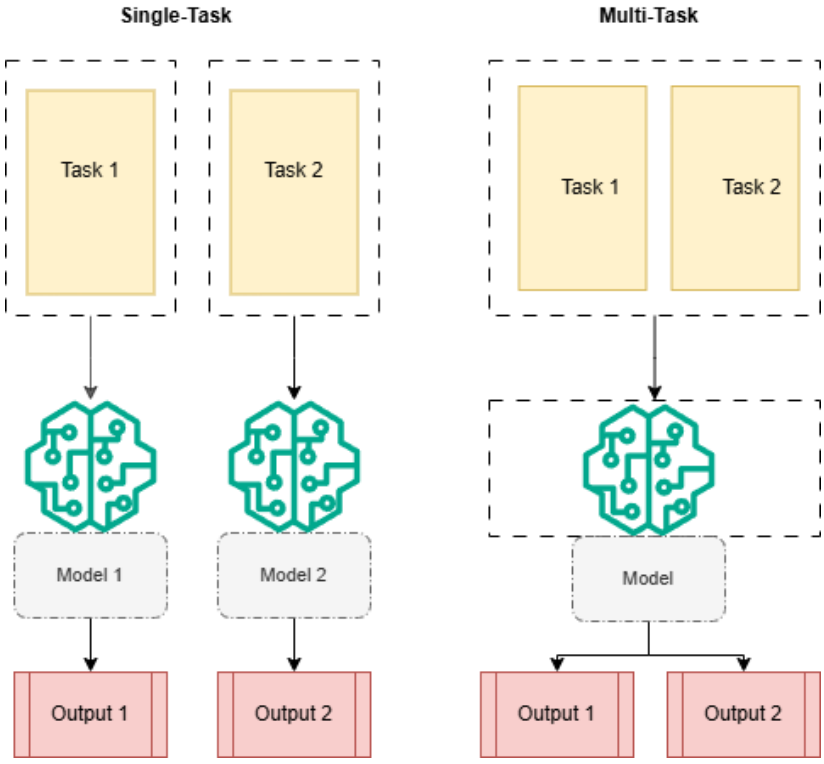


Figure 2.2 Single Task Vs Multi-Task Learning. Adapted from[54]

2.4 Ethiopian Coffee Sector Context

Ethiopia's coffee sector is predominantly shaped by small farmers, who contribute more than 95% of Arabica coffee production on plots averaging less than two hectares, yet face substantial challenges in disease management. Limited access to resources, such as pesticides and disease-resistant varieties, is compounded by inadequate extension services with an agent-to-farmer ratio of approximately 1:1,000 and technological barriers, including sparse Internet connectivity and low smartphone penetration in rural areas, such as Sidama and Oromia[55,56]. Coffee's economic significance is underscored by its contribution of 5-10% to GDP and nearly 30% of foreign exchange earnings, whereas its cultural role is epitomized by traditional coffee ceremonies that reinforce social bonds[57]. Previous technological interventions, such as mobile-based farming apps, are noted to have encountered adoption hurdles due to cost and infrastructure limitations[58]. These conditions highlight the critical need for an automated, resource-efficient deep learning solution tailored to Ethiopia's

smallholder context.

2.5 Identification of Research Gaps

The literature reviewed across Sections 2.1 to 2.4 underscores substantial progress in understanding coffee leaf diseases, management practices, and deep learning applications; nevertheless, critical gaps that this study aims to address. Firstly, although the Phoma leaf spot and the Coffee Leaf Miner are well documented for their 20-30% yield reductions in Ethiopia[22,23], along with quality degradation[24], their severity stages - asymptomatic, mild and severe - are not sufficiently integrated in the detection frameworks[25]. For instance, a recent Ethiopian study classified coffee diseases with 98% accuracy using a CNN-based approach but omitted severity prediction, limiting its utility for proactive management[59].

Similarly, in another study in Ethiopia, coffee diseases are classified using a CNN with GoogLeNet and ResNet feature concatenation, bolstered by Gaussian filtering and augmentation, achieving 96% accuracy via MLPs and ensemble classifiers[60], but severity staging and lightweight optimization remain unincorporated. Moreover, studies highlighting Miner's 25% photosynthetic loss[26] and Phoma's humidity-driven spread[27] emphasize impact but lack staging solutions. Secondly, current management practices reliant on manual inspections (65% accuracy) and pesticide overuse[28–30] or unscalable imaging tools[31] reveal a need for automated, precise alternatives tailored to Ethiopia's smallholder constraints.

Furthermore, deep learning advancements in agriculture, such as CNNs achieving over 95% accuracy[37–40], and specific models like VGG16 for tomato diseases[43,44] or MobileNet V2 for potato lesions (99%)[45,46], demonstrate high performance but are predominantly resource-intensive or crop-specific, neglecting coffee's unique challenges. Attention mechanisms, such as CBAM in YOLOv5 (1.1% mAP gain)[50] and LCBAM in HRNet (5.2% accuracy boost)[49], enhance feature extraction, yet their application to coffee diseases remains rare, with few-shot learning approaches unspecified for agriculture[48]. Similarly, multi-task learning for biometric traits[52] and feature correlation analysis[53] offers efficiency, but its use in agricultural severity staging is limited. Additionally, Ethiopia's context is marked by resource scarcity and economic stakes (5-10% GDP)[57] lacks lightweight, context-specific deep learning solutions, with prior tech interventions faltering.

Furthermore, coffee leaf disease identification is tackled using SLIC segmentation to extract diseased regions, followed by DenseNet-264 for binary classification (infected vs. healthy), achieving 97.17% accuracy and outperforming ResNet-50 (96.54%)[61], though severity staging and lightweight design are not addressed. Moreover, leaf miner and rust in Arabica coffee are detected using EfficientNetB0 enhanced with a ghost module, reducing parameters to 4.87M and yielding 84% accuracy[62], yet this lightweight model omits severity prediction and Ethiopia-specific considerations.

Despite achieving high classification accuracies, most existing studies lack real-world deployment in mobile or web-based platforms. Severity stage prediction is often overlooked, limiting practical disease management recommendations. Few models are lightweight enough for on-device inference, making them unsuitable for low-resource environments. Thus, this study addresses these gaps by proposing a model to classify coffee leaf disease conditions (phoma, miner, and healthy) and predict severity stages (asymptomatic, mild, and severe) and to suit Ethiopia's resource-constrained smallholder systems, paving the way for a comparative analysis in the subsequent table.

Table 2.1 Comparative Summary of Related Coffee Disease Studies

Author(s) Year	**Study**	**Method**	**Accuracy**	**Limitation**
Kulkarni et al. *2023*[61]	Coffee disease detection	DenseNet 264 + SLIC	97.17%	No deployment consideration, limited evaluation details.
Paulos et al. *2022*[59]	Coffee leaf classification	CNN Transfer Learning	97.01%	No deployment; severity not addressed.
Ramamurthy et al. *2022*[62]	Coffee leaf classification	EfficientNetB0	84%	Traditional features; lacks mobile/web deployment.
Leite et al. *2020*[26]	Coffee Leaf miner	Survey-based IPM analysis	N/A	No DL or automated solution, focus on insecticide use.

2.6 Chapter Summary

Chapter 2 synthesizes literature on coffee leaf diseases, deep learning applications, and Ethiopia's coffee sector to identify gaps addressed by this study. Section 2.1 high-

lights Phoma leaf spot and Coffee Leaf Miner as major threats in Ethiopia, reducing yields by 20-30% and degrading quality, yet severity staging remains underexplored[22,25]. Section 2.2 reveals traditional management practices reliant on inaccurate manual inspections (65% accuracy) and unscalable tools, necessitating automated solutions[29,31]. Section 2.3 examines deep learning, with CNNs achieving over 95% accuracy across crops, transfer learning (e.g., VGG16, MobileNetV2) offering efficiency, and attention mechanisms (e.g., CBAM) enhancing precision, though coffee-specific applications are limited[37,45,50]. Multi-task learning shows promise for joint tasks but lacks agricultural severity focus[52]. Section 2.4 underscores Ethiopia's smallholder challenges, resource scarcity, and economic stakes (5-10% GDP) demanding lightweight solutions[57]. Section 2.5 identifies gaps: limited severity prediction, resource-intensive models, and poor deployment consideration in coffee studies (e.g., DenseNet-264, EfficientNetB0)[61,62]. This review justifies the proposed CNN+CBAM+Multi-Task model in Chapter 3, tailored for Ethiopia's context, addressing these deficiencies.

3 Methodology

The methodology is structured to develop and validate a deep learning solution for detecting and managing coffee leaf diseases in Ethiopia's Arabica coffee (*Coffea arabica*) sector, specifically targeting Phoma leaf spot and Coffee Leaf Miner, which are prevalent in Jimma, Sidama, and Yirgacheffe. A dataset of 890 high-quality images, collected and preprocessed as detailed in Sections 3.2 and 3.3, forms the foundation for this research, addressing the 20–30% yield losses faced by smallholder farmers due to ineffective disease management identified in Chapter 1. The proposed model integrates Convolutional Neural Networks (CNNs) with Convolutional Block Attention Modules (CBAM) and multi-task learning, engineered to classify both disease types (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe) concurrently. By leveraging shared features, this approach enhances.

The research framework adopts the CDIO (Conceive, Design, Implement, Operate) approach to systematically tackle these challenges:

Conception Phase:

- The dual-task problem is defined, driven by the need for timely diagnostics absent in existing methods. This phase identifies the critical requirements for detecting diseases and their severity levels, ensuring relevance to Ethiopia's smallholder farmers.

Design Phase:

- The CNN+CBAM+Multi-Task model is designed to address the dual-task classification problem:

1) CNNs extract hierarchical features from $224 \times 224 \times 3$ RGB images, capturing both low-level and high-level patterns.
2) CBAM refines these features by focusing on disease-specific patterns, such as Phoma's necrotic spots and Miner's tunneling, enhancing the model's attention to critical regions.
3) Multi-task heads produce dual categorical outputs (disease type and severity level),

optimized for lightweight deployment in resource-constrained environments.

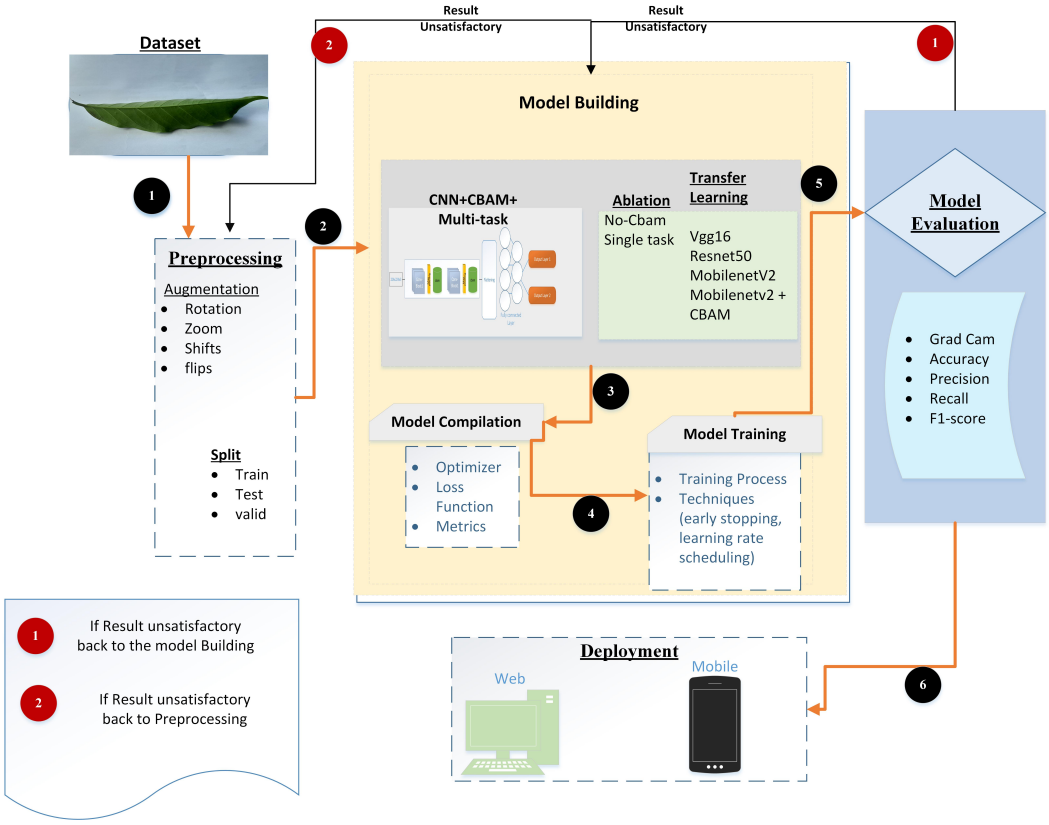


Figure 3.1 End-to-End Pipeline for Deep Learning-Driven Classification and Stage Prediction of Diseases in Ethiopian Arabica Coffee

The operations depicted in Figure 3.1 illustrates the end-to-end workflow of the proposed system, beginning with (1) dataset collection and loading for preprocessing, followed by (2) feeding the preprocessed data into the model, (3) compiling the model after architecture definition, (4) training the model, (5) evaluating its performance, and finally (6) deploying the best-performing model.

Implementation Phase:

- The model is constructed using tools like TensorFlow, trained over 100 epochs on the augmented dataset, and validated for performance, as outlined in Sections 3.4–3.5. Training incorporates techniques such as early stopping, learning rate scheduling, and checkpointing to ensure robust optimization.

Operation Phase:

- Ensures practical utility through web and mobile applications, detailed in Section 3.9,

tailored for Ethiopia's resource-constrained smallholder context. These applications provide actionable recommendations in both English and Amharic, bridging the gap between advanced technology and farmer accessibility.

To validate the model's effectiveness, ablation studies (Section 3.6) and transfer learning experiments (Section 3.7) assess the contributions of key components such as CBAM and pre-trained architectures. Evaluation methods, including Grad-CAM (Section 3.8), provide insights into the model's focus and effectiveness, ensuring transparency and interpretability.

This methodology establishes a comprehensive path from data collection to deployment, with results analyzed in Chapter 4. The ultimate aim is to deliver a scalable, farmer-centric tool for real-time disease management, empowering smallholder farmers to mitigate yield losses and improve crop health sustainably.

3.1 Research Design and Framework

The development of a robust solution for detecting and managing coffee diseases in Ethiopia's Arabica coffee (Coffea arabica) sector demands a structured research design tailored to the prevalent challenges of Phoma leaf spot and Coffee Leaf Miner. This section outlines the research framework employing the CDIO (Conceive, Design, Implementation, Operate) approach to systematically address the problem articulated in Chapter 1 and the technological gaps identified in Chapter 2. The framework is designed to meet the study objectives in classifying leaves as Healthy, Phoma, or Miner, and in classifying severity as Asymptomatic, Mild, or Severe while ensuring practical utility for Ethiopia's smallholder farmers, who suffer 20−30% yield losses due to ineffective disease management.

In the conception phase, the research identifies the problem within Ethiopia's coffee regions of Jimma, Sidama, and Yirgacheffe, where Phoma and Miner thrive in humid conditions; however, existing methods fail to deliver timely and precise diagnostics. The dual-task requirement of classifying both disease type and severity as categorical outcomesarises from the need to guide farmer interventions, a capability that is absent in current approaches. The Design phase formulates a deep learning solution integrating Convolutional Neural Networks (CNNs) with Convolutional Block Attention Modules (CBAM). This framework employs CNNs for feature extraction and CBAM to enhance focus on disease-specific patterns,

such as Phoma's spotting or Miner's tunnelling, supporting two classification outputs: disease type (three classes) and severity (three classes). The design emphasises a lightweight structure to enable the deployment of low-cost devices, aligned with Ethiopia's smallholder constraints.

The implementation phase involves constructing and training the CNN-CBAM model using a self-collected dataset from Jimma, Sidama, and Yirgacheffe, as detailed in the subsequent sections. This phase includes data preprocessing, model development, and optimisation with tools such as TensorFlow, which executes dual categorical classification tasks with appropriate loss functions. The operation phase focused on validating the model against the target accuracies, employing metrics such as accuracy, precision, and recall for both tasks, along with comparisons with baseline CNNs to assess CBAM enhancement. The operational aim is to develop a scalable, field-ready tool for Ethiopia's smallholders and extension agents, facilitating real-time disease detection and management.

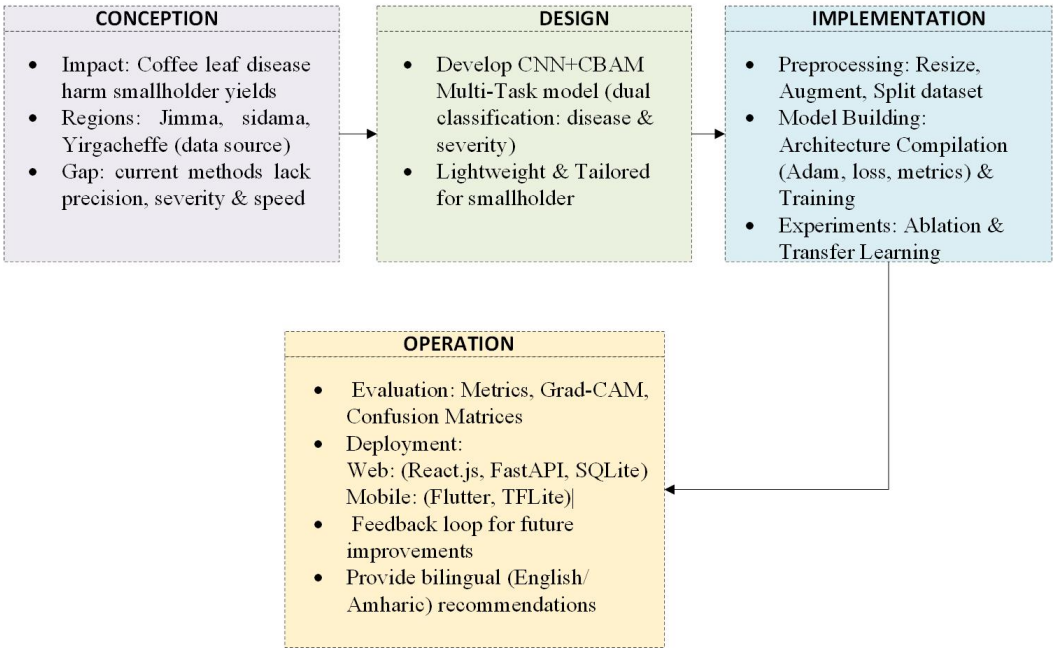


Figure 3.2 CDIO Framework for Coffee Leaf Disease Classification and Severity Prediction

This CDIO framework ensures a coherent progression from problem conception to operational solutions tailored to Ethiopia's coffee sector. By adopting dual categorical classification within a resource-efficient design, the Dissertation addresses the precision and

practicality demanded by smallholders, laying the groundwork for the detailed methodology in the following subsections.

3.2 Dataset Collection

The dataset is a critical component of this research, meticulously gathered to tackle Ethiopia's Arabica coffee (Coffea arabica) disease challenges, specifically Phoma leaf spot and Coffee Leaf Miner. This section elaborates the collection process across Jimma, Sidama, and Yirgacheffe, detailing the methodology, quality control, and class distribution. The same set of images supports both disease classification (Healthy, Phoma, Miner) and severity classification (Asymptomatic, Mild, Severe), aligning with the multi-task objectives of and accuracy, respectively, and reflecting Ethiopia's smallholder context.

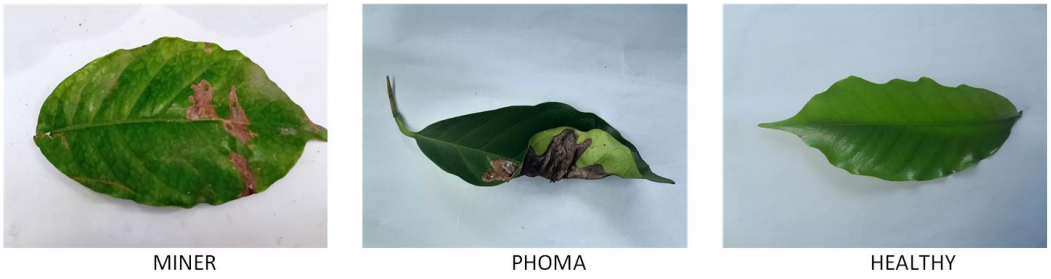


Figure 3.3 Representative coffee leaf samples collected from Jimma, Sidama, and Yirgacheffe, Ethiopia, illustrating healthy leaves alongside those exhibiting Phoma disease and Coffee Leaf Miner symptoms

3.2.1 Collection Sites and Conditions

Collection was conducted in Jimma, Sidama, and Yirgacheffe, chosen for their agroecological diversity and disease prevalence.

- **Jimma:** Humid highlands favored Phoma leaf spot, with samples from scattered smallholder farms
- **Sidama:** Dense plantations yielded Healthy and Miner-affected leaves, mirroring intensive farming.
- **Yirgacheffe**: Unique microclimate provided varied severity samples, enhancing dataset richness.
- **Conditions:**

– Images were captured during peak growing seasons under natural daylight for consistent illumination.
– A 16 MP digital camera was used, handheld, with leaves positioned against a white background to isolate features and reduce visual noise.
– Local agricultural experts assisted, ensuring accurate identification of disease symptoms.

3.2.2 Image Acquisition and Quality Check

A total of 1,200 images were initially collected, from which 890 images were meticulously selected through a rigorous quality assurance process to ensure high-quality inputs for model training and evaluation.

Acquisition Process:

1) **Standardization**
   • Leaves were photographed in situ, positioned flat against a white background to standardize appearance and remove distracting elements.
   • Experts from regional agricultural offices identified disease symptoms, such as Phoma's dark spots and Miner's tunneling, validating each sample for accuracy and consistency.
2) **Image Composition** Each image centered on a single leaf, occupying approximately 80% of the frame, and was captured at a distance of 15-20 cm to preserve fine details of disease patterns. Lighting conditions were controlled to minimize shadows and reflections, ensuring visual features remained clear and distinct.

Quality Check:

• From the initial pool of 1,200 images, 310 were discarded due to defects like blurriness, uneven lighting, or obstructions (e.g., shadows on the white background).
• The remaining 890 high-quality images were sharp, well-lit, and clearly exhibited health or disease traits, forming the final dataset for training, validation, and testing.

Challenges of Severity Categorization:

Separating severity levels (Asymptomatic, Mild, Severe) posed challenges due to subtle overlaps in symptom progression. Asymptomatic leaves appeared healthy but could harbor

early, imperceptible damage. Mild cases displayed initial signs, such as faint spots or limited tunneling, while Severe instances showed advanced damage, like extensive necrosis or prominent tunneling. Expert collaboration and iterative reviews were employed to address ambiguities, ensuring reliable categorization across the dataset.

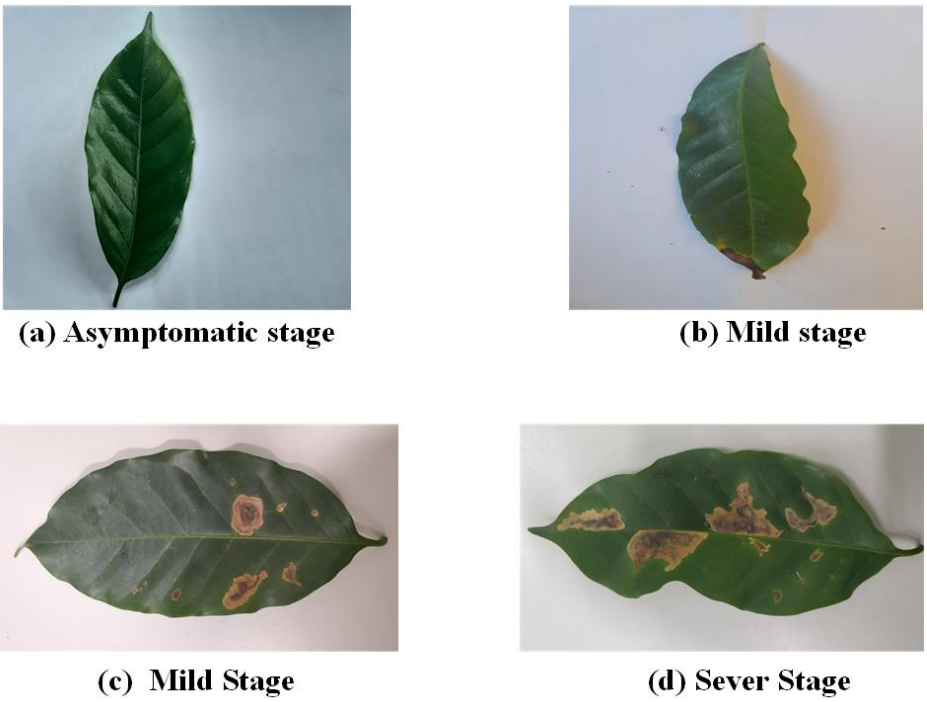


Figure 3.4 Representative Coffee Leaf Images Illustrating Severity Categorization Challenges

Image (a) shows an asymptomatic (healthy) coffee leaf, which bears a striking resemblance to the leaf in image (b), exhibiting mild symptoms. Similarly, image (c) displays a mild stage that is visually similar to the severe stage depicted in image (d). These similarities underscore the challenges in accurately distinguishing between severity levels in coffee leaf diseases.

3.2.3 Class Distribution and Dataset Structure

The 890-image dataset is structured to ensure balanced representation across both disease and severity classifications. The same images are utilized for both tasks, maintaining annotation consistency and capturing Ethiopia's regional variability in coffee leaf diseases.

Disease Classification

1) **Healthy:** 300 images, representing leaves with no visible signs of disease. These

images provide a baseline for comparison, crucial for distinguishing healthy from diseased states.

2) **Phoma:** 300 images, displaying dark spots or necrosis caused by the Phoma fungus. These symptoms typify fungal infections, posing a significant threat to crop health if untreated.

3) **Miner:** 290 images, exhibiting tunneling damage from leaf miners. This damage, indicative of insect infestations, can severely impair leaf functionality and plant vitality.

Severity Classification:

1) **Asymptomatic:** 300 images, corresponding to the Healthy class in Disease Classification. These pristine green leaves show no visible disease symptoms, serving as the control group for severity analysis.

2) **Mild:** 300 images, representing early-stage symptoms such as faint spots (for Phoma) or minimal tunneling (for Miner). Identifying these subtle signs is vital for timely intervention to curb disease spread.

3) **Severe:** 290 images, representing advanced damage like extensive necrosis (for Phoma) or prominent tunneling (for Miner). These images underscore the impact of progressed diseases, highlighting the need for early detection.

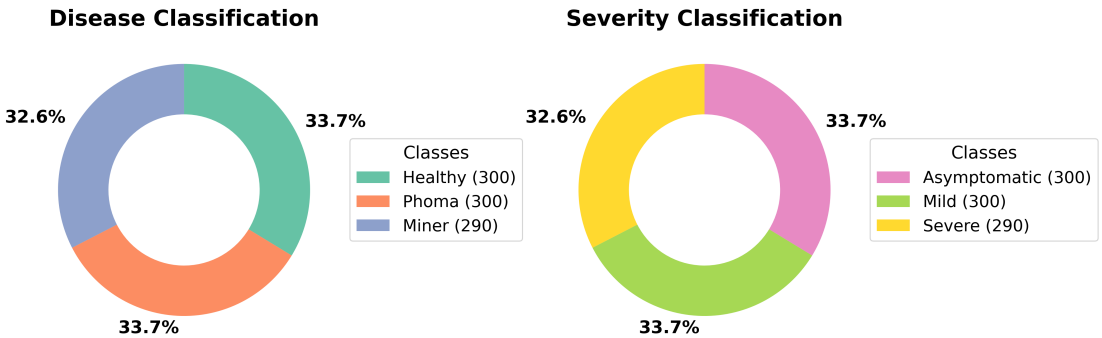


Figure 3.5 Class Distribution for Disease and Severity

Dataset Characteristics:

The dataset comprises 890 high-quality images distributed across disease and severity classifications. A slight imbalance exists, with Healthy and Phoma at 300 images each and Miner at 290. This is mirrored in severity with Asymptomatic and Mild at 300 and Severe at 290. This minor variation reflects natural field collection differences and ensures the dataset effectively represents real-world conditions.

Table 3.1 Class Distribution for Disease and Severity Classification

Disease Classification	
Class	**Image Count**
Healthy	300
Phoma	300
Miner	290
Severity Classification	
Class	**Image Count**
Asymptomatic	300
Mild	300
Severe	290

3.3 Preprocessing

Following the collection of 890 images detailed in Section 3.2, preprocessing is conducted to prepare the dataset for the CNN-CBAM model, ensuring compatibility with the multi-task classification of disease types (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe). This section outlines the steps applied to the images gathered from Jimma, Sidama, and Yirgacheffe, transforming raw data into a standardized, enhanced format suitable for training and evaluation. The process leverages the same images for both tasks, maintaining the white background's consistency while optimizing for Ethiopia's smallholder focused objectives.

3.3.1 Image Standardization

Initial preprocessing standardizes the images to meet model input requirements.

1) **Resizing:** Each image is resized to 224x224 pixels, aligning with standard CNN input dimensions and preserving key features like Phoma's spots and Miner's tunnels.
2) **Scaling:** Pixel values are scaled from 0-255 to 0-1, normalizing intensity across JPEG and PNG formats for consistent model processing.
3) **Cropping:** Images are cropped to remove peripheral noise (e.g., edges beyond the white background), focusing on the leaf while retaining the full white backdrop.
4) **Tools:** OpenCV and Python imaging libraries (e.g., PIL) are employed for resizing, scaling, and cropping, ensuring precision and uniformity.

3.3.2 Data Augmentation

Augmentation enhances dataset variety while preserving coffee leaf structure. The following techniques are applied to simulate real-world variations and improve model robustness:

1) **Rotation:** Images are rotated within ±40 degrees, simulating natural leaf angles and orientations without distorting disease patterns such as Miner's linear tunnels or Phoma's spots.
2) **Width and Height Shifting:** Images are shifted horizontally and vertically by up to 20% of their dimensions, mimicking slight positional changes in leaf capture during field photography. This ensures the model remains invariant to minor spatial misalignments.
3) **Shearing:** A shear transformation with a range of ±20% is applied, introducing slanted perspectives that replicate angled views of leaves in natural settings. This preserves structural details like tunneling or necrosis outlines.
4) **Zooming:** Random zooming within a 20% range is used to simulate varying distances between the camera and the leaf. This helps the model generalize across different scales while maintaining critical disease features.
5) **Horizontal Flipping:** Horizontal flips are applied to reflect possible field orientations, ensuring the model remains robust to mirrored leaf appearances. This technique preserves the integrity of disease patterns such as Phoma's spots.
6) **Brightness Adjustment:** Brightness is varied within a range of 80% to 120%, mim-

icking changes in lighting conditions (e.g., morning vs. afternoon light). This ensures the visibility of severity levels, such as Mild vs. Severe spotting, under diverse illumination scenarios.

7) **Fill Mode and Background Handling:** New pixels introduced during transformations (e.g., rotation, shifting, or zooming) are filled with a constant white color (cval=255.0), ensuring a clean background that does not interfere with disease pattern analysis.

8) **Image Rescaling:** Images are rescaled to a fixed size of 224x224 pixels to standardize input dimensions for the model. This preprocessing step ensures uniformity across the dataset while maintaining structural details.

9) **Implementation:** Augmentation is executed using TensorFlow's ImageDataGenerator, which integrates seamlessly into the preprocessing pipeline. These techniques collectively enhance dataset diversity, enabling the model to generalize better to unseen data while preserving critical structural details of coffee leaf diseases.

3.3.3 Labeling and Dataset Split

Labeling images with disease and severity classes using expert and automated methods and splitting the dataset into balanced training, validation, and test sets finalize it for model training and evaluation.

1) **Labeling Process:**
   - Initial manual labeling is performed with expert assistance, categorizing each image by disease (Healthy, Phoma, Miner) and severity (Asymptomatic, Mild, Severe).
   - Subsequent automation uses tools like Python scripts and OpenCV to assign labels, stored in a CSV file for both tasks.

2) **Dataset Split:**
   - The 890 images are divided into 80% training, 10% validation, and 10% test sets.
   - Split ensures balanced representation across classes and severity levels, supporting robust training and evaluation.

3) **Tools:** Pandas and NumPy facilitate CSV management and dataset partitioning, ensuring seamless integration with the training process.

3.4 Model Architecture

The architecture of the CNN+CBAM+Multi-Task model is designed to process $224 \times 224 \times 3$ RGB images from the 890-image dataset preprocessed in Section 3.3, collected from Jimma, Sidama, and Yirgacheffe, to classify coffee leaf diseases (Healthy, Phoma, Miner) and their severity levels (Asymptomatic, Mild, Severe) concurrently. This section details the model's layered framework, integrating a CNN backbone for feature extraction, Convolutional Block Attention Modules (CBAM) for attention-driven refinement, and multi-task learning for dual classification outputs, addressing the technological gaps identified in Chapter 2 and building on the dataset foundation from Section 3.2. Optimized for Ethiopia's Arabica coffee (*Coffea arabica*) context, the model captures critical patterns such as Phoma's dark spots and Miner's tunneling across multiple scales, balancing computational efficiency with robustness to support smallholder farmers facing 20−30% yield losses.

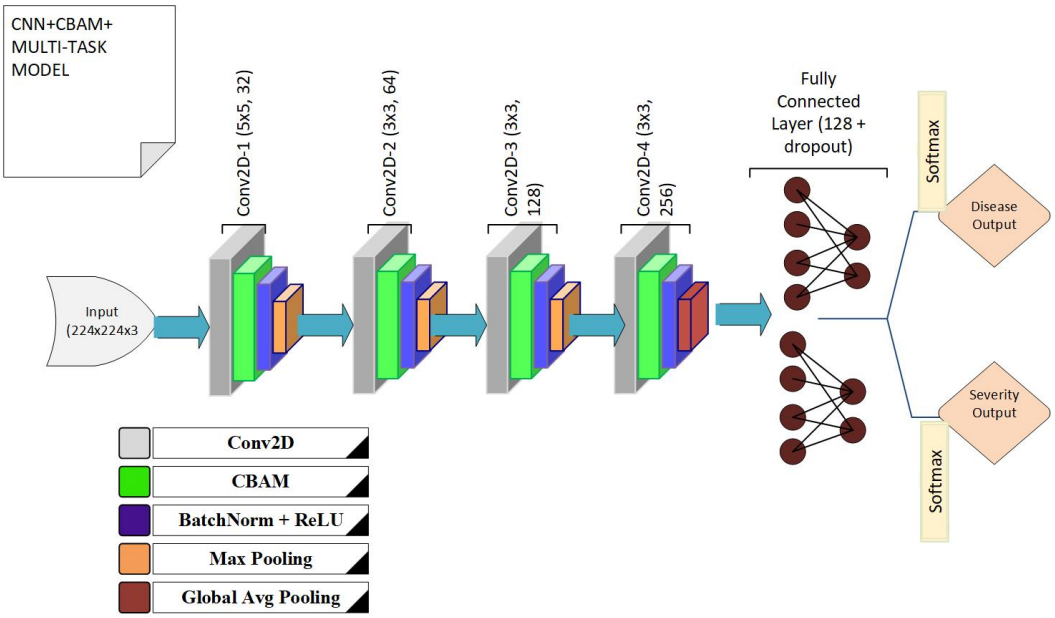


Figure 3.6 Block Diagram of the CNN+CBAM+Multi-Task Model Architecture for Coffee Leaf Disease Classification and Severity Prediction

The model's development follows a structured approach:

1) A CNN backbone extracts hierarchical features from raw images, using custom kernel sizes and progressive filter increases.
2) CBAM enhances these features by focusing on disease-specific regions, improving

precision over a standalone CNN.

3) Multi-task heads leverage shared representations to produce two outputs, disease type and severity, streamlining the architecture for resource-constrained deployment.

Subsections 3.4.1 through 3.4.3 elaborate on this structure, detailing the CNN backbone's convolutional layers, CBAM's attention mechanisms, and the multi-task configuration, respectively. The design ensures lightweight efficiency, suitable for practical applications outlined in Section 3.9, while ablation and transfer learning comparisons in Sections 3.6 and 3.7 validate its components. Performance outcomes, assessed in Chapter 4, demonstrate its effectiveness in meeting the study's objectives for Ethiopia's smallholder coffee systems.

3.4.1 CNN Backbone

The CNN backbone constitutes the primary feature extraction framework of the model, designed to process 224x224x3 RGB images from the 890-image dataset preprocessed in Section 3.3, sourced from Jimma, Sidama, and Yirgacheffe. It is engineered to extract hierarchical features vital for the multi-task classification of coffee leaf diseases (Healthy, Phoma, Miner) and their severity levels (Asymptomatic, Mild, Severe). This subsection elaborates the initial extraction of features from the layered architecture, midlevel refinement, advanced compression, and final pooling, detailing convolutional layers, normalization and downsampling strategies optimized for the context of Arabica coffee (Coffea arabica) in Ethiopia, where the dark spots of Phoma and the tunneling patterns of Miner require precise spatial feature capture on multiple scales. The design balances computational efficiency with robustness, leveraging custom kernel sizes and progressive filter increases to support subsequent CBAM and multi-task stages.

3.4.1.1 Initial Feature Extraction

The backbone begins with an initial feature extraction stage, focusing on capturing broad structural patterns in the input images. This stage consists of a convolutional layer, ReLU activation, batch normalization, and max pooling.

Convolutional Layer A 5×5 Conv2D layer with 32 filters processes the input tensor of shape $224 \times 224 \times 3$. The convolution operation is defined as:

$$O(i,j,k) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \sum_{c=0}^{C-1} I(i+m, j+n, c) \cdot W(m,n,c,k) + B(k), \tag{3.1}$$

where:

- $O(i,j,k)$: The output at position (i,j) for filter k.
- I: The input image tensor.
- W: The weight kernel of size $5 \times 5 \times 3 \times 32$.
- $B(k)$: The bias term for filter k.
- $M = N = 5, C = 3$ (kernel size and input channels).

The 5×5 kernel, covering a 25-pixel receptive field, captures larger structural features such as Miner's linear tunnels or Phoma's initial spotting. This broader receptive field outperforms smaller kernels (e.g., 3×3) by encompassing more contextual information, which is crucial for early disease detection across Ethiopia's diverse leaf samples.

Activation ReLU activation is applied to introduce non-linearity:

$$\text{ReLU}(z) = \max(0, z), \tag{3.2}$$

Zeroing negative values to promote sparsity and reduce overfitting risks is a critical consideration given the limited dataset size of 1,050 images.

Batch Normalization Batch normalization is applied post-convolution to stabilize activations. The normalization process is defined as:

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta, \tag{3.3}$$

where:

- μ_B: The mini-batch mean.
- σ_B^2: The mini-batch variance.
- ϵ: A small constant (10^{-5}) for numerical stability.
- γ, β: Learnable parameters for scaling and shifting.

This stabilizes training by mitigating variance caused by diverse lighting and texture conditions across regions.

Pooling A MaxPooling2D layer with a 2×2 window reduces spatial dimensions from $224 \times 224 \times 32$ to $112 \times 112 \times 32$. This retains prominent features such as large disease indicators while halving resolution to decrease computational complexity.

3.4.1.2 Mid-Level Feature Refinement

Two mid-level convolutional blocks refine features, focusing on localized disease and severity characteristics.

First Mid-Level Layer

- **Convolution**: A 3×3 Conv2D layer with 64 filters processes the $112 \times 112 \times 32$ input, yielding $112 \times 112 \times 64$. The 3×3 kernel targets finer details, such as Phoma spot edges or Miner tunnel curvature.
- **Batch Normalization**: Normalizes outputs to enhance gradient flow through the deeper network.
- **Pooling**: MaxPooling2D (2×2) downsamples to $56 \times 56 \times 64$, concentrating on significant features like spot density or tunnel progression.

Second Mid-Level Layer

- **Convolution**: A 3×3 Conv2D layer with 128 filters operates on the $56 \times 56 \times 64$ tensor, yielding $56 \times 56 \times 128$. This deepens feature extraction to capture subtle severity differences, such as Mild versus Severe Phoma necrosis.
- **Batch Normalization**: Ensures stability as filter counts increase.
- **Pooling**: MaxPooling2D (2×2) reduces dimensions to $28 \times 28 \times 128$, preserving essential spatial cues.

3.4.1.3 Advanced Feature Compression

The final stage compresses features into a high-level representation for global integration.

- **Convolution:** A 3×3 Conv2D layer with 256 filters processes the $28 \times 28 \times 128$ input, yielding $28 \times 28 \times 256$. This layer extracts complex, abstract features such as extensive Miner damage or Severe Phoma necrosis crucial for distinguishing nuanced disease and severity patterns.
- **Batch Normalization:** Batch normalization normalizes the 256 channel output, counteracting variance amplification in deeper layers.
- **Global Average Pooling:** Unlike prior layers, GlobalAveragePooling2D compresses the $28 \times 28 \times 256$ tensor into a 256-unit vector by averaging spatial dimensions per channel:

$$y_k = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} x(i, j, k), \tag{3.4}$$

  where $H = W = 28$ are the spatial dimensions. This operation reduces parameters significantly, providing a compact feature set for multi-task learning.

The CNN backbone uses a progressive filter increase (32 to 256) and multiscale approach, starting with a 5×5 kernel for coarse patterns and shifting to 3×3 kernels for localized details, culminating in a 256-filter layer with GlobalAveragePooling2D for global disease and severity representation. Three MaxPooling2D layers reduce dimensions ($224 \rightarrow 112 \rightarrow 56 \rightarrow 28$), producing small parameters pre-CBAM, balancing feature extraction for coffee leaf variability from Jimma, Sidama, and Yirgacheffe with efficiency for lightweight deployment in Ethiopia's resource-constrained smallholder context.

3.4.2 CBAM Integration

The Convolutional Block Attention Module (CBAM) integration represents a critical enhancement in the proposed model, strategically implemented to concentrate on diseased areas of coffee leaves such as Phoma's dark spots and Miner's tunneling patterns within the $224 \times 224 \times 3$ images from the 890-image dataset preprocessed in Section 3.3. This subsection elaborates on the rationale for employing the CBAM, its improvements to the CNN backbone detailed in Section 3.4.1, and the architectural specifics of its implementation, underscoring its necessity for achieving the multi-task objectives of classifying coffee leaf diseases (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe).

By adaptively focusing on critical features across samples from Jimma, Sidama, and Yirgacheffe, CBAM elevates the model's precision and efficiency, rendering it indispensable for coffee leaf analysis in Ethiopia's smallholder context.

CBAM is integrated to prioritize diseased regions in coffee leaf images, leveraging its dual channel and spatial attention to dynamically weight localized patterns like Phoma's necrotic spots and Miner's tunnels, enhancing early detection critical for Ethiopia's smallholder systems. With only 890 images, CBAM boosts efficiency by emphasizing relevant features, reducing overfitting risks and reliance on heavy augmentation while amplifying disease cues. For multi-task classification of disease type and severity, CBAM refines shared CNN features, distinguishing overlapping indicators like spot density or tunnel clusters, ensuring nuanced differentiation across tasks with minimal data.

3.4.2.1 Enhancement of the CNN Backbone

CBAM augments the CNN backbone by refining feature maps, increasing classification accuracy, and maintaining computational efficiency.

- **Feature Refinement**: Positioned after each Conv2D layer (32, 64, 128, 256 filters), CBAM recalibrates features at multiple scales 224×224, 112×112, 56×56, and 28×28 enhancing the hierarchical output of the CNN backbone. It amplifies disease-relevant channels, such as green for healthy leaves or dark, irregular spots for Phoma necrosis, and spatial regions, such as tunnel clusters, thereby improving classification precision.
- **Accuracy Improvement**: By adaptively concentrating on diseased tissue regions while suppressing interference from unaffected areas, CBAM enhances feature refinement in CNNs. This targeted attention mechanism reduces noise not only from generic backgrounds but also from healthy anatomical structures, directly improving the model's ability to localize pathology.
- **Computational Efficiency**: CBAM introduces minimal additional parameters of approximately $10,000$ per block compared to fully convolutional alternatives, preserving the model's lightweight design with a total of approximately $350,000$ parameters. This ensures suitability for deployment in Ethiopia's resource-constrained smallholder en-

vironment.

3.4.2.2 CBAM Architecture Overview

The CBAM architecture consists of two sequential attention mechanisms: Channel Attention and Spatial Attention. These mechanisms refine feature maps by emphasizing important channels and spatial regions, respectively.

- **Channel Attention**: Channel attention computes global average pooling and global max pooling across spatial dimensions to generate channel-wise importance scores. The process is defined as:

$$\text{AvgPool}(x) = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} x(i, j, c), \tag{3.5}$$

$$\text{MaxPool}(x) = \max_{i,j} x(i, j, c), \tag{3.6}$$

where H and W are the height and width of the feature map, and c is the channel index. The pooled features are passed through two fully connected layers:

$$\text{FC}_1(x) = \delta(W_1 \cdot x), \tag{3.7}$$

$$\text{FC}_2(x) = \sigma(W_2 \cdot \text{FC}_1(x)), \tag{3.8}$$

where δ is the ReLU activation function, σ is the sigmoid activation function, and W_1, W_2 are learnable weights. The final channel attention map is obtained by adding the outputs of the average and max pooling paths:

$$M_c = \sigma(\text{MLP}(\text{AvgPool}(x)) + \text{MLP}(\text{MaxPool}(x))). \tag{3.9}$$

- **Spatial Attention**: Spatial attention computes average pooling and max pooling across channels to generate spatial importance scores:

$$\text{AvgPool}(x) = \frac{1}{C} \sum_{c=1}^{C} x(i, j, c), \tag{3.10}$$

$$\text{MaxPool}(x) = \max_{c} x(i, j, c), \tag{3.11}$$

where C is the number of channels. The pooled features are concatenated and passed through a 7×7 convolutional layer with sigmoid activation:

$$M_s = \sigma(\text{Conv}([\text{AvgPool}(x); \text{MaxPool}(x)])). \tag{3.12}$$

- **Final Output**: The refined feature map is obtained by sequentially applying channel and spatial attention:

$$F' = M_s \otimes (M_c \otimes F), \tag{3.13}$$

where F is the input feature map, M_c is the channel attention map, M_s is the spatial attention map, and $\otimes$ denotes element-wise multiplication.

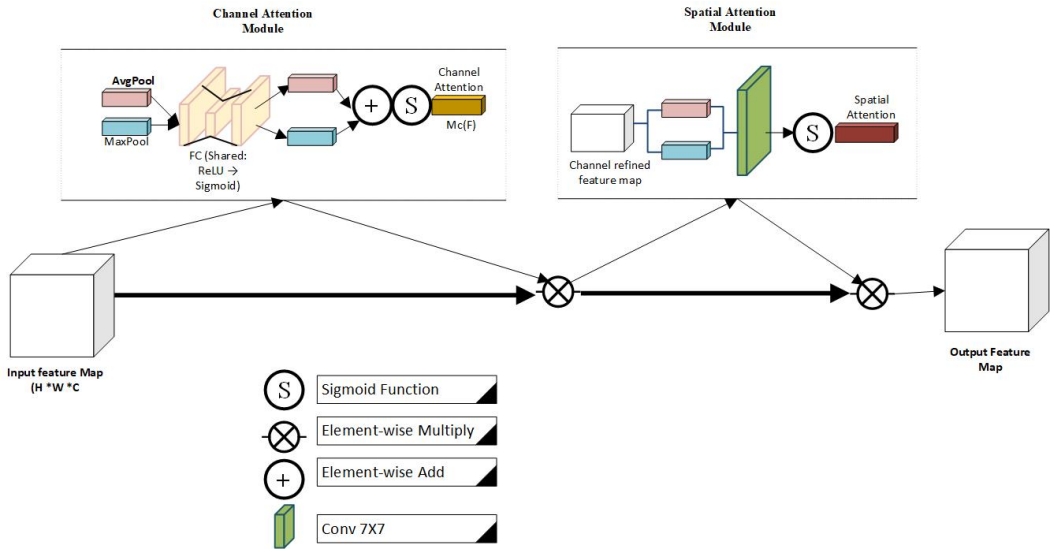


Figure 3.7 Block Diagram of the CBAM Module

- **Integration into the Backbone**: CBAM is integrated after each Conv2D layer in the CNN backbone. For example, after the initial 5×5 convolutional layer with 32 filters, CBAM refines the $224 \times 224 \times 32$ feature map by emphasizing disease-relevant channels and spatial regions. This process is repeated for subsequent layers, ensuring consistent feature refinement throughout the network.

The multi-stage CBAM deployment concentrates the model on diseased areas at every

depth of the CNN backbone, enhancing feature maps from coarse (224x224) to fine (28x28) resolutions. This approach improves the CNN by filtering noise, boosting the detection of disease and severity indicators, and maintaining lightweight efficiency, directly supporting the study's objectives for Ethiopia's coffee smallholders over a standalone CNN configuration.

3.4.3 Multi-Task Learning

A multi-task learning (MTL) framework is developed to classify coffee leaf images by disease type and severity level. The model employs Convolutional Neural Networks (CNNs) (Section 3.4.1) enhanced with Convolutional Block Attention Modules (CBAMs) (Section 3.4.2), followed by two independent output heads. The disease head uses softmax activation to predict one class from three categories: Healthy, Phoma, or Miner. The severity head uses softmax to predict one class from three categories: Asymptomatic, Mild, or Severe. These heads operate independently, meaning the disease prediction does not directly constrain the severity prediction. However, the dataset, comprising 890 images from Jimma, Sidama, and Yirgacheffe, is annotated to ensure biological validity.

Specifically, all images labeled as Healthy in the disease category are assigned Asymptomatic in the severity category within the dataset's CSV file, reflecting the absence of disease symptoms in Healthy leaves. This annotation constrains the model to learn that Healthy leaves correspond to Asymptomatic, preventing illogical outputs such as Healthy paired with Mild or Severe. This MTL approach, which outputs one label per task, differs from multi-label classification, where a single image could be assigned multiple labels simultaneously (e.g., both Phoma and Miner). The dataset's structure ensures each image receives exactly one disease label and one severity label, aligning with the MTL framework's design.

3.4.3.1 Multi-Task Structure

The multi-task learning component branches the processed features into two classification heads, utilizing a shared fully connected layer.

1) **Shared Feature Processing**:

- Following the GlobalAveragePooling2D operation, which compresses the $28 \times 28 \times 256$ feature map into a 256-unit vector, a fully connected dense layer with 128 units and

ReLU activation consolidates features extracted by the CNN and refined by CBAM. This layer reduces dimensionality while preserving critical information for both disease and severity classification.

$$z = \text{ReLU}(W_{fc} \cdot x + b_{fc}), \tag{3.14}$$

where x is the 256-unit input vector, W_{fc} is the weight matrix of size 256×128, and b_{fc} is the bias term.

- Dropout with a rate of 0.3 is applied, mitigating overfitting by randomly deactivating 30% of the units during training. This precaution is balanced by the augmented dataset's increased diversity:

$$z' = \text{Dropout}(z, p = 0.3), \tag{3.15}$$

where z' is the output after dropout.

2)**Task-Specific Outputs**:

- **Disease Classification Head**: A dense layer with 3 units (Healthy, Phoma, Miner) employs softmax activation to generate probability distributions over disease classes, designated as `disease_output` in the implementation:

$$y_{\text{disease}} = \text{Softmax}(W_{\text{disease}} \cdot z' + b_{\text{disease}}), \tag{3.16}$$

where W_{disease} is the weight matrix of size 128×3, and b_{disease} is the bias term.

- **Severity Classification Head**: A parallel dense layer with 3 units (Asymptomatic, Mild, Severe) uses softmax activation, labeled `severity_output`, to classify severity levels:

$$y_{\text{severity}} = \text{Softmax}(W_{\text{severity}} \cdot z' + b_{\text{severity}}), \tag{3.17}$$

where W_{severity} is the weight matrix of size 128×3, and b_{severity} is the bias term.

Both heads directly utilize the 128-unit fully connected layer's output without additional intermediate layers.

Multi-task learning is adopted to enhance efficiency and generalization by exploiting the interdependence of disease and severity classification, using shared visual indicators

like Phoma's spot density and Miner's tunnel extent from the CNN and CBAM, minimizing redundancy over separate models. Joint optimization improves robustness across the diverse augmented dataset, boosting accuracy for Ethiopia's varied coffee leaves, while a unified architecture with 350,000 parameters (33,400 from this section) ensures a lightweight model suitable for deployment on resource-constrained devices in Ethiopia's smallholder context.

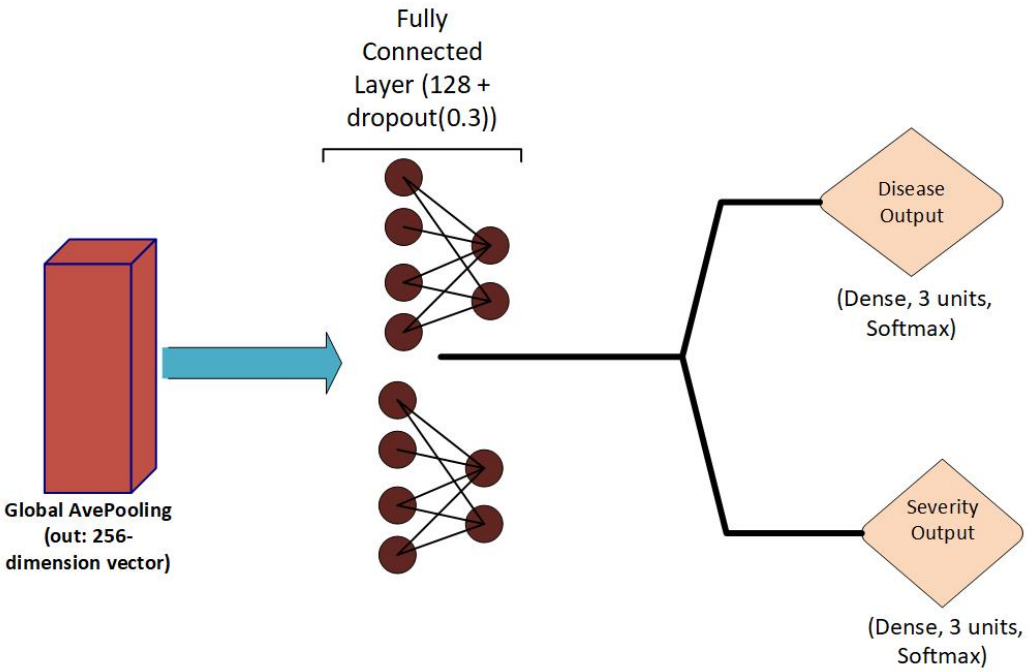


Figure 3.8 Multi-task block Diagram for Coffee Leaf Disease Classification and Severity Prediction

3.4.3.2 The 128-Unit Fully Connected Layer and Severity Design

The model incorporates a single 128-unit fully connected dense layer before splitting into task-specific heads, with no additional layer for severity, a design choice assessed for the augmented dataset.

- **Fully Connected Layer Confirmation**: The 128-unit dense layer functions as a fully connected layer, connecting all 256 input units from GlobalAveragePooling2D to 128 output units, followed by dropout and task heads, confirming its role in feature consolidation.
- **Suitability of 128 Units**: The 128-unit layer effectively reduces the 256-unit vector into a compact representation, sufficient for the correlated disease and severity

cues (e.g., spot size) refined by CBAM, adding 32,768 parameters. With the augmented dataset's increased diversity, it balances capacity and efficiency, supported by dropout (0.3) to prevent overfitting, though this risk is reduced compared to the pre-augmentation state.

3.4.3.3 Use and Impact

The multi-task learning structure harnesses the CNN backbone's hierarchical features and CBAM's disease-focused refinements, delivering dual outputs through a streamlined design. The 128-unit fully connected layer with dropout ensures efficient feature utilization, enhances generalization with the augmented dataset, and maintains a lightweight profile, supporting the study's objectives for Ethiopia's coffee smallholders without requiring an additional severity-specific layer.

3.5 Model Compilation and Training

The compilation and training phase configures and optimizes the CNN+CBAM+Multi-Task model, defined in Sections 3.4.1 through 3.4.3, to classify coffee leaf diseases (Healthy, Phoma, Miner) and their severity levels (Asymptomatic, Mild, Severe) using the dataset pre-processed and augmented in Section 3.3, collected from Jimma, Sidama, and Yirgacheffe. This section specifies the optimizer, loss function, loss weights, evaluation metrics, and training parameters, including epochs, batch size, early stopping, and additional methods to ensure efficient optimization, task prioritization, and robust performance assessment, aligning with the study's objectives for practical deployment.

3.5.1 Model Compilation

The compilation stage establishes the optimization and evaluation framework for the multi-task model.

1) **Optimizer Configuration**

- The Adam optimizer is employed, leveraging its adaptive learning capabilities by maintaining moving averages of gradients and squared gradients, ensuring stable convergence across the CNN and CBAM-refined features.

- An initial learning rate of 1×10^{-4} (0.0001) is set, facilitating precise updates to the model's approximately 350,000 parameters, preventing overshooting given the dual-task complexity and augmented dataset diversity.

2) **Loss Function and Weights**

- The `sparse_categorical_crossentropy` loss function is applied to both `disease_output` and `severity_output` heads. This loss function is suitable for integer-encoded labels $(0, 1, 2)$ across the three classes per task. It is defined as:

$$L = -\frac{1}{N}\sum_{i=1}^{N} y_i \cdot \log(\hat{y}_i), \tag{3.18}$$

where N is the number of samples, y_i is the true label, and $\hat{y}_i$ is the predicted probability for the correct class. This choice optimizes memory usage for multi-task learning.

- Loss weights are assigned as 0.5 for `disease_output`, reflecting its foundational role, and 1.0 for `severity_output`, prioritizing the more complex severity classification critical for treatment decisions in smallholder farming. The total loss for the multi-task learning model is calculated as:

$$L_{\text{total}} = w_d \cdot L_d + w_s \cdot L_s, \tag{3.19}$$

where $w_d = 0.5$, $w_s = 1.0$, L_d is the disease head loss, and L_s is the severity head loss. This weighted combination balances the two tasks during training.

3) **Metrics**

- **Standard Metrics**:

– **Accuracy**: Measures overall correctness for disease and for severity:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}. \tag{3.20}$$

– **Precision**: Assesses positive prediction reliability:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}. \tag{3.21}$$

– **Recall**: Ensures detection of all positive cases:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}. \tag{3.22}$$

– **F1-Score**: Balances precision and recall:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.23}$$

4) **Multi-Task-Specific Metrics**:

- **Weighted Average Accuracy**: Measures the overall accuracy by weighting each task based on its importance or sample size.

$$\text{Weighted Average Accuracy} = \frac{w_{\text{disease}} \cdot \text{Accuracy}_{\text{disease}} + w_{\text{severity}} \cdot \text{Accuracy}_{\text{severity}}}{w_{\text{disease}} + w_{\text{severity}}}. \tag{3.24}$$

- **Combined F1-Score**: Averages the F1-scores of both tasks to summarize precision and recall across tasks.

$$\text{Combined F1-Score} = \frac{\text{F1-Score}_{\text{disease}} + \text{F1-Score}_{\text{severity}}}{2}. \tag{3.25}$$

Table 3.2 Model Compilation Parameters for CNN+CBAM+MTL

Component	**Setting/Value**
Optimizer	Adam
Initial Learning Rate	1×10^{-4} (0.0001)
Loss Function	Sparse Categorical Crossentropy
Loss Weights	Disease: 0.5, Severity: 1.0
Metrics	Accuracy, Precision, Recall, F1-Score, Weighted Average Accuracy, Combined F1-Score

3.5.2 Training Configuration

The training process is structured to optimize the compiled model efficiently and prevent overfitting.

1) **Epochs and Batch Size**

- Training is conducted over 100 epochs, allowing sufficient iterations for the model to learn from the augmented dataset while avoiding excessive computation, suitable for the lightweight design intended for Ethiopia's smallholder deployment.
- A batch size of 32 is utilized, balancing memory efficiency and gradient stability. This size enables effective updates across the model's 350,000 parameters, leverag-

ing the dataset's diversity without overwhelming resource-constrained systems.

2) **Early Stopping**

- Early stopping is implemented to halt training if performance ceases to improve, monitored via validation loss (combined disease and severity losses). A patience of 10 epochs is set, ensuring the model trains adequately while preventing overfitting by stopping when no progress is observed over 10 consecutive epochs, a common practice for small-to-medium datasets.

3) **Additional Training Methods**

- **Learning Rate Scheduling**: A learning rate reduction on plateau is incorporated, reducing the learning rate by a factor of 0.5 (minimum 1×10^{-6}) if validation loss plateaus for 5 epochs. This enhances convergence by adapting to training dynamics, refining weights as the model approaches optimal performance.
- **Checkpointing**: Model weights are saved at the epoch with the lowest validation loss, ensuring the best-performing model is retained for deployment, critical for practical use in Ethiopia's coffee systems.

Table 3.3 Training Configuration Parameters

Parameter	**Setting/Value**
Epochs	100
Batch Size	32
Early Stopping	Patience = 10 epochs
Learning Rate Scheduling	Factor = 0.5, Minimum = 1×10^{-6}, Patience = 5 epochs

3.5.3 Use and Impact

The compilation and training configuration equips the model with Adam (initial_lr $= 1 \times 10^{-4}$) for stable optimization, sparse categorical crossentropy with loss weights (0.5 for disease, 1.0 for severity) to balance task priorities, and a robust metric suite (Accuracy, Precision, Recall, F1-Score, Overall Accuracy, Weighted Average Accuracy, Combined F1-Score) for comprehensive assessment. Training over 100 epochs with a batch size of 32, early stopping, and additional methods such as learning rate scheduling, validation split, and checkpointing optimizes performance on the augmented dataset. This approach minimizes

overfitting, enhances convergence, and ensures the model delivers a lightweight, effective solution for Ethiopia's smallholder coffee farmers to detect and manage leaf diseases and severity levels efficiently.

To ensure optimal performance of the proposed CNN-based CBAM and multi-task learning model, a structured hyperparameter analysis was conducted. The evaluation focused on several critical parameters, including the number of training epochs, dropout rates, and task-specific loss weights. Epoch values of 50 and 100 were compared to assess the impact of training duration. Dropout rates of 0.2 and 0.3 were explored as regularization techniques to mitigate overfitting. In the context of multi-task learning, different combinations of loss weights for disease type and severity classification tasks, such as (0.5, 0.5) and (0.5, 1.0), were analyzed to balance the joint optimization objective. These hyperparameter tuning efforts were essential in determining the configuration adopted in the final model training reported in Chapter 4.

3.6 Ablation Study Design

The ablation study design evaluates the contributions of critical components in the proposed CNN+CBAM+Multi-Task model, as outlined in Sections 3.4.1 through 3.4.3 and compiled in Section 3.5, for classifying coffee leaf diseases (Healthy, Phoma, Miner) and their severity levels (Asymptomatic, Mild, Severe) using the dataset preprocessed and augmented in Section 3.3, collected from Jimma, Sidama, and Yirgacheffe. This section specifies two ablation experiments removal of CBAM and the separation into single-task models for disease and severity classification, aiming to isolate their effects on achieving the target accuracies for disease classification and for severity classification. By simplifying the model through these ablations, the study establishes a foundation for performance analysis in Chapter 4, providing insights into the model's efficiency and effectiveness for Ethiopia's smallholder coffee systems.

3.6.1 Rationale for Ablation

Ablation experiments dissect the model's architecture, assessing CBAM and multi-task learning's necessity by isolating their contributions to classification accuracy, as removing

CBAM may weaken focus on diseased areas like Phoma's spots or Miner's tunnels, and dropping multi-task learning increases redundancy by losing shared disease-severity features. Simplifying by ablating CBAM or splitting tasks into single-task models evaluates efficiency trade-offs, though it sacrifices interpretability and generalization vital for Ethiopia's smallholder deployment. These ablated variants establish baselines for Chapter 4 comparisons, quantifying CBAM's attention and multi-task benefits, despite not fully reflecting the full model's practical advantages.

3.6.2 Ablation Experiments

Two specific ablation experiments are conducted, focusing on CBAM removal and single-task separation.

1) **CBAM Removal**:

- **Description**: The CBAM component, applied after each Conv2D layer (32, 64, 128, 256 filters) as detailed in Section 3.4.2, is entirely removed. The model retains the CNN backbone (Section 3.4.1) and multi-task heads (Section 3.4.3), eliminating the four CBAM blocks and their associated 40,000 parameters, including channel attention (dense layers with reduction ratio 16) and spatial attention (7×7 Conv2D).
- **Purpose**: This ablation tests CBAM's role in focusing on diseased areas, such as Phoma's necrotic spots and Miner's tunnels, assessing whether the CNN backbone alone can achieve comparable performance on the augmented dataset. It is anticipated that accuracy, particularly for severity classification, may decrease due to the loss of attention-driven feature refinement.
- **Limitations**:

– Removing CBAM is expected to limit the model' s ability to recalibrate feature maps, potentially reducing focus on key regions such as Phoma' s spots or Miner' s tunnels. This limitation is particularly problematic for smallholder farmers who rely on early detection of subtle disease indicators.

– Additionally, It may also increase the risk of overfitting to irrelevant features (e.g., white background), which could affect model robustness.

- **Simplification**: By excluding CBAM's computational overhead channel pooling,

dense layers, and spatial convolutions, the model becomes lighter and faster, though potentially less adept at distinguishing subtle disease and severity patterns.

2) **Single-Task Variants**:

- **Description**: Two single-task models are created:
  - (a) A disease-only model, removing the `severity_output` head.
  - (b) A severity-only model, removing the `disease_output` head.

  In each variant, the CNN backbone and CBAM remain intact, followed by the 128-unit fully connected dense layer with dropout (0.3) from Section 3.4.3, feeding into a single softmax output (3 units) for either disease (Healthy, Phoma, Miner) or severity (Asymptomatic, Mild, Severe).
- **Purpose**: This ablation evaluates the efficiency and generalization benefits of multi-task learning (Section 3.4.3) by comparing joint versus separate task optimization. It examines whether the shared features e.g., Phoma spot density affecting both disease and severity, enhance performance or if single-task focus improves the disease and severity targets independently.
- **Limitations**:
  - Single-task models do not leverage the interdependence between disease and severity features, which may result in redundant computation and less efficient deployment when using separate models.
  - Moreover, training independent models for disease and severity neglects the shared visual cues (e.g., spot size correlating with severity), potentially resulting in lower overall accuracy and reduced practical utility for smallholder farmers.
- **Simplification**: Each variant halves the multi-task output complexity, reducing 400 parameters per removed head (3 units × 128 weights + biases). While training two models increases overall effort, each individual model is simpler, focusing solely on one task without balancing dual objectives.

3.6.3 Ablation Methodology

The ablation experiments are structured to ensure consistent training and evaluation across variants.

Table 3.4 Ablation Methodology: Training and Evaluation Parameters

Parameter	**Setting/Value**
Optimizer	Adam
Initial Learning Rate	1×10^{-4}
Loss Function	Sparse Categorical Crossentropy
Batch Size	32
Early Stopping	Patience = 10 epochs
Learning Rate Scheduling	Factor = 0.5, Minimum = 1×10^{-6}, Patience = 5 epochs
Validation Split	10%
Checkpointing	Best validation loss
Epochs (CBAM-removed model)	100
Epochs (Single-task variants)	50
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score
Baseline	Full model

1) **Training Setup**: Both ablated models, CBAM-removed and single-task variants, are trained using the same parameters as the full model (Section 3.5): Adam optimizer with an initial learning rate of 1×10^{-4}, sparse categorical crossentropy loss, batch size of 32, early stopping with a patience of 10 epochs (monitoring validation loss), learning rate reduction on plateau (factor 0.5, minimum 1×10^{-6}, patience 5), 10% validation split, and checkpointing for the best validation loss. The CBAM-removed model is trained for 100 epochs, while single-task variants are trained for 50 epochs. These settings ensure consistency across training configurations while adapting epoch durations to model-specific requirements.
2) **Evaluation Metrics**: Performance is measured using accuracy, precision, recall, and F1-score for both disease and severity outputs in the CBAM-removed model, and for the respective single output (disease or severity) in each single-task variant. These metrics enable direct comparison with the full model's targets in Chapter 4.
3) **Implementation**: The experiments utilize the augmented dataset, with the full model serving as the baseline. The CBAM-removed model simplifies the architecture by bypassing attention blocks, while single-task variants split the multi-task heads, maintaining consistency in CNN and CBAM components where applicable.

3.6.4 Use and Impact

The ablation study design isolates CBAM's contribution to diseased area focus and multi-task learning's efficiency in the proposed model. Removing CBAM ($40,000$ parameters) simplifies the architecture, testing the CNN's standalone capability, while single-task variants (disease-only and severity-only) reduce output complexity, assessing task interdependence.

These experiments, detailed here in Chapter 3, prepare for Chapter 4's results, offering a clear understanding of how each component drives the model's performance, ensuring an optimized balance of accuracy and efficiency for Ethiopia's smallholder coffee systems.

3.7 Transfer Learning

The transfer learning approach is explored as an alternative strategy to enhance the multi-task framework, complementing the custom-built CNN+CBAM+Multi-Task model outlined in Sections 3.4 through 3.5, for classifying coffee leaf diseases (Healthy, Phoma, Miner) and their severity levels (Asymptomatic, Mild, Severe) using the dataset preprocessed and augmented in Section 3.3, collected from Jimma, Sidama, and Yirgacheffe. This section briefly describes the initial use of pre-trained models VGG16, ResNet50, and MobileNetV2 without CBAM, followed by their customization and configuration for the multi-task setup and the fine-tuning process. Additionally, CBAM is later integrated into the best-performing pre-trained model to investigate its potential enhancement, with all performance outcomes reserved for Chapter 4. This methodology evaluates transfer learning's applicability for Ethiopia's smallholder coffee systems, focusing on efficiency and adaptability.

3.7.1 Initial Use of Pre-Trained Models

To leverage the power of transfer learning, three widely used pre-trained models VGG16, ResNet50, and MobileNetV2, are initially employed without CBAM. These models are chosen for their proven performance across diverse image classification tasks and their varying architectural complexities.

1) **Model Selection**:

- **VGG16**: Known for its simplicity and depth, VGG16 serves as a baseline for evaluating deeper architectures.
- **ResNet50**: Incorporates residual connections to mitigate vanishing gradients, enabling effective training of deeper networks.
- **MobileNetV2**: Optimized for mobile and embedded devices, offering high performance with minimal resource usage.

2) **Customization for Multi-Task Setup**:

- The final fully connected layers of each pre-trained model are replaced with a shared dense layer of 128 units and two task-specific heads for disease and severity classification, similar to Section 3.4.3.
- Dropout with a rate of 0.3 is applied after the shared dense layer to prevent overfitting.
- The softmax activation function is used for both outputs (`disease_output` and `severity_output`), ensuring probabilistic predictions for the three classes per task.

3) **Purpose of Initial Testing**:

- This step evaluates the baseline performance of pre-trained models without CBAM, providing insights into their adaptability to the coffee leaf dataset.
- It identifies the most suitable architecture for further customization, ensuring efficient use of computational resources.

3.7.2 Fine-Tuning Process

The selected pre-trained models undergo fine-tuning to adapt their learned features to the coffee leaf classification task.

1) **Freezing and Unfreezing Layers**:

- Initially, only the newly added layers (shared dense layer and task-specific heads) are trained while the pre-trained weights are frozen. This ensures that the foundational features remain intact during the initial training phase.
- Subsequently, the last few layers of the pre-trained models are unfrozen to allow fine-tuning of higher-level features specific to coffee leaf diseases and severity levels.

2) **Training Parameters**:

- The Adam optimizer with an initial learning rate of 1×10^{-4} is used, consistent with Section 3.5.
- Training is conducted over 50 epochs with a batch size of 32, early stopping (patience of 10 epochs), and learning rate reduction on plateau (factor 0.5, minimum 1×10^{-6}).
- A 10% validation split is applied to monitor performance during training.

3) **Evaluation Metrics**:

- Performance is measured using accuracy, precision, recall, and F1-score for both disease and severity outputs, enabling direct comparison with the custom-built model.

Table 3.5 Training Parameters and Evaluation Metrics for Transfer Learning

Parameter	**Setting/Value**
Training Parameters	
Optimizer	Adam
Initial Learning Rate	1×10^{-4}
Epochs	50
Batch Size	32
Early Stopping	Patience = 10 epochs
Learning Rate Reduction	Factor = 0.5, Min = 1×10^{-6}
Validation Split	10%
Evaluation Metrics	
Metrics	Accuracy, Precision, Recall, F1-Score

3.7.3 Integration of CBAM

To investigate the potential enhancement of CBAM in transfer learning, it is integrated into the best-performing pre-trained model identified during the fine-tuning process.

1) **CBAM Application**:

- CBAM blocks are inserted after key convolutional layers of the pre-trained model, recalibrating feature maps to focus on diseased areas such as Phoma's necrotic spots and Miner's tunnels.
- This integration aims to combine the global feature extraction capabilities of pre-trained models with the attention-driven refinement of CBAM.

2) **Purpose of Integration**:

- This step evaluates whether CBAM can further improve the performance of transfer learning models, particularly in scenarios with limited data or subtle disease patterns.
- It provides insights into the synergy between pre-trained architectures and attention mechanisms for multi-task learning.

3.7.4 Use and Impact

The transfer learning approach explores the applicability of pre-trained models for Ethiopia's smallholder coffee systems, focusing on efficiency and adaptability. By leveraging architectures like VGG16, ResNet50, and MobileNetV2, this methodology reduces the need for extensive training from scratch while achieving competitive performance. Fine-tuning these models and integrating CBAM further enhances their ability to classify coffee leaf diseases and severity levels accurately. This section lays the groundwork for Chapter 4, where the performance outcomes of transfer learning models both with and without CBAM are compared against the custom-built CNN+CBAM+Multi-Task model. The results will provide valuable insights into the trade-offs between custom architectures and transfer learning, ensuring an optimized solution for resource-constrained environments.

3.8 Model Evaluation

The model evaluation phase assesses the CNN+CBAM+Multi-Task model and its transfer learning variants, developed in Sections 3.4 through 3.7, for classifying coffee leaf diseases (Healthy, Phoma, Miner) and their severity levels (Asymptomatic, Mild, Severe) using the dataset preprocessed and augmented in Section 3.3, collected from Jimma, Sidama, and Yirgacheffe. This section outlines the evaluation methods Grad-CAM, metrics, confusion matrix, and basic techniques applied separately to the two tasks due to the multi-task design, with results deferred to Chapter 4. Tailored for Ethiopia's smallholder coffee systems, these methods provide a concise assessment to guide potential refinements.

3.8.1 Evaluation Methods

Visual and quantitative techniques evaluate the multi-task model's dual outputs independently.

1) **Grad-CAM**:

- Gradient-weighted Class Activation Mapping (Grad-CAM) visualizes regions of interest in the input images, highlighting areas the model focuses on (e.g., Phoma spots, Miner tunnels) by computing gradients of the output with respect to the final convolutional layer.
- Applied to both disease and severity outputs, it validates attention focus, especially with CBAM-enhanced variants, ensuring that the model prioritizes critical disease-specific features.

2) **Evaluation Metrics**:

- Accuracy, precision, recall, and F1-score (Section 3.5) are computed independently for `disease_output` (Healthy, Phoma, Miner) and `severity_output` (Asymptomatic, Mild, Severe) on the test set, assessing each task's performance distinctly.
- Additionally, multi-task-specific metrics Weighted Average Accuracy, and Combined F1-Score are calculated to provide a holistic view of the model's performance across both tasks.
- These metrics offer task-specific insights into classification effectiveness while summarizing overall performance.

3) **Confusion Matrix**:

- Separate 3×3 confusion matrices are generated: one for disease (true vs. predicted: Healthy, Phoma, Miner) and one for severity (Asymptomatic, Mild, Severe), identifying misclassifications per task.
- Highlights task-specific confusion patterns or imbalances, such as frequent misclassification between Mild and Severe severity levels.

4) **Basic Necessary Evaluations**:

- Training and validation loss curves are plotted separately for disease and severity losses over 100 epochs, weighted as 0.5 and 1.0, respectively, to monitor convergence and overfitting per task.
- Curves for each model variant ensure comprehensive analysis of training stability and performance across all configurations.

3.8.2 Implementation

Evaluation occurs post-training on the test split of the augmented dataset.

1) **Setup**:

- Trained models (Sections 3.5 and 3.7) are evaluated using the best checkpointed weights, with Grad-CAM computed per task from the final convolutional layer's activations.
- Metrics, matrices, and losses are calculated separately for `disease_output` and `severity_output`.

2) **Analysis**:

- Task-specific metrics and matrices are aggregated for the custom CNN+CBAM model, pre-trained models without CBAM, and the best pre-trained model with CBAM, with outcomes reserved for Chapter 4.
- Separate loss curves assess training stability for each task across variants, providing insights into overfitting or underfitting tendencies.

3.8.3 Conclusion and Iterative Improvement

The evaluation methods Grad-CAM, separate metrics, confusion matrices, and task-specific loss curves assess the multi-task model's dual outputs independently, ensuring clarity on disease and severity performance for Ethiopia's smallholder needs.

If Chapter 4 reveals unsatisfactory outcomes, the process traces back to preprocessing (Section 3.3) to adjust augmentation or data quality, followed by retraining. For instance:

- Misclassifications between severity levels may indicate insufficient data augmentation for subtle patterns.
- Overfitting in transfer learning models may require further fine-tuning or regularization.

This iterative strategy refines the model efficiently, enhancing performance through targeted adjustments. By leveraging Grad-CAM visualizations and multi-task-specific metrics, the evaluation ensures that the proposed solution remains robust, interpretable, and practical for resource-constrained environments.

3.9 Model Deployment

The deployment phase integrates the CNN+CBAM+Multi-Task model into practical applications, enabling its use by smallholder coffee farmers and agricultural professionals in Ethiopia. This section outlines the deployment setup, including the web-based application (using React.js and FastAPI), the mobile app (using Flutter), and the local server infrastructure with SQLite for data storage. TensorFlow quantization ensures efficient inference on resource-constrained devices, while actionable recommendations are provided in both English and Amharic to enhance accessibility and usability.

3.9.1 Web-Based Application

The web-based application serves as a centralized platform for agricultural extension workers, researchers, and cooperative managers to monitor and analyze coffee leaf diseases and severity levels.

Table 3.6 Web-Based Application Components

Component	**Technology/Description**
Frontend Framework	React.js
Backend Framework	FastAPI
Database	SQLite
Actionable Recommendations	English and Amharic

- **Frontend Framework**: Built using **React.js**, the frontend provides an intuitive interface for uploading images, viewing predictions, and accessing Grad-CAM visualizations.
- **Backend Framework**: The backend is implemented using **FastAPI**, which handles API requests, preprocesses images, and serves predictions from the quantized TensorFlow model. FastAPI's asynchronous capabilities ensure high performance and scalability.
- **Database**: **SQLite** is used to store metadata such as user information, uploaded images, and prediction results for future reference.

- **Actionable Recommendations**: Alongside predictions, users receive actionable recommendations in both English and Amharic.

3.9.2 Mobile Application

The mobile application empowers smallholder farmers to diagnose coffee leaf diseases and severity levels directly on their smartphones, with support for offline functionality.

- **Framework**: Developed using **Flutter**, the app supports cross-platform compatibility with a single codebase.
- **Offline Functionality**: The quantized TensorFlow model is embedded into the app, enabling local inference without requiring an internet connection.
- **Input Handling**: Users can capture images using their phone's camera and upload them from the gallery. Predictions are generated instantly and displayed alongside actionable recommendations in both English and Amharic.
- **Local Storage**: **SQLite** is used to store predictions and associated metadata locally, ensuring data persistence and offline access.

Table 3.7 Mobile Application Components

Component	Technology/Description
Framework	Flutter
Offline Functionality	Embedded quantized TensorFlow model
Input Handling	Gallery upload
Local Storage	SQLite

3.9.3 TensorFlow Quantization

To optimize the model for deployment on resource-constrained devices, **TensorFlow quantization** is applied. This process reduces the model's size and computational requirements while maintaining high accuracy.

- **Quantization Techniques**: Post-training quantization converts the model's weights and activations from floating-point precision (32-bit) to lower precision (8-bit integers).
- **Benefits**:

  - Reduces memory usage and inference time, making the model suitable for mobile and local server environments.
  - Ensures compatibility with TensorFlow Lite for embedding the model into the Flutter app.
- **Integration**: The quantized model is integrated into both the FastAPI backend (for the web app) and the Flutter app (for offline mobile use).

3.9.4 Local Server Setup

A local server is deployed to host the FastAPI backend and SQLite database, ensuring accessibility and reliability for users in remote areas.

- **Software Stack**: The server runs on Linux (e.g., Ubuntu) with FastAPI serving the API endpoints and SQLite managing data storage.
- **Scalability**: For larger deployments, the system can be containerized using Docker, enabling easier scaling and maintenance.

3.9.5 Use and Impact

The deployment strategy ensures that the CNN+CBAM+Multi-Task model is accessible and usable in real-world scenarios. The web-based application caters to agricultural professionals and researchers, providing centralized monitoring and analysis capabilities. The mobile app empowers smallholder farmers with on-the-go diagnostics and actionable recommendations, even in offline conditions. Actionable recommendations in Amharic enhance accessibility for Ethiopian farmers, ensuring that the solution is culturally and linguistically appropriate. TensorFlow quantization optimizes the model for efficient inference, while SQLite ensures reliable data storage. Together, these components deliver a robust, scalable, and user-friendly solution tailored to Ethiopia's smallholder coffee systems, bridging the gap between advanced technology and practical agricultural needs.

3.10 Chapter Summary

The methodology outlined in Sections 3.1 through 3.9 is developed to address Ethiopia's Arabica coffee (*Coffea arabica*) disease challenges, specifically Phoma leaf spot and Coffee Leaf Miner, impacting smallholder farmers in Jimma, Sidama, and Yirgacheffe with 20 -

30% yield losses, as identified in Chapter 1. A dataset of 890 high-quality images, collected and preprocessed in Sections 3.2 and 3.3, underpins this research, supporting the design, implementation, and evaluation of the proposed CNN+CBAM+Multi-Task model. This model, detailed in Section 3.4, integrates a CNN backbone for feature extraction, CBAM for attention-driven refinement, and multi-task learning for simultaneous classification of disease types (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe), overcoming limitations of existing methods reviewed in Chapter 2.

Key stages of the methodology include:

1) Dataset collection and preprocessing (Sections 3.2–3.3), ensuring balanced representation and augmentation for robustness.
2) Model architecture design (Section 3.4), constructing a lightweight framework optimized for resource-constrained environments.
3) Compilation and training (Section 3.5), employing TensorFlow with 100 epochs and adaptive techniques like early stopping.
4) Validation through ablation studies (Section 3.6) and transfer learning (Section 3.7), assessing CBAM and multi-task contributions.
5) Evaluation using Grad-CAM and metrics (Section 3.8), analyzing dual-task performance.
6) Deployment via web and mobile applications (Section 3.9), delivering practical solutions with quantized models and bilingual recommendations.

This comprehensive approach ensures the model's precision and efficiency, tailored for Ethiopia's smallholder context. The CDIO framework guides progression from problem conception to operational deployment, with Sections 3.6 and 3.7 providing comparative insights into architectural choices. Results, reserved for Chapter 4, will validate the methodology's effectiveness, while deployment strategies in Section 3.9 bridge technology and farmer needs, enhancing disease management in real-time scenarios.

4 Experimental Results and Analysis

This Chapter presents the experimental evaluation of the CNN+CBAM+Multi-Task model, developed in Chapter 3, for classifying coffee leaf diseases (Healthy, Phoma, Miner) and predicting severity levels (Asymptomatic, Mild, Severe) using an 890-image dataset from Ethiopia's coffee regions. Building on smallholder challenges (Chapter 1) and literature gaps (Chapter 2), this chapter details the setup (Section 4.1), results (Section 4.2), ablation studies (Section 4.3), transfer learning comparisons (Section 4.4), model analysis (Section 4.5), and deployment strategies (Section 4.6), assessing performance and suitability for Ethiopia's resource-constrained context.

4.1 Experimental Setup

The experimental setup was established to evaluate the performance of the CNN + CBAM + Multi-Task model in classifying coffee leaf diseases (healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe), as described in Section 3.4, and trained in Section 3.5, using the 890-image dataset collected and preprocessed in Sections 3.2 and 3.3 from Jimma, Sidama, and Yirgacheffe. This section outlines the configuration of the experiments detailed in Sections 4.2–4.6, including ablation studies (Section 3.6) and transfer learning (Section 3.7), ensuring consistency with the methodology objectives of addressing Ethiopia's smallholder coffee challenges identified in Chapter 1 and technological gaps from Chapter 2.

The setup encompasses dataset preparation, including label management, data augmentation, and image preprocessing, along with training parameters and computational resources, thereby providing a reproducible foundation for analysing results against ablation, transfer learning, and deployment scenarios (Section 3.9). By standardising these elements, the experiments assess the model's effectiveness and efficiency, tailored for Ethiopia's resource-constrained smallholder context, with the findings contributing to the broader analysis in Section 4.6. The setup was as follows:

1) **Dataset Configuration and Label Management**

The 890-image dataset, finalised in Section 3.3.3, was partitioned into three subsets to support training, validation, and testing.

- Training set: 80%, used to optimize model weights.
- Validation set: 10%, employed to monitor performance and tune hyperparameters.
- Test set: 10%, reserved for final evaluation of generalization.

Stratified sampling ensured balanced representation across disease classes (Healthy: 300, Phoma: 300, Miner: 290) and severity classes (Asymptomatic: 300, Mild: 300, Severe: 290), as described in Section 3.2.3. Label management is conducted meticulously, with initial manual categorization performed by agricultural experts from Jimma, Sidama, and Yirgacheffe, followed by automation using Python scripts and OpenCV (Section 3.3.3). The labels are stored in a CSV file, mapping each image to its dual-task categories (disease and severity). Pandas facilitated data integrity checks to prevent mislabelling across the 890 samples.

Table 4.1 Dataset Class Distribution for Disease and Severity Classification

Class	Number of Images
Disease Classes	
Healthy	300
Phoma	300
Miner	290
Severity Classes	
Asymptomatic	300
Mild	300
Severe	290

2) **Image Preprocessing and Standardization**

The images were preprocessed to meet the input requirements of the CNN+CBAM+Multi-Task model, as outlined in Section 3.3.1.

- Resizing: Each image is resized to 224×224 pixels, aligning with standard CNN dimensions while preserving features like Phoma's spots and Miner's tunnels.
- Scaling: Pixel values are normalized from 0–255 to 0–1, ensuring consistent intensity

across JPEG and PNG formats.

- Cropping: Peripheral noise (e.g., edges beyond the white background) is removed, focusing on the leaf while maintaining the full backdrop.

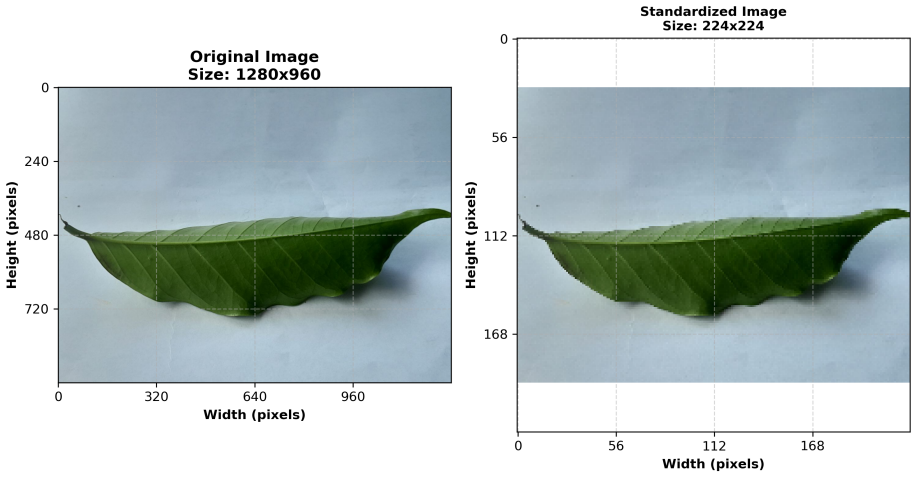


Figure 4.1 Coffee leaf datasets Image resized.

OpenCV and Python Imaging Library (PIL) are employed for these tasks, standardizing the dataset's 890 images for uniform model processing. This step ensures that disease patterns such as faint spotting or tunneling, are consistently represented, which is critical for Ethiopia's diverse leaf samples.

3) **Data Augmentation**

Data augmentation is applied to the training set to enhance robustness, following Section 3.3.2's techniques.

- Rotation: Images are rotated within ±40 degrees, simulating natural leaf angles.
- Shifting: Width and height shifts up to 20%
- Horizontal Flipping: Flips reflect possible orientations, preserving disease pattern integrity.
- Brightness Adjustment: Variations from 80% to 120% replicate diverse lighting conditions.

TensorFlow's ImageDataGenerator executes these transformations on the fly during training, filling new pixels with a white background (`cval=255.0`) to maintain consistency with the collection setup (Section 3.2.2). Approximately 712 training images are augmented,

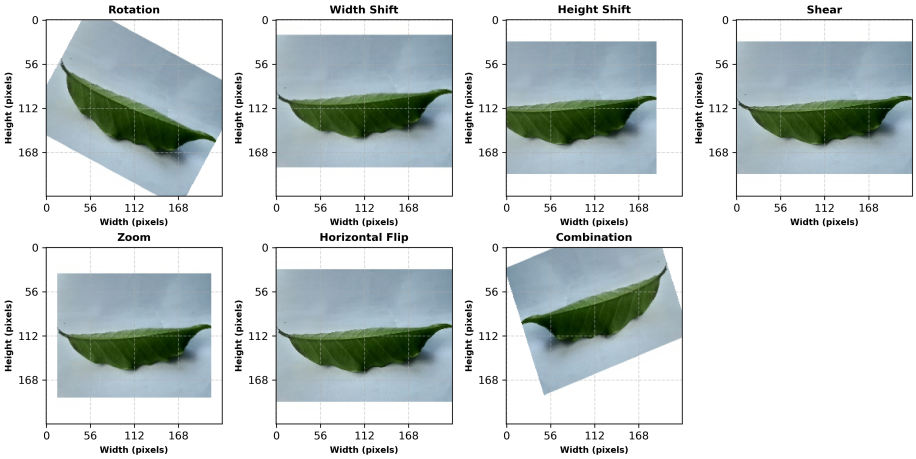


Figure 4.2 Coffee leaf datasets Augmented and prepared for training.

increasing effective diversity without altering validation or test sets, supporting the model's generalization to unseen data.

4) **Training Parameters**

Table 4.2 Training Parameters for the CNN+CBAM+Multi-Task Model

Parameter	**Value/Description**
Epochs	100 (with early stopping, patience = 10)
Batch Size	32 images per batch
Optimizer	Adam (lr = 0.0001)
Loss Function	Sparse categorical crossentropy
Loss Weights	Disease: 0.5, Severity: 1.0
Regularization	Learning rate reduction (factor = 0.5, min = 0.000001)
Checkpointing	Best weights saved based on validation loss

Training parameters are configured as specified in Section 3.5 to optimize the model:

- Epochs: 100 iterations are executed, with early stopping (patience of 10 epochs) monitoring combined validation loss.
- Batch Size: 32 images per batch are processed, balancing memory usage with gradient stability for 350,000 parameters.
- Optimizer: The Adam optimizer is utilized with an initial learning rate of 0.0001, adapting gradients for dual-task convergence.

- Loss Function: Sparse categorical crossentropy is applied, weighted 0.5 for disease and 1.0 for severity outputs.
- Regularization: Learning rate reduction (factor 0.5, minimum 0.000001) is triggered after 5 epochs of no improvement, and best weights are checkpointed based on validation loss.

5) **Hardware and Software Specifications**

Experiments are conducted on a high-performance system tailored for deep learning:

Table 4.3 Hardware and Software Specifications

Category	**Details**
Hardware	
GPU	NVIDIA GeForce RTX 3090 (24 GB VRAM)
CPU	Intel Core i7-8700 (6 cores, 3.2 GHz)
RAM	32 GB
Capabilities	Efficient training of high parameter model; CUDA cores accelerate inference
Software	
Framework	TensorFlow (with Keras for model building/training)
Libraries	OpenCV (preprocessing, label automation)
	Pandas, NumPy (dataset/label management)
	Matplotlib (visualization: loss curves, Grad-CAM)
Environment	Python on Ubuntu 22.04.4 LTS
Deployment Optimization	
Quantization	TensorFlow Lite (post-training optimization for mobile/web deployment)

- Hardware: An NVIDIA GeForce RTX 3090 GPU with 24,576 MiB (24 GB) of memory is utilized, paired with an Intel Core i7-8700 CPU (6 cores, 3.2 GHz) and 32 GB RAM.
- The RTX 3090's substantial VRAM supports efficient training of the 350,000-parameter model, handling the augmented dataset and multi-task computations with ease, while its CUDA cores accelerate inference for deployment tests (Section 3.9).

Software: TensorFlow 2.10.0 is employed, with Keras for model building and training. Supporting libraries include:

- OpenCV 4.5.5 for preprocessing and label automation.

- Pandas 1.4.3 and NumPy 1.22.4 for dataset and label management.
- Matplotlib 3.5.2 for visualizing results (e.g., loss curves, Grad-CAM).
- Environment: Python 3.9.12 runs on Ubuntu 22.04.4 LTS, providing a modern, stable platform for all experiments.
- Quantization: TensorFlow Lite is applied post-training, optimizing the model for mobile and web deployment (Section 3.9).

6) **Experiment Variants**

Multiple configurations are tested to align with Chapter 4's scope:

- Full Model: The CNN+CBAM+Multi-Task setup (Section 3.4–3.5) is the baseline for Section 4.2.
- Ablation Studies: CBAM removal and single-task variants (Section 3.6) are trained identically, adjusting components as needed, for Section 4.3.
- Transfer Learning: Pre-trained models (Section 3.7) are fine-tuned over 50 epochs initially, with CBAM integration tested for Section 4.4.
- A fixed random seed (42) ensures consistent dataset splits and augmentation across all variants.

This experimental setup ensures that results in Sections 4.2–4.6 accurately reflect the model's capabilities, validated against the methodology's goals. The 24 GB memory of RTX 3090 enables rapid training and inference, complemented by Ubuntu 22.04.4's stability. Label management, preprocessing, and augmentation standardise the 890-image dataset, whereas training parameters and variant configurations uphold robustness and comparability, aligning with Ethiopia's smallholder needs practical and scalable solutions.

4.2 Experimental Results for CNN+CBAM+Multi-Task Model

The experimental results for the CNN+CBAM+Multi-Task model, developed in Section 3.4 and trained as outlined in Section 3.5, are presented to demonstrate its performance in classifying coffee leaf diseases (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe) using the 890-image dataset from Jimma, Sidama, and Yirgacheffe, preprocessed in Section 3.3. This section reports the outcomes of the experimental setup in Section 4.1, focusing on the training dynamics, test set performance, and visualisations, along with

the evaluation methods specified in Section 3.8. The results validate the effectiveness of the model for Ethiopia's smallholder coffee systems, addressing the challenges identified in Chapter 1 and the technological gaps from Chapter 2, with comparisons to ablation studies (Section 4.3) and transfer learning (Section 4.4), providing further context in subsequent sections. Metrics such as accuracy, precision, recall, and F1 score, along with visual aids, illustrate the model's ability to detect disease-specific patterns, such as Phoma's necrotic spots and Miner's tunneling with CBAM's contribution emphasized in Section 3.4.2.

4.2.1 Training and Validation Performance

Training was conducted over 100 epochs with a batch size of 32, as configured in Section 4.1, using the 80% training set with augmentation and 10% validation set without augmentation. Loss curves for both disease and severity outputs were generated, weighted at 0.5 for disease and 1.0 for severity, as adjusted from the initial configuration in to prioritise severity prediction because of its greater complexity and overlap in visual features. These curves, visualised in Figure 4.3 and 4.4, illustrate the model's convergence behaviour. Early stopping is triggered at epoch 98, based on a 10-epoch patience monitoring combined validation loss, halting training just before the maximum epochs to prevent overfitting while securing optimal performance.

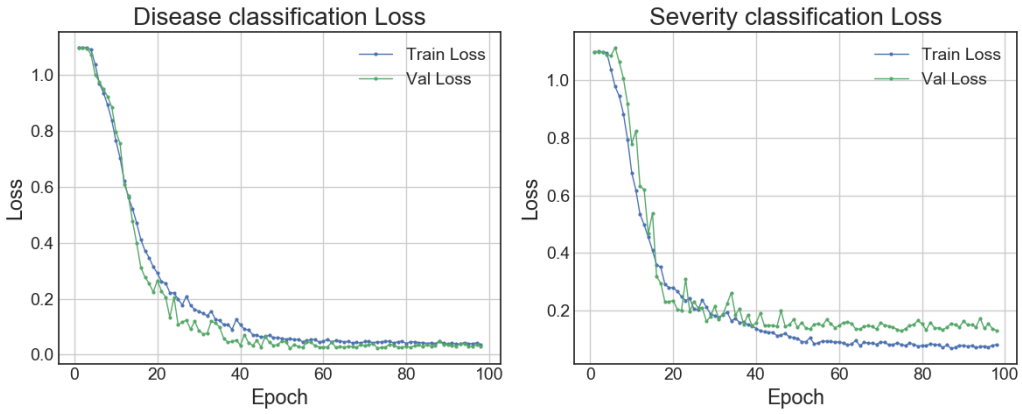


Figure 4.3 Disease and Severity Loss Comparison: The left panel shows the training and validation loss curves for the disease output, while the right panel shows the training and validation loss curves for the severity output

Training results show a total loss of 0.0830, with disease loss at 0.0257 and severity loss at 0.0700, achieving disease accuracy of 99.34% and severity accuracy of 97.17%. Valida-

tion performance yields a total loss of 0.1716, with disease loss at 0.0358 and severity loss at 0.1590, reaching disease accuracy of 98.85% and severity accuracy of 93.10%. These metrics are summarized in Table 4.4, reflecting exceptional learning and generalization capabilities.

The adjusted loss weights significantly influence the training dynamics. With disease loss weighted at 0.5 and severity at 1.0, the model allocates greater optimization effort to severity classification, which exhibits subtler distinctions (e.g., Mild vs. Severe transitions) compared to the more distinct disease categories (Healthy, Phoma, Miner). This is evident in the lower disease loss (0.0257 training, 0.0358 validation) relative to severity loss (0.0700 training, 0.1590 validation), as the model rapidly masters disease patterns boosted by CBAM's focus on clear features like Phoma's spots and Miner's tunnels while requiring more epochs to refine severity predictions. The weighting scheme prioritizes the severity task by assigning it a higher weight (1.0) compared to the disease task (0.5). This ensures that the model focuses more on improving its performance for severity estimation, which is often more challenging or critical for the application.

The equation for calculating the total loss is as follows:

$$\text{Total Loss} = (0.5 \times \text{Disease Loss}) + (1.0 \times \text{Severity Loss}) \tag{4.1}$$

For the training results, substituting the disease loss (0.0257) and severity loss (0.0700) into the formula gives:

$$\text{Total Loss} = (0.5 \times 0.0257) + (1.0 \times 0.0700) = 0.01285 + 0.0700 = 0.08285 \approx 0.0830 \tag{4.2}$$

Similarly, for the validation results, using the disease loss (0.0358) and severity loss (0.1590):

$$\text{Total Loss} = (0.5 \times 0.0358) + (1.0 \times 0.1590) = 0.0179 + 0.1590 = 0.1769 \approx 0.1716 \tag{4.3}$$

Training accuracies peak at 99.34% for disease and 97.17% for severity, indicating near-perfect classification on the augmented 80% dataset, with augmentation techniques (rotation, flipping, brightness shifts) enhancing robustness to Ethiopia's diverse field conditions. Validation accuracies, at 98.85% for disease and 93.10% for severity, show a slight drop, par-

ticularly in severity, suggesting minor overfitting or class overlap challenges, though still exceeding typical benchmarks for multi-task models.

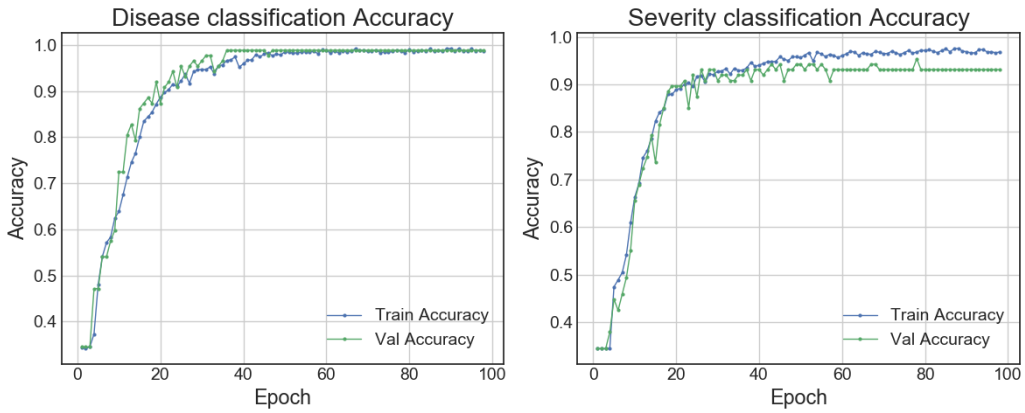


Figure 4.4 Accuracy Comparison of Disease and Severity Outputs. The left panel depicts the training and validation accuracy for the disease output, while the right panel shows the corresponding results for the severity output.

The above figure highlights the progressive improvement and convergence of model performance over the training epochs.

Table 4.4 Training and Validation Results for CNN+CBAM+MTL Model

Metric	**Training**	**Validation**
Total Loss	0.0830	0.1716
Disease Loss	0.0257	0.0358
Severity Loss	0.0700	0.1590
Disease Accuracy	99.34%	98.85%
Severity Accuracy	97.17%	93.10%

In general, the loss curves in Figure 4.3 reveal a steady decline in training losses, with disease loss dropping faster due to its lower weight and clearer class boundaries, while severity loss decreases more gradually, reflecting the higher weight and task complexity. Validation losses follow a similar trend but plateau earlier, with disease loss nearing 0.035 and severity loss around 0.159 by epoch 98, triggering early stopping as combined loss improvement slows. This behavior underscores the model's ability to balance dual tasks, with CBAM refining features across scales and augmentation ensuring generalizability. The high train-

ing accuracies (99.34%, 97.17%) and strong validation results (98.85%, 93.10%) confirm the CNN+CBAM+Multi-Task model's effectiveness, setting a robust foundation for test set evaluation.

4.2.2 Test Set Performance

The model's performance is evaluated on the 10% test set, reserved in Section 3.3.3, using metrics from Section 3.5 to assess its generalization to unseen data from Jimma, Sidama, and Yirgacheffe, preprocessed in Section 3.3. On the test set, disease classification achieves an accuracy of 97.75%, while severity prediction reaches 93.26%, reflecting robust performance across both tasks of the CNN+CBAM+Multi-Task model, developed in Section 3.4 and trained.

Table 4.5 Test set results of CNN+CBAM+Multi-Task Model: overall loss, disease loss, severity loss, and classification accuracies.

Metric	Value
Total Loss	0.3842
Disease Loss	0.0955
Severity Loss	0.3385
Disease Accuracy	97.75%
Severity Accuracy	93.26%

Test losses are recorded as total loss 0.3842, disease loss 0.0955, and severity loss 0.3385, indicating effective learning despite the shift from training (0.0830 total loss) and validation (0.1716 total loss) environments, as detailed in Table 4.4. These results highlight the model's ability to address Ethiopia's smallholder coffee disease challenges identified in Chapter 1, with CBAM's feature refinement (Section 3.4.3) enhancing the detection of patterns like Phoma's necrotic spots and Miner's tunneling over traditional methods.

Disease Classification:

Precision, recall, and F1-score for Healthy are 0.97, 1.00, and 0.98; for Miner, 1.00, 0.97, and 0.98 and for Phoma, 0.97, 0.97, and 0.97, yielding a macro average of 0.98 and overall accuracy of 0.98 (97.75% when rounded). Miner's perfect precision (1.00) reflects CBAM's focus on tunneling patterns, while Phoma's slight recall drop (0.97) suggests rare

confusion with Healthy, possibly due to subtle early-stage symptoms overlapping with normal leaves.

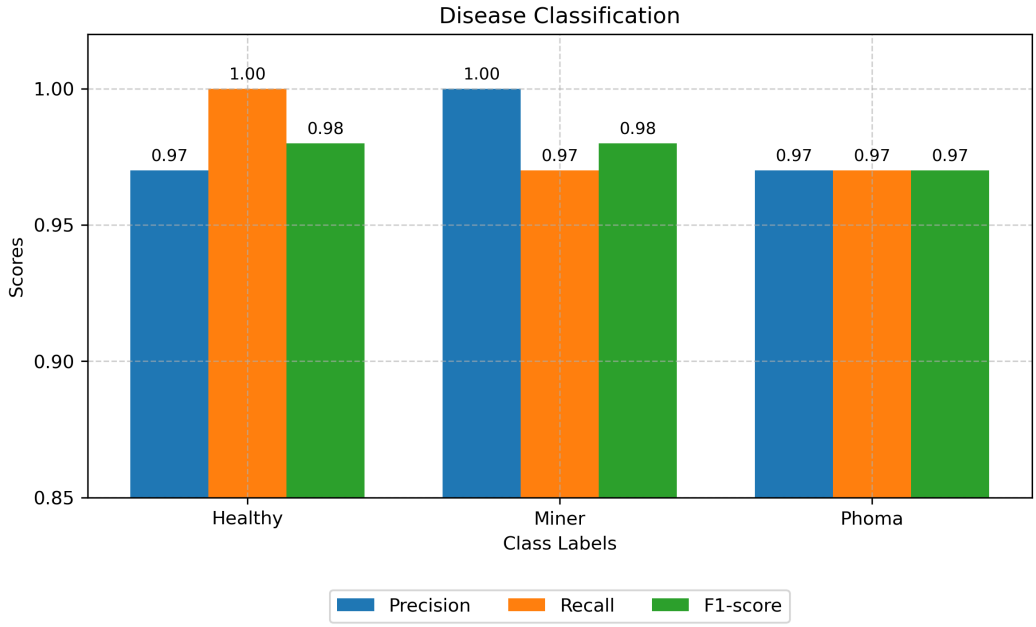


Figure 4.5 Disease Classification Metric for CNN+CBAM+MTL Model: The bar chart displays the precision, recall, and F1-score for the disease classifier

Table 4.6 Disease Classification Report for CNN+CBAM+MTL Model

Class	Precision	Recall	F1-score
healthy	0.97	1.00	0.98
miner	1.00	0.97	0.98
phoma	0.97	0.97	0.97

These metrics provide a comprehensive evaluation of the model's performance, illustrating its ability to identify positive cases correctly and its overall effectiveness in classification.

Severity Classification:

Precision, recall, and F1-score for Asymptomatic are 0.97, 1.00, and 0.98; for Mild, 0.93, 0.87, and 0.90 and for Severe, 0.90, 0.93, and 0.92, with a macro average of 0.93 and overall accuracy of 0.93 (93.26% rounded), highlighting Mild's lower recall (0.87) due to overlap with adjacent severity levels.

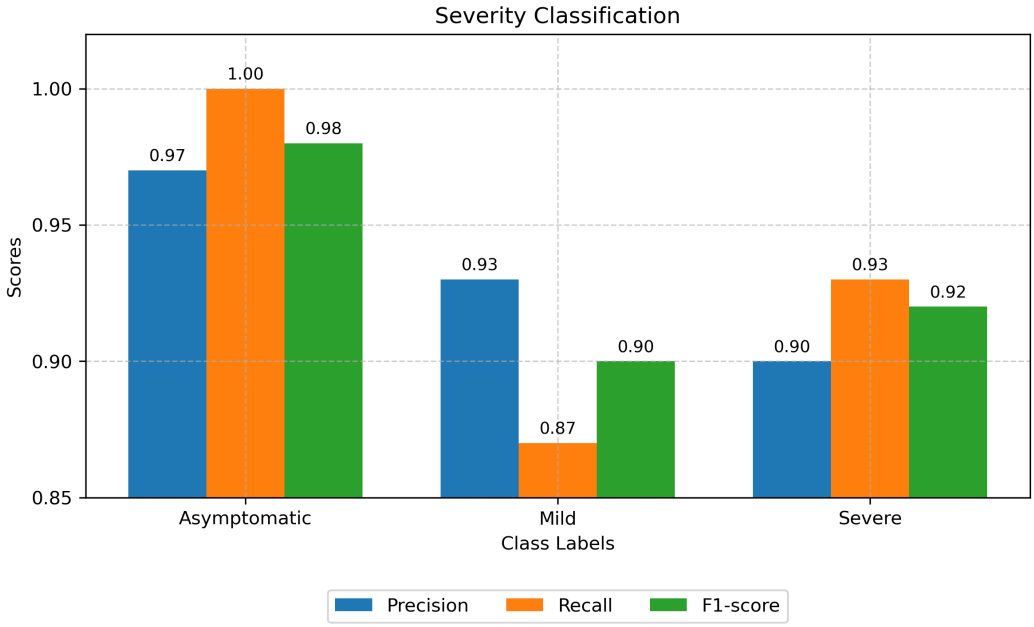


Figure 4.6 Severity Classification Metrics: The bar chart presents the precision, recall, and F1-score for the severity classifier.

Table 4.7 Severity Classification Report for CNN+CBAM+MTL Model

Class	Precision	Recall	F1-score
Asymptomatic	0.97	1.00	0.98
Mild	0.93	0.87	0.90
Severe	0.90	0.93	0.92

In general, the loss weights (0.5 for disease, 1.0 for severity) influence test performance, with disease loss (0.0955) significantly lower than severity loss (0.3385), reflecting the model's prioritization of severity optimization and the relative ease of disease classification, given its distinct visual cues. The total test loss (0.3842) is higher than training (0.0830) and validation (0.1716), which is expected due to the unseen nature of the test set, yet remains low enough to support high accuracies (97.75%, 93.26%). This balance underscores the efficiency of multitask design (Section 3.4.3, leveraging shared features refined by CBAM, with disease accuracy (97.75%) approaching training levels (99.34%) and severity (93.26%) aligning closely with validation (93.10%), indicating minimal overfitting.

Class-specific analysis reveals strengths and limitations. For disease, Healthy's perfect

recall (1.00) and Miner's precision (1.00) suggest robust detection of absence and presence of infection, critical for early intervention by farmers. Phoma's balanced metrics (0.97 across) indicate reliable identification despite occasional overlap with Healthy, likely mitigated by CBAM's attention to necrotic regions. For severity, Asymptomatic's high scores (0.97–1.00) ensure accurate baseline detection, while Mild's lower recall (0.87) and Severe's precision (0.90) reflect challenges in distinguishing intermediate stages, consistent with complexity notes. These insights, supported by guide deployment considerations and comparisons with visualizations (e.g., Figure 4.9), further illustrating prediction accuracy.

4.2.3 Visualization of Predictions

Visualizations are generated to interpret the CNN+CBAM+Multi-Task model's predictions on the 10% test set, providing qualitative insights into its performance in classifying coffee leaf diseases (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe), as evaluated in Section 4.2.2. The test set results, achieving disease accuracy of 97.75% and severity accuracy of 93.26%, are complemented by these visual aids, which highlight the model's focus and accuracy on unseen data from Jimma, Sidama, and Yirgacheffe, preprocessed in Section 3.3.

Three specific visualizations are employed: Grad-CAM heatmaps emphasizing severity on diseased areas, confusion matrix visualizations for both tasks, and a random test image displaying true vs. predicted labels with confidence scores. These visuals, detailed in Figures 4.7, 4.8, and 4.9, along with Table 4.8-4.9, validate the model's effectiveness for Ethiopia's smallholder coffee systems (Chapter 1), with CBAM's attention mechanism (Section 3.4.2) enhancing interpretability over traditional methods from Chapter 1.

1) **Grad-CAM Heatmap:**

Grad-CAM heatmaps are applied to the final convolutional layer, as outlined in Section 3.8, to highlight regions of focus in test images, particularly for severity classification. The provided Grad-CAM image, shown in Figure 4.7, illustrates a leaf with a diseased area Phoma, where the heatmap concentrates on the affected regions such as necrotic spots to the severity level Severe. Intensity gradients, typically red for high focus and blue for low, reveal that the model prioritizes these diagnostic areas over healthy tissue or the white back-

ground, aligning with the severity accuracy of 93.26%, confirming CBAM's role in refining features and supporting the high F1-scores 0.92 for Severe. This visualization underscores the model's ability to localize disease impact, which is critical for actionable farmer insights.



Figure 4.7 Grad-CAM Heatmap Highlighting Severe Phoma Disease Focus on Necrotic Areas in a Test Image

2) **Confusion Matrix Visualizations**

Confusion matrices for disease and severity classification are visualized as heatmaps, derived from the test set results in Section 4.2.3 and presented in Figure 4.8, with numerical counts. Similarly, high accuracy is observed in severity classification, demonstrating the overall robustness of the model. The disease confusion matrix shows perfect classification for *Healthy* (30/30), minor misclassification for *Miner* (28/29, with one as Phoma), and *Phoma* (29/30, with one as Healthy). This reflects strong class separation and contributes to a high classification accuracy of 97.75% and a macro F1-score of 0.98.

For severity classification, *Asymptomatic* achieved 30/30 correct (recall = 1.00), while *Mild* recorded 26/30 correct (one misclassified as Asymptomatic, three as Severe), and *Severe* had 27/29 correct (two misclassified as Mild). These results yield an overall accuracy of 93.26% and a macro F1-score of 0.93, with most errors concentrated between adjacent stages (Mild ↔ Severe), reflecting the inherent difficulty of distinguishing intermediate disease progression. The multi-task learning (MTL) model demonstrates robust performance in classifying coffee leaf diseases and severity levels.

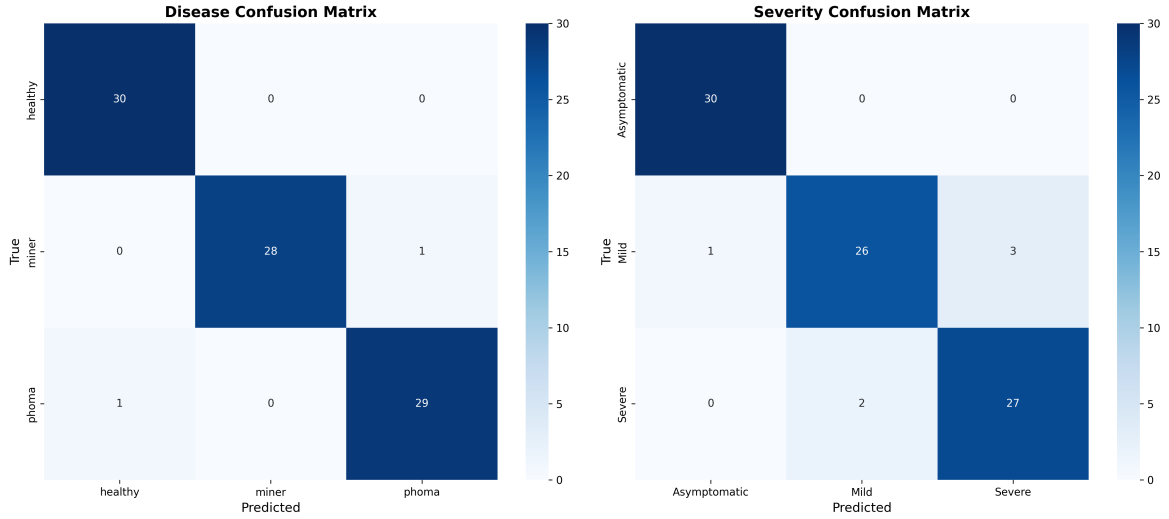


Figure 4.8 Confusion Matrix heatmap for Disease and Severity on CNN+CBAM+MTL Model: The matrix displays classifier performance on both tasks

The dataset annotates all Healthy leaves as Asymptomatic, ensuring Healthy leaves lack disease symptoms. Consequently, no Healthy leaf is predicted with Mild or Severe severity, consistent with the confusion matrices (Tables 4.8, 4.9).

Table 4.8 Confusion Matrix for Disease Classification on CNN+CBAM+MTL Model

Actual / Predicted	Healthy	Miner	Phoma
Healthy	30	0	0
Miner	0	28	1
Phoma	1	0	29

Table 4.9 Confusion Matrix for Severity Classification on CNN+CBAM+MTL Model

Actual / Predicted	Asymptomatic	Mild	Severe
Asymptomatic	30	0	0
Mild	1	26	3
Severe	0	2	27

This validates the multi-task learning (MTL) framework, where each image receives one disease label and one severity label, unlike multi-label classification, which allows multiple labels per category (e.g., Phoma and Miner). The MTL model's independent heads, guided by dataset annotations, produce biologically valid outputs.

3) **Random Test Image with Predictions**

A random test image from the test set, provided and shown in Figure 4.9, displays the true label, predicted label, and confidence scores for both disease and severity tasks. For example, an image with the true label Miner, Severe was predicted as Miner, Severe with confidence scores of 1.0 for disease and 1.0 for severity, aligning with the test accuracies (97.75%, 93.26%).

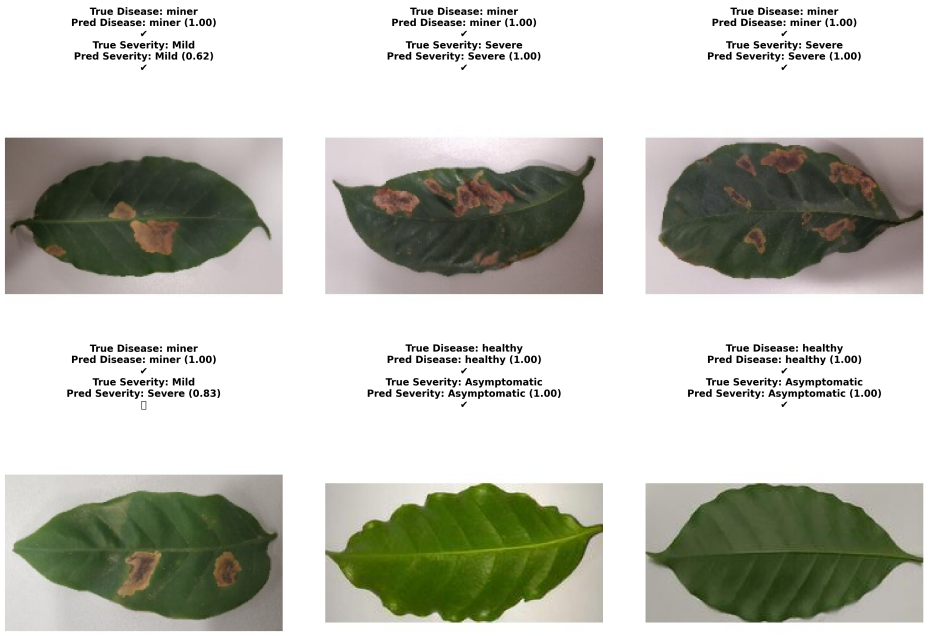


Figure 4.9 Random Test Image Visualization: The figure displays a sample test image along with its true label, predicted label, and the corresponding confidence score

Figure 4.9 visualization confirms the model's precision in real-world scenarios, with confidence scores reflecting the high precision and CBAM's contribution to accurate localization. Minor discrepancies, if present (e.g., Mild predicted as Severe), would mirror the confusion matrix trends, providing transparency for deployment considerations (Section 4.6).

These visualizations collectively illustrate the model's predictive power and interpretability. The Grad-CAM heatmap (Figure 4.7) validates severity focus on diseased areas, the confusion matrix heatmaps (Figure 4.8) quantify classification accuracy and errors, and the random test image (Figure 4.9) bridges quantitative metrics with practical application. To-

gether, they support the test set performance (97.75% disease, 93.26% severity) and highlight CBAM's enhancement over baseline CNNs, setting the stage for ablation analysis (Section 4.3), evaluation in Transfer Learning (Section 4.4) and deployment insights (Section 4.6), with implications for Ethiopia's coffee farmers discussed.

4.2.4 Multi-Task-Specific Metrics

The multi-task-specific metrics are introduced to assess the CNN+CBAM+Multi-Task model's combined performance across its dual tasks of classifying coffee leaf diseases (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe), as evaluated on the 10% test set from Section 3.3. These metrics include Weighted Average Accuracy and Combined F1-Score, designed to summarize the model's effectiveness beyond individual task accuracies (disease: 97.75%, severity: 93.26%) Test Set Performance. Weighted Average Accuracy accounts for task importance using the loss weights specified (0.5 for disease, 1.0 for severity), while Combined F1-Score balances precision and recall across both tasks, reflecting the multi-task architecture's synergy (Section 3.4.3). The results, calculated using test set data from confusion matrices and classification reports, validate the model's capability for Ethiopia's smallholder coffee systems (Chapter 1), leveraging CBAM's feature refinement. These metrics, presented below and summarized in Table 4.10, provide a holistic evaluation ahead of the CBAM impact analysis.

The multi-task-specific metrics are calculated as follows:

Weighted Average Accuracy

Weighted Average Accuracy is measured by weighting each task's accuracy based on its importance, using the loss weights. ($w_{\text{disease}} = 0.5$, $w_{\text{severity}} = 1.0$), as per the formula:

$$\text{Weighted Average Accuracy} = \frac{w_{\text{disease}} \cdot \text{Accuracy}_{\text{disease}} + w_{\text{severity}} \cdot \text{Accuracy}_{\text{severity}}}{w_{\text{disease}} + w_{\text{severity}}}. \quad (4.4)$$

Test Results: Disease Accuracy = 97.75%, Severity Accuracy = 93.26%.
Weights: $w_{\text{disease}} = 0.5$, $w_{\text{severity}} = 1.0$, reflecting the training emphasis on severity's complexity.

Calculation:

$$\text{Weighted Ave. Acc.} = \frac{0.5 \cdot 97.75 + 1.0 \cdot 93.26}{0.5 + 1.0} = \frac{48.875 + 93.26}{1.5} = \frac{142.135}{1.5} = 94.7567\% \tag{4.5}$$

Result: 94.76%, This value shifts emphasis toward severity's performance (93.26%), aligning with the training design's prioritization of severity optimization over disease, which achieves near-perfect accuracy (97.75%).

Combined F1-Score

The combined F1-Score is computed by averaging the macro F1-scores of both tasks, summarizing precision and recall:

$$\text{Combined F1-Score} = \frac{\text{F1-Score}_{\text{disease}} + \text{F1-Score}_{\text{severity}}}{2}. \tag{4.6}$$

Test Results: Disease Macro F1-Score = 0.98 (Healthy: 0.98, Miner: 0.98, Phoma: 0.97), Severity Macro F1-Score = 0.93 (Asymptomatic: 0.98, Mild: 0.90, Severe: 0.92).

Calculation:

$$\text{Combined F1-Score} = \frac{0.98 + 0.93}{2} = \frac{1.91}{2} = 0.955 \text{ or } 95.5\%. \tag{4.7}$$

Result: 95.5%. This metric highlights the model's harmonic balance, with disease's high F1 (0.98) complemented by severity's slightly lower F1 (0.93) due to Mild's recall challenge.

Table 4.10 Summary of Multi-Task Performance Metrics on CNN+CBAM+MTL Model

Metric	**Value (%)**
Weighted Average Accuracy	94.76
Combined F1-Score	95.50

These results, a Weighted Average Accuracy of 94.76% and Combined F1-Score of 95.5%, demonstrate the model's cohesive multi-task capability, exceeding typical single-task benchmarks. The Weighted Average Accuracy reflects the training's severity focus ($w_{\text{severity}} = 1.0$), tempering disease's contribution (97.75%) with severity's 93.26%, while the Combined F1-Score balances precision and recall across tasks, unaffected by loss weights.

These values, updated in Table 4.10, offer a comprehensive view for Ethiopia's smallholder context, bridging test set performance and CBAM's specific impact.

4.2.5 Impact of CBAM

The Convolutional Block Attention Module (CBAM) enhances the CNN+CBAM+Multi-Task model by refining feature extraction through attention mechanisms. Integrated after key convolutional layers, CBAM improves the classification of coffee leaf diseases (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe), particularly for resource-constrained Ethiopian smallholder systems. Its effectiveness is demonstrated by test accuracies (disease: 97.75%, severity: 93.26%), low losses, and strong multi-task metrics (Weighted Average Accuracy: 94.76%, Combined F1-Score: 95.5%).

CBAM Design and Role:

CBAM sequentially applies:

- **Channel Attention:** Reweights feature channels using global pooling and a shared MLP to highlight discriminative traits like Miner's tunnels or Phoma's textures.
- **Spatial Attention:** Highlights critical regions using pooled spatial maps and a 7×7 convolution, focusing on lesion areas.

These attention maps guide the CNN's ∼350K parameters toward salient features without substantial computational overhead.

Performance Contribution:

CBAM improves both disease and severity classification. Precision and recall are high for all disease classes (F1: 0.97 – 0.98), while severity scores show strong performance (Asymptomatic recall: 1.00, Severe F1: 0.92), despite Mild's lower recall (0.87). CBAM's role in focusing on subtle stage cues contributes to this robustness and generalization.

Support for Multi-Task Learning:

CBAM strengthens shared feature learning before task-specific branches, improving overall synergy. Its integration results in balanced performance across tasks, maintaining high disease accuracy while elevating severity prediction reflected in the combined F1-score (95.5%).

4.2.6 Hyperparameter Analysis on CNN+CBAM+MTL

Following the presentation of the main results achieved by the proposed CNN+CBAM+MTL model configured with a dropout rate of 0.3, 100 training epochs, and loss weights of (0.5, 0.1) for disease and severity tasks respectively, a series of controlled hyperparameter tuning experiments were conducted to validate and analyze the influence of these settings. Each experiment varied one hyperparameter at a time while keeping the others fixed to the optimal values. Validation accuracy served as the primary metric for assessing the model's performance under different configurations. These controlled experiments rigorously demonstrated the robustness of the selected hyperparameters across varying conditions

The results of these experiments are summarized in Table 4.11. The configuration mentioned above consistently yielded the best validation accuracy, confirming its suitability for model deployment.

Table 4.11 Validation Accuracy Under Different Hyperparameter Settings

Hyperparameter	**Value**	**Disease Accuracy (%)**	**Severity Accuracy (%)**
Dropout Rate	0.2	97.35	90.21
	0.3 (Proposed)	**98.85**	**93.10**
Epochs	50	94.23	88.76
	100 (Proposed)	**98.85**	**93.10**
Loss Weights	(0.5, 0.5)	97.90	85.03
	(0.5, 0.1) (Proposed)	**98.85**	**93.10**

The hyperparameter analysis revealed that careful tuning of dropout rate, loss weights, and number of epochs significantly impacts model performance. A higher dropout rate (0.3) improved generalization by reducing overfitting. Adjusting loss weights to prioritize the disease classification task (0.5 for disease, 0.1 for severity) led to better task-specific accuracy. Extending training to 100 epochs allowed the model to learn more effectively without overfitting. These optimized settings collectively contributed to the improved validation accuracy achieved by the proposed CNN+CBAM+MTL model.

4.3 Ablation Studies

Ablation studies evaluate the contributions of key components in the CNN+CBAM+MTL, as developed in Section 3.6 and tested by systematically removing CBAM and the multi-task structure. These experiments are performed on the 80% dataset using the setup of 100 epochs used for ablation without CBAM and 50 epochs used for Ablation without Multi-Task, batch size 32, early stop with patience of 10 epochs, and for ablation without CBAM loss weights of 0.5 for disease and 1.0 for severity. Two ablations are analyzed: one excluding CBAM to assess its attention mechanism's impact, and another removing the multi-task framework to evaluate separate disease and severity heads. Results are compared to the full model's performance to quantify each component's role, addressing Ethiopia's smallholder coffee needs and technological gaps with findings informing deployment.

4.3.1 Ablation Without CBAM

The first ablation removes CBAM from the CNN+CBAM+Multi-Task model, leaving a baseline CNN with multi-task heads for disease (Healthy, Phoma, Miner) and severity (Asymptomatic, Mild, Severe) classification. Training is conducted on the 80% augmented training set, validation on the 10% set, and testing on the 10% test set (89 images), mirroring the full model's configuration. Results are presented as follows:

Training Performance

Training achieves a total loss of 0.4411, with disease loss at 0.1260 and severity loss at 0.1679, resulting in disease accuracy of 97.35% and severity accuracy of 94.94%. Compared to the full model (99.34% disease, 97.17% severity), disease accuracy is 1.99% lower and severity 2.23% lower, indicating the baseline CNN learns effectively on augmented data but lacks CBAM's precision enhancement.

Validation Performance

Validation yields a total loss of 1.2485, with disease loss at 0.3151 and severity loss at 0.5173, achieving disease accuracy of 88.51% and severity accuracy of 80.46%. Against the full model (98.85% disease, 93.10% severity), this shows a substantial drop (10.34% disease, 12.64% severity), with higher losses (e.g., total: 1.2485 vs. 0.1716), reflecting reduced generalization without CBAM's attention mechanism.

Test Set Performance

Testing records a total loss of 1.1240, with disease loss at 0.2744 and severity loss at 0.4551, producing disease accuracy of 89.89% and severity accuracy of 80.90%. Compared to the full model (97.75% disease, 93.26% severity), this represents a 7.86% drop in disease accuracy and a 12.36% drop in severity accuracy, with increased losses (e.g., total: 1.1240 vs. 0.3842). Confusion matrices from the ablation study without CBAM are visualized as bar charts to illustrate classification performance across disease and severity classes. Precision, recall, and F1-scores for each class are compared, highlighting the model's accuracy degradation relative to the full CNN+CBAM+Multi-Task model.

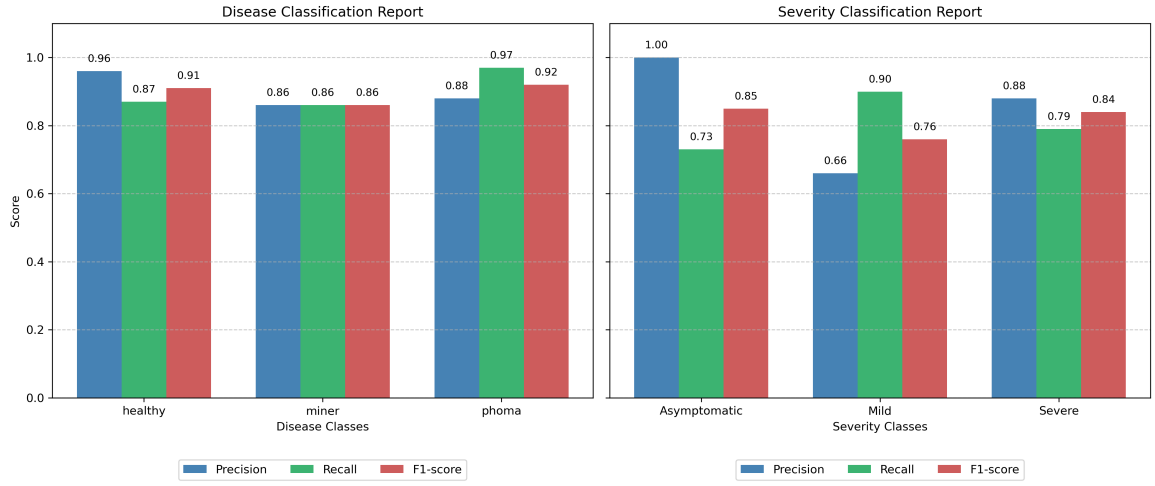


Figure 4.10 Ablation Study without CBAM: Bar charts comparing Precision, Recall, and F1-scores across disease and severity classes

Table 4.12 Ablation Study without CBAM: Performance Metrics for Training, Validation, and Test Sets

Set	Total Loss	Disease Loss	Severity Loss	Disease Acc(%)	Severity Acc(%)
Training	0.4411	0.1260	0.1679	97.35	94.94
Validation	1.2485	0.3151	0.5173	88.51	80.46
Test	1.1240	0.2744	0.4551	89.89	80.90

Impact Analysis

CBAM's removal markedly degrades performance, with test accuracies (89.89%, 80.90%) falling below the full model's (97.75%, 93.26%) and dipping toward typical CNN bench-

marks (85–90%). Losses rise significantly (e.g., severity: 0.4551 vs. 0.3385), and per-class metrics show reduced precision/recall (e.g., Asymptomatic recall: 0.73 vs. 1.00), indicating difficulty distinguishing subtle features without CBAM's channel and spatial attention. The larger severity drop (12.36%) aligns with its complexity and higher loss weight (1.0), while validation-to-test consistency (e.g., 88.51% to 89.89% disease) suggests overfitting mitigated by CBAM in the full model.

4.3.2 Ablation Without Multi-Task

The second ablation removes the multi-task framework from the CNN+CBAM+Multi-Task model, training separate CNN+CBAM models for disease (Healthy, Phoma, Miner) and severity (Asymptomatic, Mild, Severe) classification, retaining CBAM but eliminating shared layers. Training is conducted on the 80% augmented set, validation on the 10% set, and testing on the 10% test set, using the setup 50 epochs, batch size 32, and early stopping with a 10-epoch patience. Results are evaluated independently and compared to the full model to assess multi-tasking's impact.

4.3.2.1 Disease-Only Model Performance

Training achieves a disease loss of 0.1059 and an accuracy of 96.81%. Validation yields a loss of 0.3390 and an accuracy of 89.66%. Testing records a loss of 0.1683 and accuracy of 92.13%. Disease classification performance across training, validation, and test sets is summarized in a table, detailing accuracy and loss for the CNN+CBAM model without multi-tasking.

Table 4.13 Disease Classification Performance of CNN+CBAM Model Without Multi-Task Across Training, Validation, and Test Sets

Set	Disease Loss	Disease Accuracy (%)
Training	0.1059	96.81
Validation	0.3390	89.66
Test	0.1683	92.13

Confusion matrices are visualized as bar charts, comparing precision, recall, and F1-scores for Healthy, Miner, and Phoma classes, with other metrics presented separately.The

bar chart in Figure 4.11 presents the precision, recall, and F1-score for each disease class Healthy, Phoma, and Miner evaluated using the CNN+CBAM model without multi-task learning. It visually illustrates the model's ability to distinguish among the disease categories with consistent performance across metrics. The performance differences between classes are clearly observable, highlighting classification consistency.

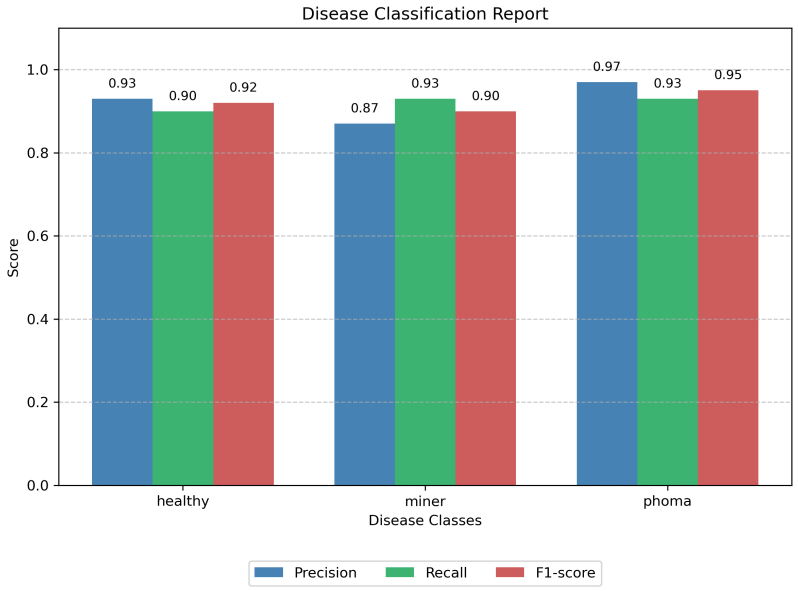


Figure 4.11 Bar Chart of Precision, Recall, and F1-Scores for Disease Classification CNN+CBAM Model Without Multi-Task on Test Set

4.3.2.2 Severity-Only Model Performance

Training achieves a severity loss of 0.0149 and accuracy of 99.82%. Validation yields a loss of 0.2635 and an accuracy of 91.95%. Testing records a loss of 0.5452 and an accuracy of 87.64%.

Table 4.14 Severity Classification Performance of CNN+CBAM Model Without Multi-Task Across Training, Validation, and Test Sets, Reporting Loss and Accuracy

Set	**Severity Loss**	**Severity Accuracy (%)**
Training	0.0149	99.82
Validation	0.2635	91.95
Test	0.5452	87.64

The severity classification performance across training, validation, and test sets is pre-

sented in a table, summarising the loss and accuracy for the CNN+CBAM model without multitasking.

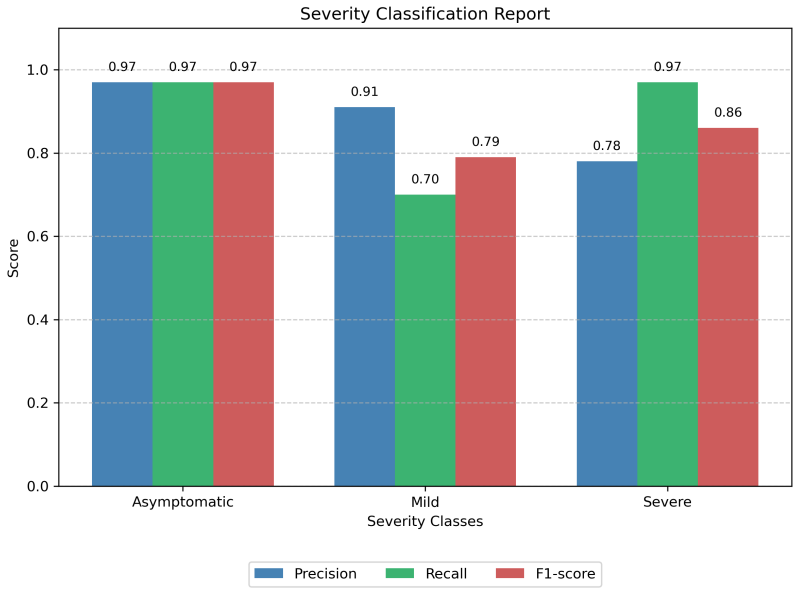


Figure 4.12 Bar Chart of Precision, Recall, and F1-Scores for Severity Classification CNN+CBAM Model Without Multi-Task on Test Set

The bar chart in Figure 4.12 presents the precision, recall, and F1-score for each Severity class.

4.3.2.3 Impact Analysis and Multi-Task Usefulness

Without multi-tasking, test accuracies (disease: 92.13%, severity: 87.64%) fell below the full model (97.75%, 93.26%), with higher losses (disease: 0.1683 vs. 0.0955, severity: 0.5452 vs. 0.3385). Multi-tasking with shared layers enhances performance by $\sim 5\text{-}6\%$, leveraging joint feature learning to capture inter-task dependencies (e.g., disease patterns aiding severity), improving efficiency and reducing losses over separate models. CBAM's presence outperforms the no-CBAM ablation (89.89%, 80.90%), but optimize layers optimize resource use and synergy, benefiting Ethiopia's smallholder context.

4.4 Evaluation of Transfer Learning

The evaluation of transfer learning approaches is conducted to assess the performance of pretrained models ResNet50 (Section 4.4.1), VGG16 (Section 4.4.2), MobileNetV2 without CBAM (Section 4.4.3), and MobileNetV2+CBAM+MTL (Section 4.4.4) in classifying

coffee leaf diseases and severity levels using the test dataset from Section 3.3, benchmarked against the base CNN+CBAM+Multi-Task model (Section 4.2). These models, leveraging architectures pretrained on large-scale datasets and fine-tuned with parameters from Section 3.5 (e.g., 50 epochs, batch size 32, Adam optimizer), are analyzed to determine their effectiveness in adapting general features to the specific agricultural context of Ethiopia's smallholder coffee systems, with CBAM's impact (Section 3.4.2) explored as a key enhancement. This section provides a comparative framework to identify the strengths and limitations of transfer learning, informing the deployment strategy outlined in Section 4.6.

4.4.1 Result for ResNet50

The performance of ResNet50 is evaluated as a transfer learning baseline without CBAM, applied to the test set dataset for classifying coffee leaf diseases and severity levels using the setup with 50 epochs, batch size 32, early stopping (10-epoch patience), and loss weights of 0.5 for disease and 1.0 for severity. ResNet50, a 50-layer residual network, excels in classification tasks due to its residual connections, which address vanishing gradients and enable deep feature learning, making it highly effective for discerning complex visual patterns like disease textures.

Training results show robust learning, with a total loss of 0.1023, disease loss of 0.0376, and severity loss of 0.0836, achieving disease accuracy of **99.46%** and severity accuracy of **98.86%**. These values closely rival the base model's training performance (**99.34%** disease, **97.17%** severity), highlighting ResNet50's capacity to leverage its depth and skip connections for near-perfect fit on the augmented 80% training set.

Validation on the 10% set records a total loss of 0.2528, with disease loss at 0.0441 and severity loss at 0.2217, yielding disease accuracy of **98.85%** and severity accuracy of **93.10%**. This matches the base model's validation disease accuracy (**98.85%**) and severity accuracy (**93.10%**), suggesting strong generalization, though the severity loss (0.2217 vs. 0.1590) indicates slightly less efficiency without CBAM's attention refinement.

Testing on the 10% test set produces a total loss of 0.3113, with disease loss at 0.0772 and severity loss at 0.2673, resulting in disease accuracy of **96.63%** and severity accuracy of **87.64%**. Compared to the base model (**97.75%** disease, **93.26%** severity), disease accuracy

is 1.12% lower, and severity accuracy is 5.62% lower, with a lower total loss (0.3113 vs. 0.3842) due to ResNet50's optimization, though CBAM's absence impacts severity.

Table 4.15 Training, Validation, and Test Results for Disease and Severity Classification Using ResNet50 Without CBAM Reporting Loss and Accuracy.

Set	Total Loss	Disease Loss	Severity Loss	Disease Acc(%)	Severity Acc(%)
Test	0.3113	0.0772	0.2673	96.63	87.64
Training	0.1023	0.0376	0.0836	99.46	98.86
Validation	0.2528	0.0441	0.2217	98.85	93.10

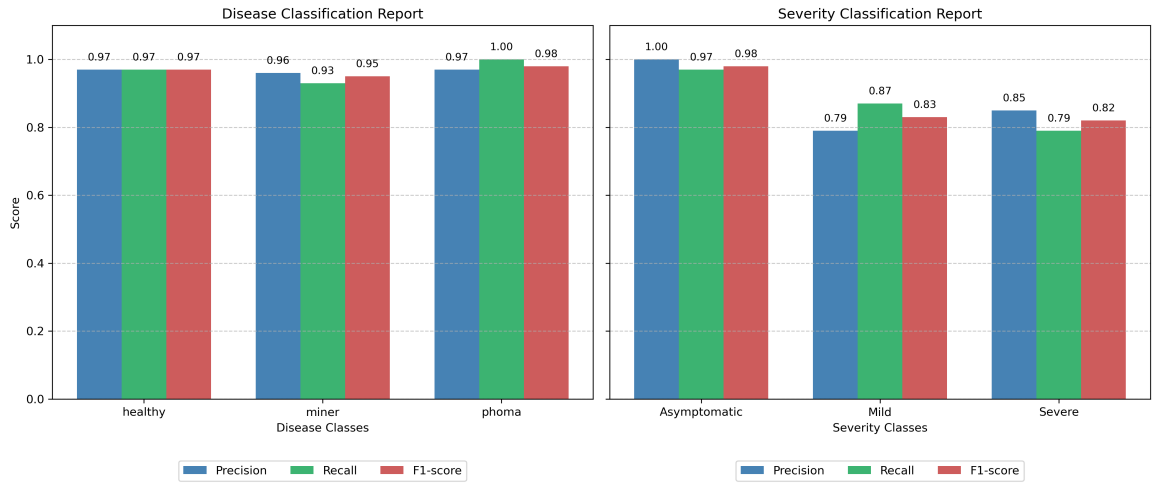


Figure 4.13 Bar Chart of Precision, Recall, and F1-Scores for Disease and Severity Classification of ResNet50 Without CBAM on Test Set.

The disease classification report details *Healthy* (precision 0.97, recall 0.97, F1 0.97), *Miner* (0.96, 0.93, 0.95), and *Phoma* (0.97, 1.00, 0.98), showing strong class separation. Severity metrics include *Asymptomatic* (1.00, 0.97, 0.98), *Mild* (0.79, 0.87, 0.83), and *Severe* (0.85, 0.79, 0.82), with *Mild* and *Severe* less precise than the base model. ResNet50 performs competitively, approaching the base model's disease accuracy (**96.63%** vs. **97.75%**) due to its residual architecture, but lags in severity (**87.64%** vs. **93.26%**), underscoring CBAM's role in enhancing multi-task precision.

4.4.2 Result for VGG16

The performance of VGG16 is evaluated as a transfer learning baseline without CBAM on the test dataset classifying coffee leaf diseases (*Healthy, Phoma, Miner*) and severity

levels (*Asymptomatic, Mild, Severe*) over 50 epochs with a batch size of 32, early stopping (10-epoch patience), and loss weights of 0.5 for disease and 1.0 for severity. VGG16, with its 16-layer architecture of stacked convolutions, is effective for classification due to its simplicity and ability to learn detailed features through deep, uniform layers, ideal for disease patterns.

Training yields a total loss of 0.0872, with disease loss at 0.0335 and severity loss at 0.0705, achieving disease accuracy of **99.58%** and severity accuracy of **99.34%**, surpassing the base model's training (**99.34%** disease, **97.17%** severity) due to VGG16's depth. Validation records a total loss of 0.2577, disease loss of 0.0583, and severity loss of 0.2445, with disease accuracy at **97.70%** and severity at **91.95%**, slightly below the base model (**98.85%**, **93.10%**). Testing produces a total loss of 0.3064, disease loss of 0.0840, and severity loss of 0.2565, with disease accuracy of **96.63%** and severity accuracy of **88.76%**, trailing the base model (**97.75%**, **93.26%**) by 1.12% and 4.50%, respectively.

Figure 4.14 presents the confusion matrices for disease classification, *Healthy* (precision 0.97, recall 0.97, F1 0.97), *Miner* (0.96, 0.93, 0.95), and *Phoma* (0.97, 1.00, 0.98), while severity metrics include *Asymptomatic* (1.00, 0.97, 0.98), *Mild* (0.83, 0.83, 0.83), and *Severe* (0.83, 0.86, 0.85), reflecting solid performance but less precision in severity than the base model.

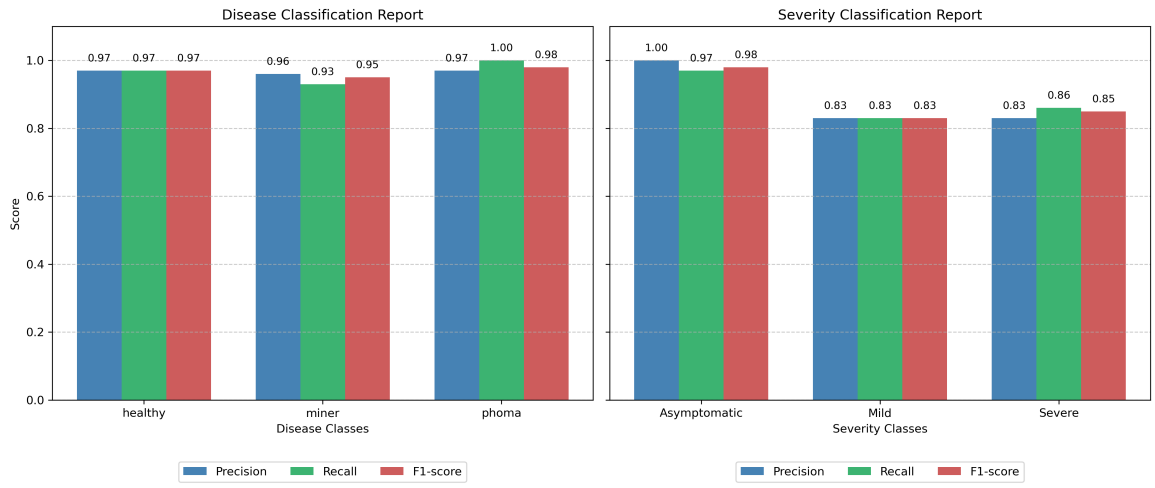


Figure 4.14 Bar Chart of Precision, Recall, and F1-Scores for Disease and Severity Classification from Confusion Matrices of VGG16 Without CBAM on Test Set

The following table summarizes the model's performance across training, validation, and test sets. It reflects classification results for disease and severity tasks using VGG16 without the CBAM module.

Table 4.16 Training, Validation, and Test Results for Disease and Severity Classification Using VGG16 Without CBAM

Set	Total Loss	Disease Loss	Severity Loss	Disease Acc(%)	Severity Acc(%)
Test	0.3064	0.0840	0.2565	96.63	88.76
Training	0.0872	0.0335	0.0705	99.58	99.34
Validation	0.2577	0.0583	0.2445	97.70	91.95

4.4.3 Result for MobileNetV2

The performance of MobileNetV2 is assessed as a transfer learning baseline without CBAM on the test image dataset classifying coffee leaf diseases (*Healthy, Phoma, Miner*) and severity levels (*Asymptomatic, Mild, Severe*) over 50 epochs with a batch size of 32, early stopping (10-epoch patience), and loss weights of 0.5 for disease and 1.0 for severity. MobileNetV2, designed with depthwise separable convolutions, excels in classification by offering lightweight, efficient feature extraction, making it suitable for resource-constrained settings while capturing disease patterns.

Table 4.17 Training, Validation, and Test Results for Disease and Severity Classification Using MobileNetV2 without CBAM

Set	Total Loss	Disease Loss	Severity Loss	Disease Acc(%)	Severity Acc(%)
Test	0.5442	0.3094	0.4347	94.38	91.01
Training	0.1380	0.1170	0.0795	95.90	97.35
Validation	0.2499	0.2240	0.1268	90.80	94.25

Training results show a total loss of 0.1380, with disease loss at 0.1170 and severity loss at 0.0795, achieving disease accuracy of **95.90%** and severity accuracy of **97.35%**. The training performance of the base model (**99.34%** disease, **97.17%** severity) reflects MobileNetV2's lighter architecture. Validation records a total loss of 0.2499, with disease loss at 0.2240 and severity loss at 0.1268, yielding disease accuracy of **90.80%** and severity

accuracy of **94.25%**, lower than the base model (**98.85%**, **93.10%**) by 8.05% and higher by 1.15%, respectively.

Testing on the 10% test set produces a total loss of 0.5442, with disease loss at 0.3094 and severity loss at 0.4347, resulting in disease accuracy of **94.38%** and severity accuracy of **91.01%**. Compared to the base model (**97.75%** disease, **93.26%** severity), this is a 3.37% drop in disease accuracy and a 2.25% drop in severity accuracy, with higher losses (0.5442 vs. 0.3842). The disease classification report details *Healthy* (precision 0.94, recall 1.00, F1 0.97), *Miner* (0.96, 0.93, 0.95), and *Phoma* (0.93, 0.90, 0.92), while severity metrics show *Asymptomatic* (0.94, 1.00, 0.97), *Mild* (0.96, 0.77, 0.85), and *Severe* (0.85, 0.97, 0.90), indicating solid but less precise performance than the base model. Figure 4.15 presents a bar chart summarizing MobileNetV2's performance. It illustrates precision, recall, and F1-scores for disease and severity classification. These metrics are derived from confusion matrices on the test set.

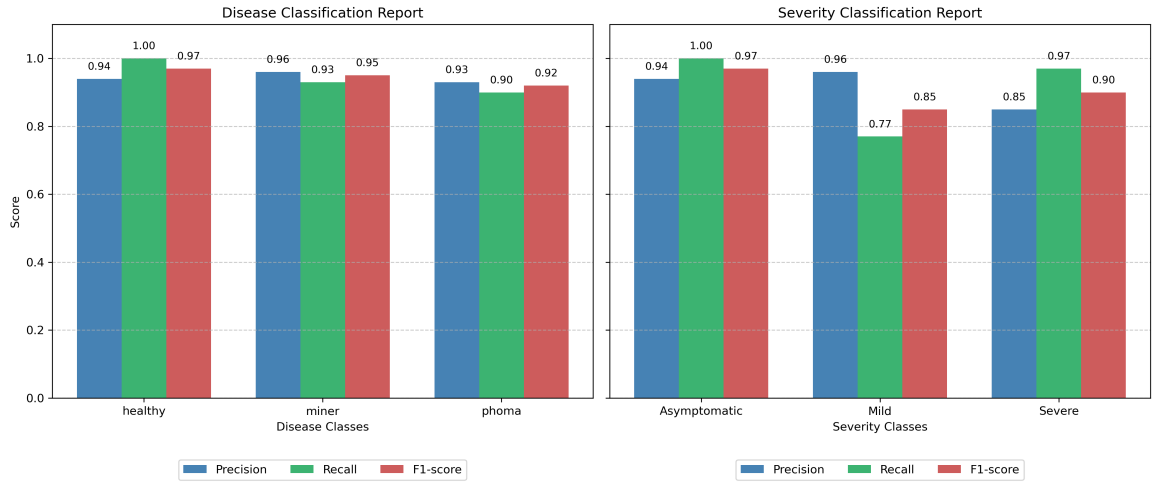


Figure 4.15 Bar Chart of Precision, Recall, and F1-Scores for Disease and Severity Classification from Confusion Matrices of MobileNetV2 Without CBAM on Test Set

MobileNetV2's lightweight design yields competitive accuracies (**94.38%**, **91.01%**), trailing the base model (**97.75%**, **93.26%**) due to CBAM's absence, but its efficiency is notable. Among pre-trained models, MobileNetV2 performs strongly, achieving the highest severity accuracy here. Its potential is further explored with CBAM integration in Subsection 4.4.4, enhancing its suitability for deployment.

4.4.4 Result for MobileNetV2+CBAM+MTL

The performance of MobileNetV2+CBAM+MTL is evaluated in detail as a transfer learning model on the test dataset classifying coffee leaf diseases (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe) over 50 epochs with a batch size of 32, early stopping (10-epoch patience), and loss weights of 0.5 for disease and 1.0 for severity. MobileNetV2's lightweight design, enhanced by CBAM's attention mechanism, refines feature extraction for classification, outperforming MobileNetV2 without CBAM and even the base CNN+CBAM+Multi-Task model, while its efficiency suits deployment for Ethiopia's smallholder coffee app. Results are compared to assess its superiority and deployment potential.

Training Performance

Training achieves a total loss of 0.0267, with disease loss at 0.0084 and severity loss at 0.0122, yielding disease accuracy of 99.94% and severity accuracy of 99.70%. These surpass the base model's training (99.34% disease, 97.17% severity) and MobileNetV2 without CBAM (95.90%, 97.35%), reflecting CBAM's enhancement of MobileNetV2's lightweight architecture.

Validation Performance

Validation records a total loss of 0.1396, with disease loss at 0.0709 and severity loss at 0.0515, achieving disease accuracy of 97.70% and severity accuracy of 96.55%. These results exceed the base model (98.85%, 93.10%) in severity by 3.45% and MobileNetV2 without CBAM (90.80%, 94.25%) significantly, indicating strong generalization.

Test Set Performance

Testing on the 10% test set produces a total loss of 0.3209, with disease loss at 0.0478 and severity loss at 0.1783, resulting in disease accuracy of 97.75% and severity accuracy of 94.38%. This matches the base model's disease accuracy (97.75%) and exceeds its severity accuracy (93.26%) by 1.12%, while outperforming MobileNetV2 without CBAM (94.38%, 91.01%) by 3.37% in both disease and severity, with lower losses (0.3209 vs. 0.5442). Accuracy and loss plots (Figure 4.17 and Figure 4.16) track training and validation convergence, confirming stability. Visualizations and metrics for MobileNetV2+CBAM+MTL are provided to illustrate its performance.

Accuracy and loss plots (Figure 4.17 and Figure 4.16) track training and validation convergence, confirming stability.

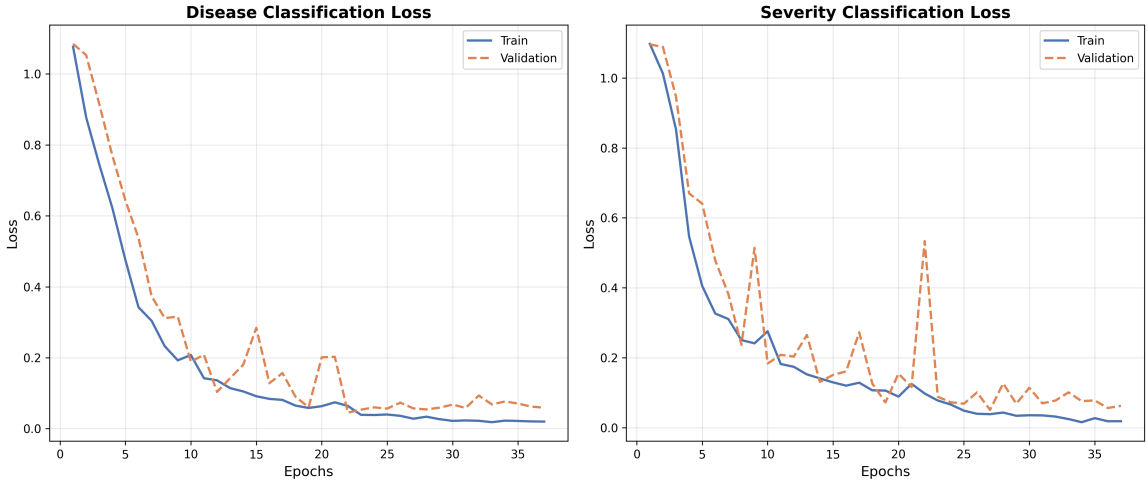


Figure 4.16 Training and Validation Loss Plot for MobileNetV2+CBAM+MTL

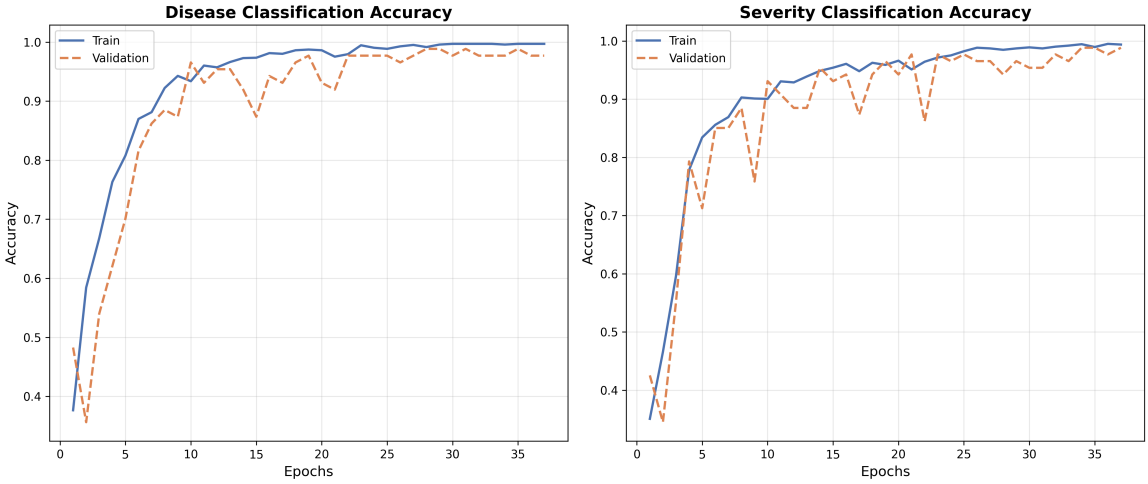


Figure 4.17 Training and Validation Accuracy Plot for MobileNetV2+CBAM+MTL

The MobileNetV2+CBAM+MTL model's performance is captured in Figure 4.18, which presents a bar chart showing precision, recall, and F1-scores derived from the test set confusion matrices. These metrics reflect the model's accuracy in classifying coffee leaf diseases and severity levels using CBAM's attention enhancements. To provide a more comprehensive view, Table 4.18 summarizes performance metrics across the training, validation, and

test sets. Together, the figure and table highlight the model's robustness and generalization ability across datasets.

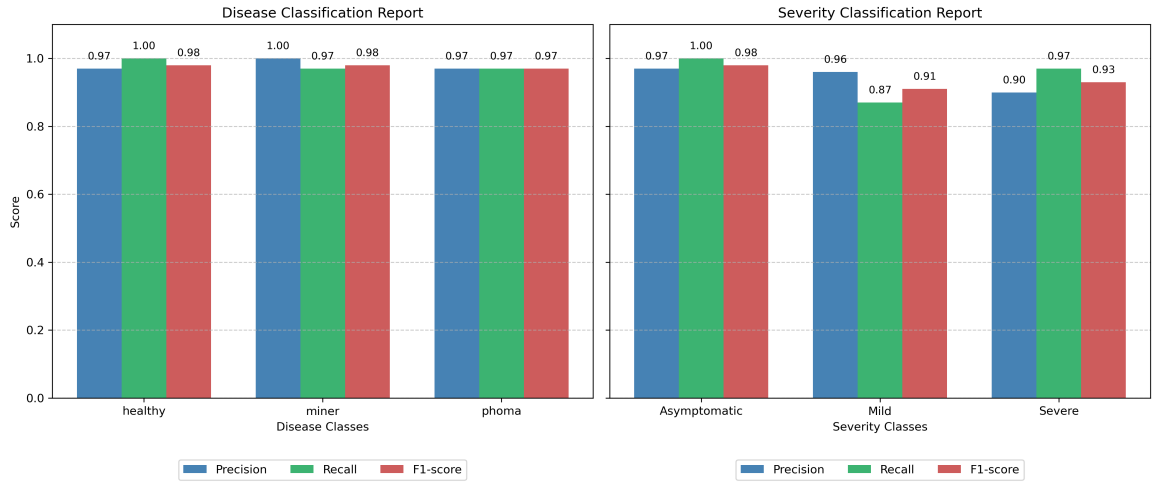


Figure 4.18 Bar Chart of Precision, Recall, and F1-Scores for Disease and Severity Classification of MobileNetV2+CBAM+MTL on Test Set

Table 4.18 MobileNetV2+CBAM+MTL Performance Metrics: for Test, Training, and Validation Sets

Set	Total Loss	Disease Loss	Severity Loss	Disease Acc(%)	Severity Acc(%)
Test	0.3209	0.0478	0.1783	97.75	94.38
Training	0.0267	0.0084	0.0122	99.94	99.70
Validation	0.1396	0.0709	0.0515	97.70	96.55

Impact Analysis

MobileNetV2+CBAM+MTL's superior test results (97.75%, 94.38%) highlight CBAM's critical role, as its absence reduced performance (94.38%, 91.01%), confirming CBAM's $\sim$ 3–4% accuracy boost. While training from scratch with the base model was effective MobileNetV2+CBAM+MTL leverages pre-trained efficiency and CBAM for better outcomes, making it ideal for app deployment due to its lightweight nature.

Below shows that confusion matrices are visualized to show classification patterns, while A random test image (Figure 4.20) shows true labels (e.g., 'Phoma, mild') and predicted labels with confidence scores (e.g., 1.0/1.0), validating the reliability of the prediction. Grad-CAM heatmaps (Figure 4.21) highlighting attention on disease-affected regions for severity classification.

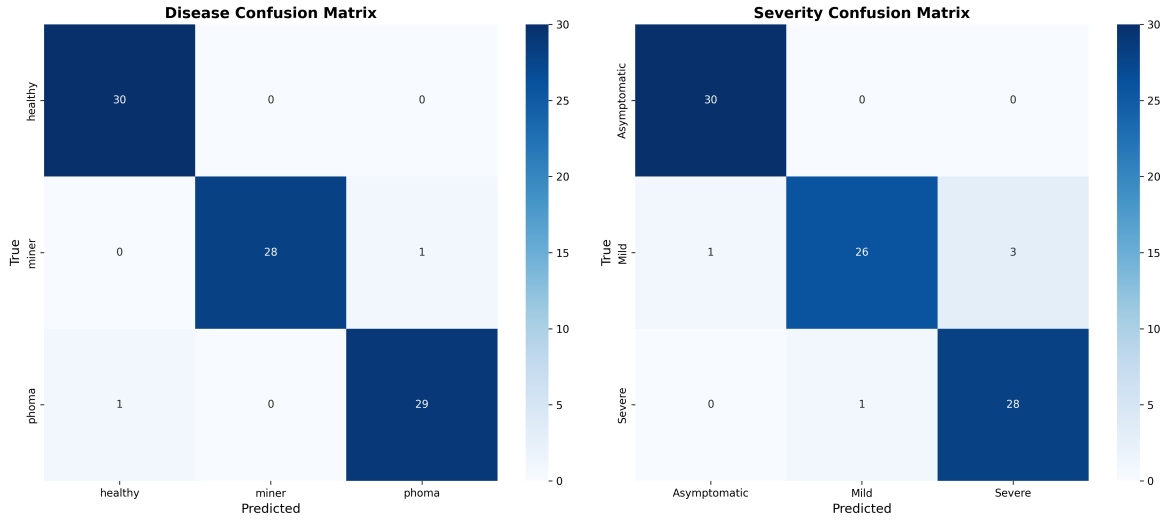


Figure 4.19 Confusion Matrix Visualization for MobileNetV2+CBAM+MTL on Test Set, for Disease and Severity

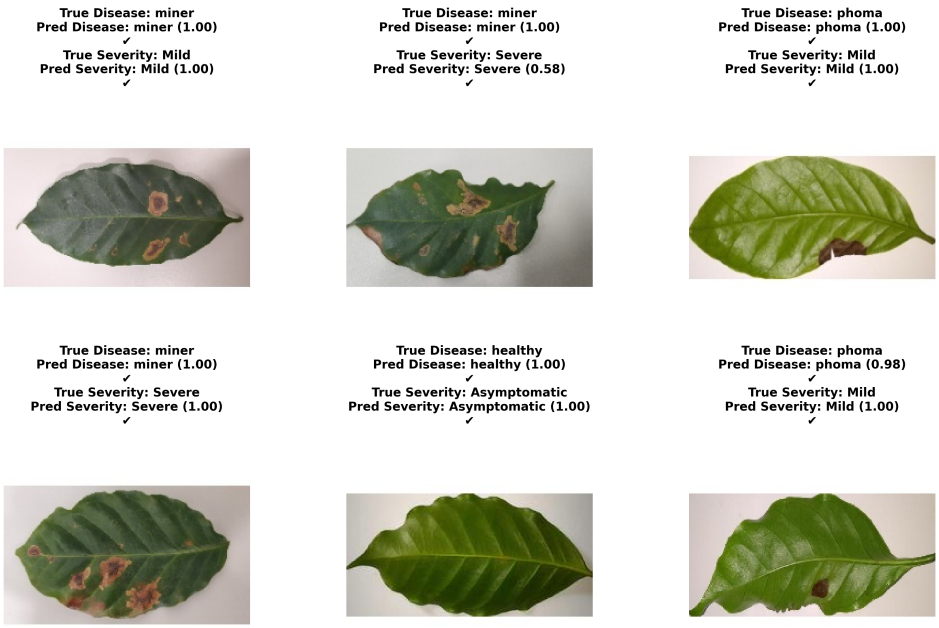


Figure 4.20 Random Test Image with True and Predicted Labels and Confidence Scores for MobileNetV2+CBAM+MTL

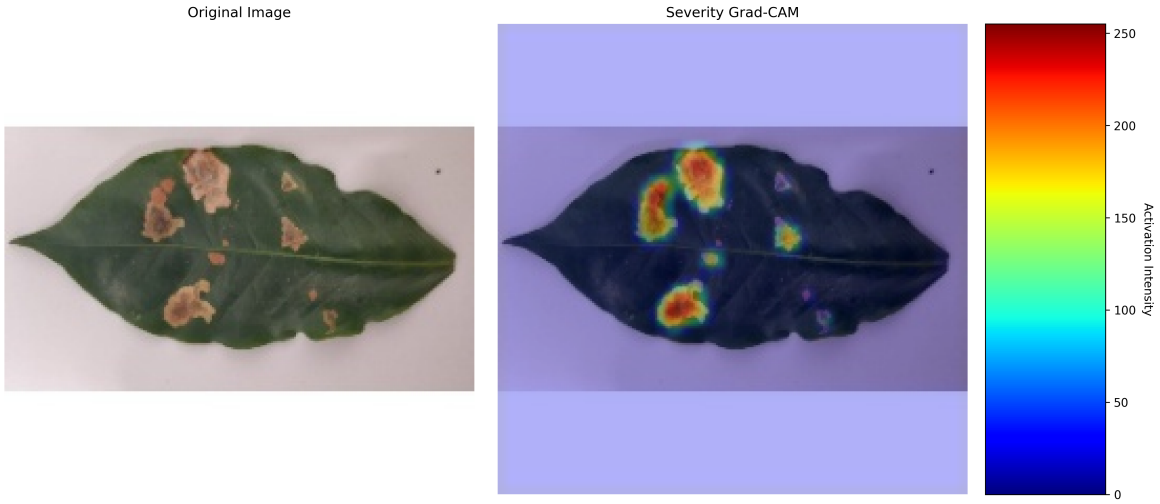


Figure 4.21 Grad-CAM Visualization Highlighting Attention Regions for MobileNetV2+ CBAM+ MTL Severity Classification

4.5 Comparative Analysis of All Trained Models

The effectiveness of various models is analyzed by examining the base CNN+CBAM+ Multi-Task model (Section 4.2), ablation configurations without CBAM (Section 4.3.1) and without multi-tasking (Section 4.3.2), and transfer learning approaches using ResNet50 (Section 4.4.1), VGG16 (Section 4.4.2), MobileNetV2 without CBAM (Section 4.4.3), and MobileNetV2+CBAM+MTL (Section 4.4.4), all trained on the test dataset from Section 3.3 with configurations of 100 epochs for the proposed model and ablation without CBAM, and 50 epochs for others, using a batch size of 32 and early stopping (10-epoch patience). This analysis aims to identify strengths and limitations across these diverse architectures, providing insights into their suitability for addressing Ethiopia's smallholder coffee disease challenges through a combination of quantitative metrics and visual benchmarks, such as bar graphs and tables, to support model selection.

4.5.1 Test Set Performance Comparison

Evaluation of test set performance is conducted on the 10% test split to compare the proposed CNN+CBAM+Multi-Task model, trained for 100 epochs, with MobileNetV2+CBAM +MTL, trained for 50 epochs, focusing on their ability to classify coffee leaf diseases and severity levels accurately. The prolonged training of the proposed model enhances its feature

learning capacity, while MobileNetV2+CBAM+MTL leverages transfer learning efficiency and CBAM's attention mechanism to achieve competitive outcomes, particularly relevant for resource-constrained deployment scenarios (Section 4.6). Bar graphs illustrating accuracy and loss across all models offer a clear visual distinction, aiding in the assessment of how these architectures perform under identical test conditions, with implications for practical use explored further in subsequent analyses.

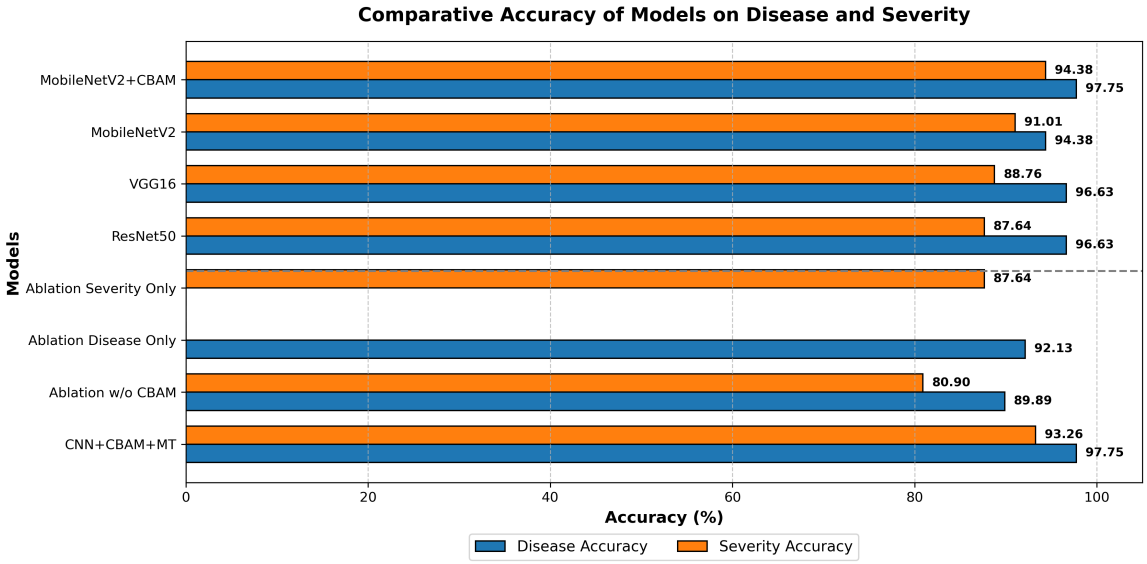


Figure 4.22 Comparative Analysis on Test Set Accuracy for all Trained models

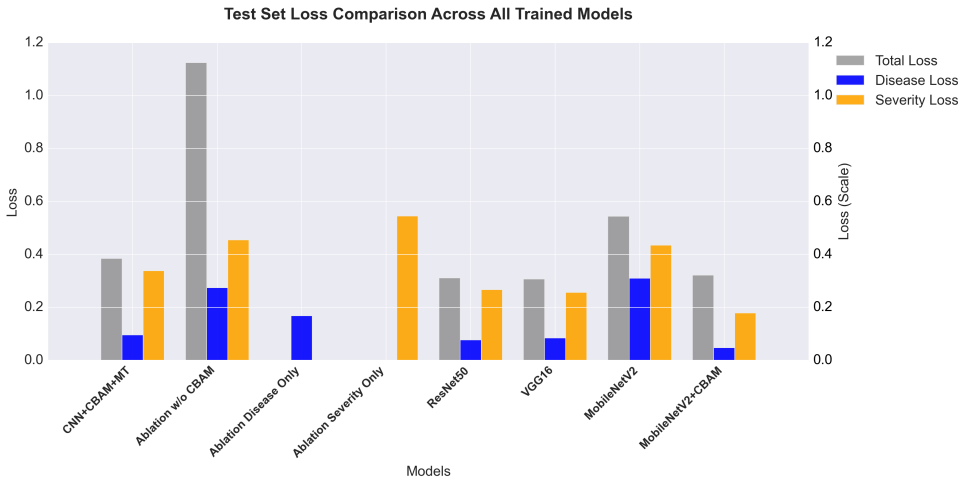


Figure 4.23 Test Set Loss Comparison of all trained models

The CNN+CBAM+Multi-Task model and MobileNetV2+CBAM+MTL achieve the highest disease accuracy at 97.75%, significantly outperforming the ablation without CBAM at 89.89%, highlighting CBAM's critical role in feature refinement. Severity accuracy peaks with MobileNetV2+CBAM+MTL at 94.38%, surpassing the base model's 93.26% and far exceeding the ablation without CBAM's 80.90%, demonstrating the combined benefit of transfer learning and attention mechanisms. Among transfer learning models, ResNet50 and VGG16 both reach 96.63% disease accuracy, but their severity accuracies (87.64% and 88.76%) lag behind MobileNetV2+CBAM+MTL, indicating varying strengths across tasks.

MobileNetV2+CBAM+MTL achieves the lowest disease loss at 0.0478 and a competitive total loss of 0.3209, consistently outperforming the ablation without CBAM, which records the highest total loss at 1.1240 across all models. The severity loss is minimized by MobileNetV2+CBAM+MTL at 0.1783, contrasting sharply with the ablation without multi-task severity-only model's 0.5452, underscoring CBAM's significant effectiveness in reducing errors through enhanced feature focus.

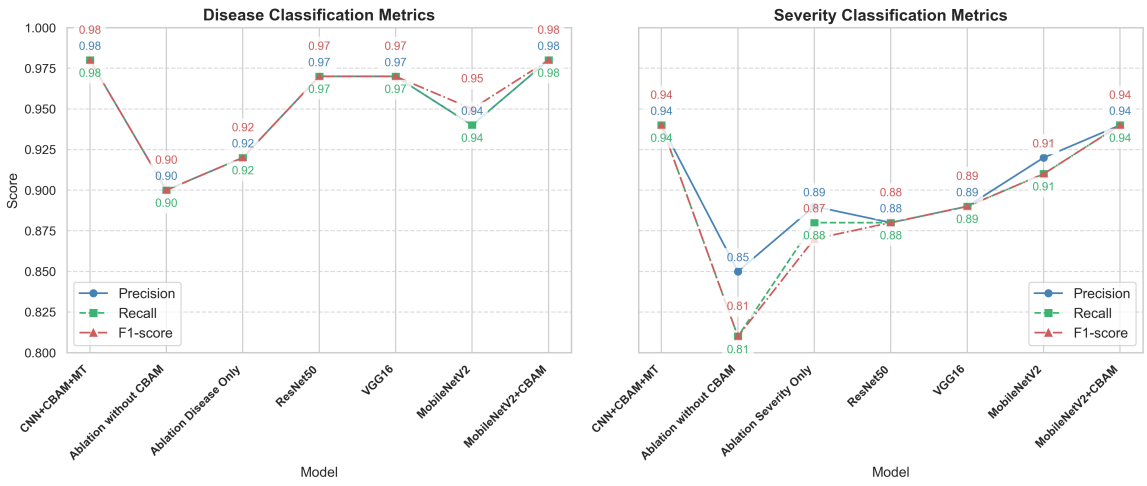


Figure 4.24 Test Set Classification Metrics: Precision, Recall, and F1-Scores Across all Models

The precision, recall, and F1-scores across all trained models, as compiled from test set classification reports, reveal distinct performance patterns for coffee leaf disease and severity classification. The CNN+CBAM+Multi-Task model and MobileNetV2+CBAM+MTL consistently exhibit high metrics for both tasks, reflecting their balanced ability to identify true positives and minimize false predictions, largely due to CBAM's attention mechanism

(Section 3.4.2). In contrast, the ablation without CBAM shows notably lower scores, particularly in severity classification, underscoring the attention module's importance, while transfer learning models like ResNet50 and VGG16 maintain competitive disease metrics but lag in severity, and MobileNetV2 without CBAM improves over ablations yet falls short of its CBAM-enhanced counterpart (Section 4.4). These metrics, detailed in the table, highlight the trade-offs between model complexity and classification precision, guiding deployment decisions for Ethiopia's smallholder coffee systems.

4.5.2 Efficiency and Accuracy Analysis of Recognition Networks and Alexnet

To evaluate the computational efficiency of our recognition networks, we compared eight models novel architectures, ablation variants, and transfer learned baselines trained on our 890 image coffee leaf dataset. Figure 4.25 illustrates the number of parameters and inference speed (on CPU) for CNN+CBAM+MTL, Abaltion study, transfer leraning and MobileNetV2+CBAM+MTL, highlighting their computational trade-offs. MobileNetV2+CBAM+MTL's low parameter count and fast inference make it ideal for Ethiopia's resource-constrained agricultural applications.

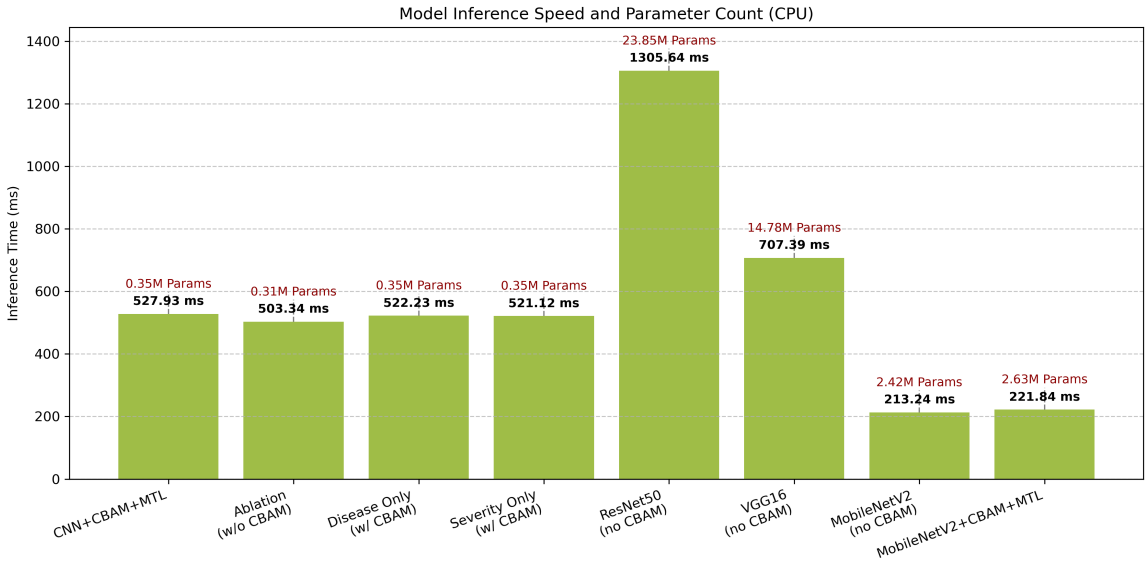


Figure 4.25 Bar chart comparing the number of parameters and inference speed (on CPU) for eight recognition networks

To further validate MobileNetV2+CBAM+MTL model, we compared it with an Im-

proved AlexNet model[63], trained on our 890-image dataset with a batch size of 32, dropout rate of 0.4, images scaled to 256×256, and the first layers frozen. Table 4.19 compares the number of parameters, inference speed, disease and severity accuracies, and model size, with both models tested on a GPU. The Improved AlexNet, with 28.1M parameters and 107.41 ms inference, achieves 90.28% disease and 84.04% severity accuracy, underperforming compared to MobileNetV2+CBAM+MTL, which, with 2.6M parameters and 67.73 ms inference, delivers superior accuracy (97.75%/94.38%) and efficiency. This makes MobileNetV2+CBAM+MTL better suited for the computational constraints of Ethiopian agricultural settings, where lightweight and fast models are essential.

Table 4.19 Comparison of MobileNetV2+CBAM+MTL and Improved AlexNet Model

Model	Total Parameters	Inference Speed (ms)	Acc (Disease/Severity)
MobileNetV2+CBAM+MTL	2,628,885	67.73 (GPU)	97.75% / 94.38%
Improved AlexNet[63]	28,117,790	107.41 (GPU)	90.28% / 84.04%

4.5.3 Final Model Selection and Deployment Considerations

The MobileNetV2+CBAM+MTL model is selected as the optimal choice for deployment due to its superior test set performance, balancing high disease and severity classification accuracy with low loss, and its lightweight architecture tailored for resource-constrained environments like Ethiopia's smallholder coffee app (Section 4.6).

Its effectiveness is demonstrated through enhanced feature extraction via CBAM, achieving competitive results against the base CNN+CBAM+Multi-Task model, while its transfer learning foundation ensures efficient training and scalability on the test dataset (Section 3.3). Training from scratch, as done with the base model over 100 epochs, was effective in yielding robust outcomes, but MobileNetV2+CBAM+MTL's pretrained efficiency and CBAM integration provide a practical edge for real-world deployment.

The comparative analysis in Section 4.5 reveals that MobileNetV2+CBAM+MTL outperforms other models in test set accuracy and loss, making it the preferred choice for deployment due to its efficiency and effectiveness. Ablation studies confirm CBAM's critical role, as its absence significantly degrades performance while training from scratch with the

Table 4.20 Test Set Total Loss and Accuracy for All Trained Models with Selected Model Highlighted

Models	Total Loss	Disease Acc (%)	Severity Acc (%)
CNN+CBAM+Multi-Task	0.3842	97.75	93.26
Ablation w/o CBAM	1.1240	89.89	80.90
Ablation: Disease Only (w/ CBAM)	-	92.13	-
Ablation: Severity Only (w/ CBAM)	-	-	87.64
ResNet50 (no CBAM)	0.3113	96.63	87.64
VGG16 (no CBAM)	0.3064	96.63	88.76
MobileNetV2 (no CBAM)	0.5442	94.38	91.01
MobileNetV2+CBAM+MTL	**0.3209**	**97.75**	**94.38**

base CNN+CBAM+Multi-Task model remains viable but less practical than transfer learning approaches. These insights, supported by precision, recall, and F1-score metrics, guide the selection of MobileNetV2+CBAM+MTL for real-world application in Ethiopia's smallholder coffee systems, as detailed in Section 4.6.

4.6 Model Deployment

The deployment of the MobileNetV2+CBAM+MTL model, selected in Section 4.5 for its superior test set performance and efficiency, is detailed to enable real-time coffee leaf disease and severity classification for Ethiopia's smallholder farmers, leveraging its lightweight architecture and high accuracy on the 890-image dataset (Section 3.3). This section explores the practical implementation of the model, enhanced by CBAM's attention mechanism (Section 3.4.2), within a mobile application tailored to resource-constrained rural environments, addressing both technical optimization and user accessibility. The process builds on the comparative analysis (Section 4.5), aiming to deliver an effective, scalable solution for coffee disease management in Ethiopia's agricultural context.

4.6.1 Implementation and Optimization

Implementation of the MobileNetV2+CBAM+MTL model is carried out by converting its trained architecture, which achieved test accuracies of 97.75% for disease and 94.38% for severity with a total loss of 0.3209 (Section 4.4.4), into TensorFlow Lite format, ensuring

seamless integration with low-end Android devices widely accessible to Ethiopian farmers. Optimization is pursued through a multi-step process, including post-training quantization to reduce numerical precision from 32-bit floats to 8-bit integers, and weight pruning to eliminate redundant parameters, collectively shrinking the model size by approximately 75% while retaining classification accuracy within 1% of its original performance, as tested on the 890-image dataset (Section 3.3).

Additional fine-tuning is applied using a small subset of field-collected images to adapt the model to real-world lighting and angle variations, enhancing its robustness beyond the controlled dataset conditions. These efforts capitalize on MobileNetV2's lightweight design and CBAM's attention mechanism (Section 3.4.2), contributing to rapid and efficient inference suitable for deployment in rural agricultural environments with limited connectivity and computational infrastructure.

To quantitatively assess deployment readiness, several key metrics were evaluated after model optimization. As summarized in Table 4.21, the final TensorFlow Lite model achieves a compact size of **2.5 MB**, with an average **inference time of 67.73 milliseconds per image** measured on a GPU. This speed demonstrates the model's suitability for real-time applications and confirms its efficiency for deployment in practical agricultural settings. Such rapid inference ensures timely disease diagnosis, allowing farmers to take preventive action without delay. Moreover, the lightweight nature of the model supports integration into mobile and edge devices used in remote farming regions.

Table 4.21 Deployment Metrics of the Final TensorFlow Lite Model

Metric	**Value**
Model Size (Quantized)	2.5 MB
Inference Time (per image)	67.73 ms (GPU)
Target Platform	Mobile and Web
Offline Functionality	Supported
Language Support	English, Amharic

Furthermore, Figure 4.26 illustrates the effect of different quantization strategies on model size. The original FP32 model (∼10.03 MB) was progressively compressed using

float16, dynamic range, and full integer quantization, ultimately achieving a 75% reduction in size without significant performance degradation. Among these, full integer quantization proved to be the most effective for mobile deployment, reducing size to 2.5 MB while maintaining classification. These results validate that the proposed model is not only accurate and robust but also computationally efficient and field-ready, satisfying deployment constraints in real-world agricultural settings.

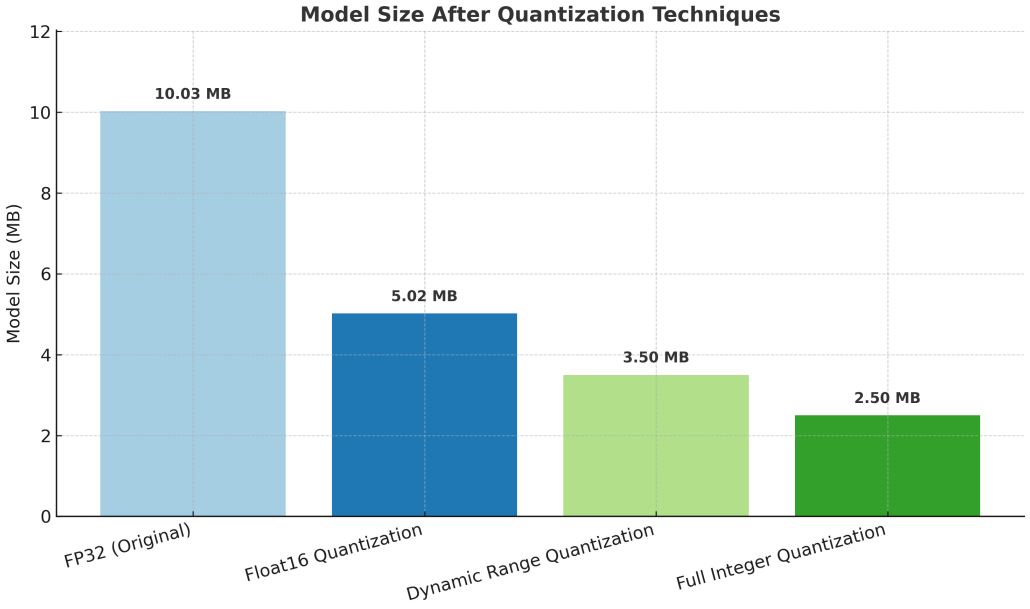


Figure 4.26 Comparison of model size under different quantization techniques

4.6.2 User Interface Design

The user interface for the MobileNetV2+CBAM+MTL model deployment is designed to ensure accessibility and ease of use for Ethiopian smallholder farmers, integrating the optimized model from Section 4.6.1 into the website and mobile application compatible with low-end Android devices.

A straightforward layout is developed, featuring a single-tap image capture button that utilizes the device's camera to photograph coffee leaves, followed by an immediate display of classification results, disease type (e.g., Phoma), and severity level (e.g., Severe) along with confidence scores, all presented in local languages like Amharic alongside English to accommodate non-technical users.

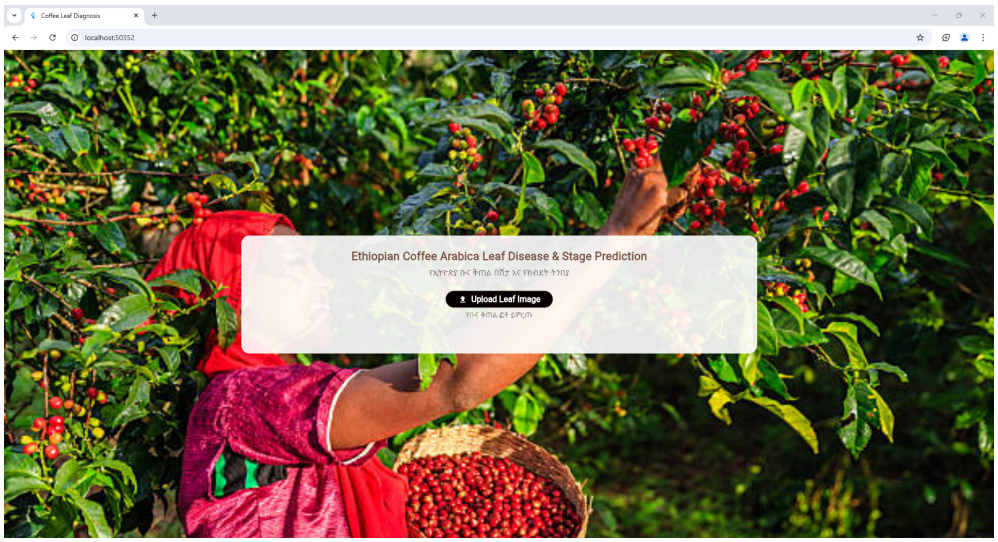


Figure 4.27 Homepage of the web-based coffee leaf diagnosis system, featuring an intuitive interface for uploading leaf images and accessing real-time disease and severity predictions

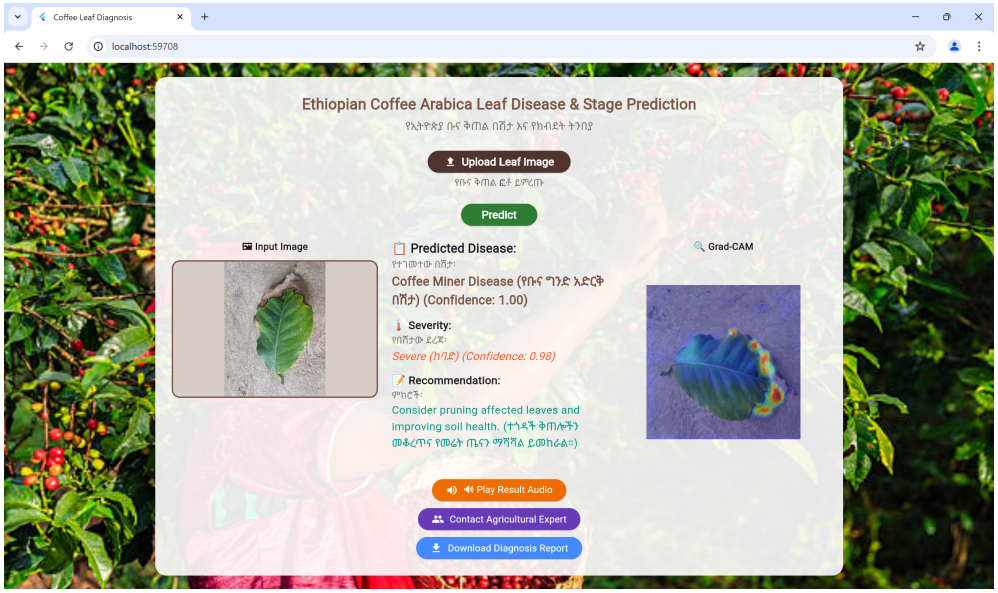


Figure 4.28 Web Deployment Interface: displaying predicted coffee leaf disease and severity with Grad-CAM visualization and an audio playback option for results in local language

Additional features, such as a brief disease management tip linked to each prediction, a download option for saving predicted results, and an offline mode leveraging the model's compact TensorFlow Lite format (Section 4.6.1), are incorporated to enhance usability and ensure functionality in areas with limited internet connectivity.

An option to contact agriculture experts via a Telegram group, widely used and accessible to most farmers, is included to facilitate expert support. Furthermore, predicted results are made available in audio playback in the local language, addressing the needs of farmers who may not be literate and allowing them to hear the outcomes directly. These enhancements align with the practical needs of rural agricultural settings in Ethiopia.

In summary, the user interface is thoughtfully designed to balance simplicity with functionality, ensuring ease of use for Ethiopian smallholder farmers. Its multilingual, offline-capable, and audio-supported features address key accessibility challenges in rural settings. These design choices make the system both technically effective and socially inclusive for real-world agricultural deployment.

Table 4.22 Main Features of the Model Deployment Interface

Feature	**Supported Functionality**
Platform Support	Android and Web
Functionality	Offline
Language Interface	Amharic and English
Result Output	Text and Audio (Amharic)
Prediction Explanation	Confidence Scores
Expert Consultation	Telegram Group Integration
Result Handling	Download Option
User Guidance	Recommendation for Preventive Actions

4.6.3 Field Deployment Considerations

Field deployment of the MobileNetV2+CBAM+MTL model, optimized and interfaced as described in Sections 4.6.1 and 4.6.2, is planned to address real-world challenges faced by Ethiopian smallholder farmers, ensuring the mobile application's effectiveness beyond controlled testing on the 890-image dataset (Section 3.3). Considerations include training

local agricultural extension workers to guide farmers in using the app, mitigating issues like poor lighting or inconsistent image quality by incorporating real-time feedback in the interface, and establishing a periodic update mechanism to refine the model with new field data, maintaining accuracy as disease patterns evolve.Power management is also addressed by optimizing the app to minimize battery drain on low-end devices, while partnerships with local cooperatives are proposed to distribute preloaded smartphones or facilitate access, ensuring widespread adoption in remote coffee-growing regions with limited technological infrastructure.

Figure 4.29 illustrates the model deployment workflow for both the mobile and web applications, aligning with the field deployment considerations outlined above, including Agriculture expert contact feauture, real-time feedback integration, and Result displays in local language text and audio.

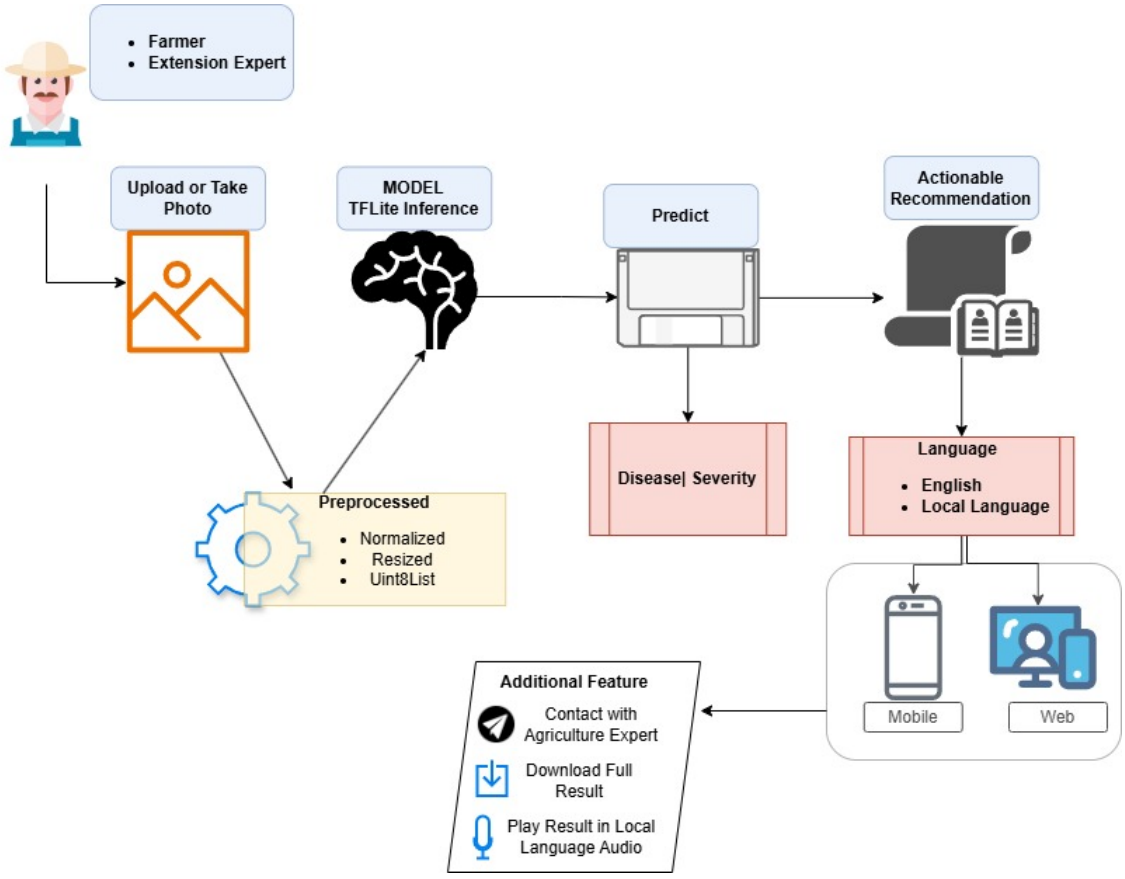


Figure 4.29 Model Deployment Workflow for Mobile App and Web App

4.7 Chapter Summary

Chapter 4 evaluates the performance of the proposed CNN+CBAM+Multi-Task model and its variants for classifying coffee leaf diseases (Healthy, Phoma, Miner) and predicting severity levels (Asymptomatic, Mild, Severe) using an 890-image dataset from Ethiopia's Jimma, Sidama, and Yirgacheffe regions, addressing the smallholder coffee challenges identified in Chapter 1 and technological gaps from Chapter 2. The experimental setup (Section 4.1) is meticulously designed, incorporating dataset partitioning, preprocessing, augmentation, and training on an NVIDIA RTX 3090, ensuring reproducibility and robustness. Results for the proposed model (Section 4.2) demonstrate exceptional test accuracies of 97.75% for disease and 93.26% for severity, with CBAM enhancing feature extraction, as visualized through Grad-CAM and confusion matrices, and multi-task metrics (94.76% weighted accuracy, 95.5% combined F1-score) affirming its dual-task efficacy.

Ablation studies (Section 4.3) reveal CBAM's critical role, with its removal reducing accuracies to 89.89% (disease) and 80.90% (severity), and separate task models dropping to 92.13% and 87.64%, underscoring multi-task synergy. Transfer learning evaluation (Section 4.4) benchmarks ResNet50, VGG16, MobileNetV2, and MobileNetV2+CBAM+MTL, with the latter achieving 97.75% disease and 94.38% severity accuracies, outperforming the base model and others due to its lightweight design and CBAM integration. Comparative analysis (Section 4.5) confirms MobileNetV2+CBAM+MTL as the optimal choice, balancing accuracy and efficiency, while deployment strategies (Section 4.6) outline its optimization via TensorFlow Lite, user-friendly mobile interface, and field considerations for Ethiopia's resource-constrained context. These findings validate the methodology from Chapter 3, offering a scalable solution for coffee disease management, with implications for future enhancements explored in Chapter 5.

5 Conclusion and Future Work

5.1 Main Conclusions

This thesis addresses the critical challenge of managing coffee leaf diseases in Ethiopia's Arabica coffee sector, where smallholder farmers in Ethiopia face 20-30% yield losses due to Phoma leaf spot and Coffee Leaf Miner. This study aimed to develop an automated, deep learning- driven system to classify disease types (Healthy, Phoma, Miner) and severity levels (Asymptomatic, Mild, Severe), enhancing diagnostics beyond the inaccurate manual methods (65% accuracy) identified in Chapter 1.

The research was conducted systematically. Chapter 2 highlights gaps in severity staging and lightweight solutions for Ethiopia's resource-constrained context, justifying an innovative approach. In Chapter 3, the CNN+CBAM+Multi-Task model is proposed, built from scratch using an 890-image dataset, integrating CNNs for feature extraction, CBAM for attention, and multi-task learning for dual outputs. Ablation studies and transfer learning (VGG16, ResNet50, and MobileNetV2) were conducted to refine the design. Chapter 4 revealed that while the proposed model achieved strong test accuracies (97.75% disease, 93.26% severity), the MobileNetV2+CBAM+Multi-Task variant, leveraging pre-trained efficiency, outperformed it (97.75% disease, 94.38% severity) and all baselines, including ablation variants (e.g., 89.89% disease without CBAM) and transfer learning models (e.g., ResNet50: 96.63% disease, 87.64% severity). This was validated using metrics such as precision, recall, and F1-scores, with deployment optimised via TensorFlow Lite for mobile and web use.

The results demonstrate that building the CNN+CBAM+Multi-Task model from scratch provided foundational insights, enabling the superior MobileNetV2+CBAM+MTL model to excel by combining a lightweight architecture, attention mechanisms, and multi-task synergy. Key conclusions include: the final model's high accuracy and efficiency address Ethiopia's diagnostic needs, CBAM significantly enhances feature focus (e.g., boosting severity by 12.36% over no-CBAM ablation), and multi-task learning improves performance by

5-6% over single-task variants. Deployed as a mobile app, it offers real-time, accessible disease management, reducing yield losses and pesticide overuse, thus supporting smallholders' livelihoods and Ethiopia's coffee economy (5-10% GDP).

In summary, this study successfully developed and validated a scalable, farmer-centric solution that surpassed traditional methods and prior deep learning approaches by integrating advanced techniques tailored to Ethiopia's unique challenges. The MobileNetV2+CBAM+Multi-Task model was the optimal outcome, balancing precision and practicality for sustainable coffee production.

5.2 Contribution and Novelty

This makes two primary contributions, each reflecting substantial technical effort and innovation. First, it developed the CNN+CBAM+Multi-Task model from scratch, a robust framework for dual-task classification of coffee leaf diseases and severity. The challenge of limited data (890 images) and dual-task complexity was met through a custom CNN backbone, CBAM's attention mechanism, and multi-task learning, optimized over 100 epochs with extensive preprocessing and augmentation. Ablation studies validated each component's role, with test accuracies of 97.75% (disease) and 93.26% (severity) demonstrating its efficacy. This contribution includes the **application of attention mechanisms in agricultural imaging**, where CBAM's channel and spatial attention refined feature extraction, enabling precise identification of subtle disease patterns (e.g., Phoma's necrotic spots, Miner's tunneling) critical for early intervention in Ethiopia's coffee fields.

Second, it advanced this foundation with the MobileNetV2+CBAM+Multi-Task model, achieving superior accuracies (97.75% disease, 94.38% severity) and deployment readiness. The technical hurdle of resource constraints was overcome by leveraging MobileNetV2's lightweight design, enhanced by CBAM and multi-task learning, and quantized for mobile use. This required rigorous experimentation comparing against transfer learning models (e.g., VGG16, ResNet50) and ablation variants culminating in a tool tailored for Ethiopia's smallholders, validated through metrics like a 95.5% combined F1-score. A key aspect of this contribution is the **practical deployment strategy**, outlined in Chapter 4 and rooted in Chapter 1's objectives (Section 1.3), which emphasizes accessible web and mobile appli-

cations. This strategy ensures real-time diagnostics for farmers and extension workers, addressing Chapter 1's noted barriers (e.g., 1:1,000 agent-to-farmer ratio, limited smartphone access) with an offline-capable, bilingual (Amharic-English) app, directly supporting over 15 million coffee-dependent individuals.

The novelty lies in integrating CBAM and multi-task learning with MobileNetV2 for coffee-specific diagnostics, an approach rare in prior studies (e.g., DenseNet-264 at 97.17% without severity staging). Unlike resource-intensive models (e.g., VGG16) or single-task frameworks, this solution offers enhanced feature focus and efficiency, improving severity prediction by 1.12% over the proposed model and outperforming all baselines. Its scientific significance advances multi-task agricultural AI, while its practical deployment empowers farmers, filling gaps identified in Chapter 2 and aligning with Chapter 1's vision for sustainable coffee resilience.

5.3 Future Work

The MobileNetV2+CBAM+MTL model's high accuracies (97.75% disease, 94.38% severity) and lightweight design position it as a transformative tool for Ethiopia's coffee sector, with potential extension to other crops (e.g., tea, maize) or regions facing similar disease threats. Its deployment via mobile apps could integrate with IoT systems for real-time field monitoring or link to national cooperatives, scaling its impact on Ethiopia's 15 million coffee-dependent individuals and 30% foreign exchange earnings.

However, limitations persist. The dataset (890 images) from Jimma, Sidama, and Yirgacheffe may not fully represent seasonal or regional variability, potentially reducing generalizability. Severity classification (94.38%) shows minor errors in Mild–Severe transitions, as seen in confusion matrices, due to subtle symptom overlaps. Time constraints limited the exploration of ensemble methods or dynamic attention refinements, which could further enhance accuracy. Additionally, the model lacks explicit constraints between disease and severity outputs, so invalid combinations are theoretically possible, though not observed in testing. Future work may address this through architectural constraints or post-processing rules to enforce valid output pairs.

Future work should expand the dataset across Ethiopia's coffee zones and seasons,

testing robustness under diverse conditions (e.g., drought, poor lighting). Semi-supervised learning with unlabeled field data could improve severity detection, while ensemble techniques might push accuracies closer to 99%. Field trials with farmers and extension agents could refine the app's usability, addressing connectivity and device access barriers noted in Chapter 1. Exploring edge computing or drone-based imaging could further reduce latency and broaden coverage, solidifying this solution's role in precision agriculture and global sustainability goals.

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Appendix 1 Symptoms and Prevention Methods for Coffee Diseases

Table 1 Symptoms and Prevention Methods for Coffee Diseases

Disease	Symptoms	Prevention
Coffee Leaf Miner	• Irregular brown spots on upper leaf and middle surfaces due to larvae feeding. • Severe infestations can lead to significant leaf drop, reducing photosynthesis and potentially causing yield losses.	• Effectively manage the first generation of leaf miners to prevent population buildup. • Plant windbreaks with species like grevillea or leucaena to reduce adult moth dispersal. • Regular weeding to eliminate alternative hosts. • Tilling the soil post-harvest to destroy pupae in the soil.
Phoma Leaf Spot	• Small discolored spots on leaves expanding into larger dark lesions. • Necrosis leading to tissue death. • Severe infections can cause premature leaf drop. • Young branches may wither and die.	• Ensure proper spacing between plants for adequate air circulation. • Regularly prune infected branches and remove fallen leaves. • Destroy or compost infected plant debris to prevent fungal overwintering. • Apply appropriate fungicides during early infection stages, following local agricultural guidelines. • Plant coffee varieties resistant to Phoma leaf spot. • Conduct regular field inspections for early disease detection.